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वार्षिक प्रतिवेदन



Learning Strategies (%)

85.78

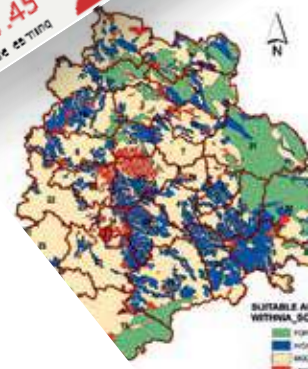
Waiting access

83.33

Research & development

27.45

Selective testing



भा.कृ.अनु.प.— नार्म

ICAR-NAARM

2017-18

ANNUAL REPORT



भाकअनुप-राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी
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Participants of Workshop on 'Let's Listen to Farmers' with Hon'ble MoS for Agriculture and Farmers' Welfare, Shri G.S. Shekhawat



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Abbreviations

ABF	Agri-Biotech Foundation
ABM	Agribusiness Management
AERA	Agricultural Economics Research Association
AICTE	All India Council for Technical Education
a-IDEA	Association for Innovation Development of Entrepreneurship in Agriculture
ANGRAU	Acharya N. G. Ranga Agricultural University
APMC	Agricultural Produce Market Committee
ASCI	Administrative Staff College of India
ASRB	Agricultural Scientists Recruitment Board
ASTI	Agricultural Science and Technology Indicators
ATARI	Agricultural Technology Application Research Institutes
ATMA	Agriculture Technology Management Agency
AUs	Agricultural Universities
CAB	College of Agricultural Banking
CAI	Centre for Agri-Innovation
CGIAR	Consultative Group for International Agricultural Research
CIFA	Central Institute of Freshwater Aquaculture
CIFE	Central Institute of Fisheries Education
CIMMYT	International Maize and Wheat Improvement Center
CIWA	Central Institute for Women in Agriculture
COLLAge	Centre for Open and Lifelong Learning in Agriculture
CRIDA	Central Research Institute for Dryland Agriculture
CRISP	Centre for Remote Imaging, Sensing and Processing
CTCRI	Central Tuber Crops Research Institute
CVI	Cumulative Vulnerability Index
DBT	Department of Biotechnology
DFI	Doubling Farmers Income
DKMA	Directorate of Knowledge Management in Agriculture
DST	Department of Science and Technology
EDP	Executive Development Program
e-NAM	National Agriculture Market
FAI	Fertiliser Association of India
FET	Field Experience Training
FFP	Farmer FIRST Programme
FIRST	Farm, Innovations, Resources, Science and Technology
FOCARS	Foundation Course on Agricultural Research Management
FOCAU	Foundation Course for Faculty of Agricultural Universities
FOSS	Free and Open Source Software
GIS	Geographic Information System
GoI	Government of India
GPS	Global Positioning System

GST	Goods and Services Tax
GTWG	Global Technology Watch Group
IARI	Indian Agricultural Research Institute
IASRI	Indian Agricultural Statistics Research Institute
ICAR	Indian Council of Agricultural Research
ICBS	International Conference on Biocontrol for sustainable insect Pest Management
ICRISAT	International Crops Research Institute for the Semi-Arid Tropics
ICT	Information and Communication Technology
IFAF	Indian Fisheries and Aquaculture Forum
IFPRI	International Food Policy Research Institute
IGKV	Indira Gandhi Krishi Vishwavidyalaya
IGNFA	Indira Gandhi National Forest Academy
IIHR	Indian Institute of Horticultural Research
IIM	Indian Institute of Management
IIMR	Indian Institute of Millets Research
IIRR	Indian Institute of Rice Research
IISASTR	International Institute of SAS Training and Research
ILWIS	Integrated Land and Water Information System
IPE	Institute of Public Enterprise
IPNI	International Plant Names Index
ISB	Indian School of Business
ISHS	International Society of Horticulture Science
ITK	Indigenous Technical knowledge
JNKVV	Jawaharlal Nehru Krishi Vishwa Vidyalaya
KVAFSU	Karnataka Veterinary, Animal and Fisheries Sciences University
KVKs	Krishi Vigyan Kendras
MANAGE	National Institute of Agricultural Extension Management
MBA	Master of Business Administration
MCCIA	Maharashtra Chamber of Commerce, Industry & Agriculture
MDP	Management Development Programme
MOOC	Massive Open Online Courses
MoU	Memorandum of Understanding
MPCC	Monthly Per Capita Consumption
MWDS	Mean Weighted Discrepancy Scores
NAARM	National Academy of Agricultural Research Management
NAAS	National Academy of Agricultural Sciences
NAIF	National Agriculture Innovation Fund
NAPCC	National Action Plan for Climate Change
NARES	National Agricultural Research and Education System
NARS	National Agricultural Research Systems
NASC	National Agricultural Science Complex

NBA	National Board of Accreditation
NBP	Need-based programmes
NBSSLUP	National Bureau of Soil Survey and Land Use Planning
NCDEX	National Commodity & Derivatives Exchange Limited
NDRI	National Dairy Research Institute
NDVI	Normalized Difference Vegetation Index
NFCL	Nagarjuna Fertilizers and Chemicals Limited
NFDB	National Fisheries Development Board
NGOs	Non-Government Organizations
NIAP	National Institute of Agricultural Economics and Policy Research
NIPHM	National Institute of Plant Health Management
NIRD & PR	National Institute of Rural Development and Panchayat Raj
OER	Open Educational Resources
PCs	Producer Companies
PGDMA	Post Graduate Diploma in Management- Agriculture
PGDTMA	Post Graduate Diploma in Technology Management in Agriculture
PJTSAU	Professor Jayashankar Telangana State Agricultural University
PME	Priority setting, Monitoring and Evaluation
PRSI	Public Relations Society of India
PVNRTVU	P.V. Narasimha Rao Telangana Veterinary University
QRT	Quinquennial Review Team
RGMVP	Rajiv Gandhi Mahila Vikas Paryojana
RKVY	Rashtriya Krishi Vikas Yojana
SAIDP	State Agriculture Infrastructure Development Plans
SAMETI	State Agricultural Management and Extension Training Institute
SAU's	State Agricultural Universities
SEVI	Socio-Economic Vulnerability Index
SMD	Subject Matter Division
SSRN	Social Science Research Network
STEM	Society for Technology Management
TAAS	Trust for Advancement of Agricultural Sciences
TANUVAS	Tamil Nadu Veterinary and Animal Sciences University
TBI	Technology Business Incubator
TElAgE	Technology Enhanced Learning in Agricultural Education
TIFAC	Technology Information, Processing and Assessment Council
TISS	Tata Institute of Social Sciences
TNAU	Tamil Nadu Agricultural University
UoH	University of Hyderabad
VNMAU	Vasantrao Naik Marathwada Krishi Vidyapeeth



निदेशक की कलम से



डा. सि.एच. श्रीनिवास राव
निदेशक

भारतीय कृषि अनुसंधान परिषद (ICAR) के अंतर्गत सभी संस्थानों में भाकृअनुप – राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी (ICAR - NAARM), हैदराबाद का द्वितीय स्थान है। अर्थशास्त्र के सभी सेक्टरों में मानव संसाधन विकास और क्षमता को मजबूती प्रदान करने की बढ़ रही जरूरतों की पृष्ठभूमि में अकादमी, राष्ट्रीय कृषि अनुसंधान एवं शिक्षा प्रणाली (NARES) के अनुसंधान पेशेवरों के क्षमता निर्माण कार्य में निरन्तर सक्रिय है। इस परिप्रेक्ष्य में, मुझे वर्ष 2017-18 के दौरान कृषि के क्षेत्र में अनुसंधान, क्षमता निर्माण, शिक्षा एवं उद्यमशीलता विकास के सभी आयामों पर अकादमी की प्रगति एवं उपलब्धियों को प्रस्तुत करते हुए खुशी का अनुभव हो रहा है। यह वर्ष भाकृअनुप – राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी (ICAR - NAARM), हैदराबाद के लिए अत्यधिक लाभप्रद रहा जैसा कि इस वर्ष अकादमी द्वारा न केवल बड़ी संख्या (54) में प्रशिक्षण कार्यक्रमों का आयोजन करके वरन् अकादमी के छः रणनीतिपरक विषयी क्षेत्रों में कार्यक्रमों की क्षमता में विविधीकरण करके अपनी सीमाओं को एक नया विस्तार दिया गया है। कुल मिलाकर, विभिन्न कार्यक्रमों के आयोजन से देश के विभिन्न भागों के लगभग 2500 प्रतिभागी लाभान्वित हुए। भाकृअनुप – राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी (ICAR - NAARM), हैदराबाद द्वारा 84वें नव नियुक्त एआरएस प्रोबेशनर्स के लिए दो फोकार्स (106 एवं 107) कार्यक्रम आयोजित किए गए। फोकार्स की तर्ज पर, कृषि विश्वविद्यालयों के संकाय सदस्यों के लिए एक-एक महीने के दो सेतु कार्यक्रम (Bridge Programme) भी आयोजित किए गए। अपने नियमित प्रशिक्षण कार्यक्रम यथा एमडीपी, ईडीपी, पुनश्चर्या पाठ्यक्रम, जरूरत आधारित कार्यक्रम एवं ऑफ-कैम्पस प्रशिक्षण कार्यक्रम के अलावा, अकादमी मैसिव ओपन ऑन-लाइन कोर्स (MOOC) के माध्यम से ऐसे पेशेवरों की जरूरतों को पूरा करने में सफल रहा जो कि ऑन-कैम्पस प्रशिक्षण कार्यक्रमों में भाग लेने में असमर्थ हैं।

साथ ही अकादमी द्वारा अनुसंधान गतिविधियों को भी पर्याप्त महत्व दिया जा रहा है, जानकारी आधार को शामिल करके अनुसंधान गतिविधियों के परिणामों को क्षमता निर्माण कार्यक्रमों में एकीकृत किया जा रहा है। अकादमी का संकाय स्टाफ रिपोर्टाधीन अवधि में विभिन्न सार्वजनिक एवं निजी एजेंसियों से अनुबंध एवं परामर्शी परियोजनाओं के माध्यम से बड़ी मात्रा में धनराशि हासिल करने में सफल रहा। ये परियोजनाएं सामयिक प्रकृति की थीं। इसका प्रमाण उच्च प्रभाव वाली राष्ट्रीय एवं अंतर्राष्ट्रीय पत्रिकाओं में प्रकाशित गुणवत्ता अनुसंधान प्रकाशन, नीति संक्षेप और अनुसंधान परियोजनाएं हैं। इनसे अकादमी की पहुंच और विजिबिलिटी में सुधार लाने में मदद मिली है।

उच्चतर शिक्षा कार्यक्रम से भावी छात्रों के साथ साथ रोजगारकर्ता भी अत्यधिक आकर्षित हुए। पूर्णकालिक दो वर्षीय कृषि प्रबंधन में स्नातकोत्तर डिप्लोमा (PGDMA) में बड़ी संख्या में छात्र आकर्षित हुए और अपनी परम्परा के अनुसार अति सम्माननीय वेतन पैकेज के साथ प्रतिष्ठित कृषि व्यवसाय कम्पनियों में 100 प्रतिशत छात्रों को रोजगार मिला। कृषि प्रबंधन में स्नातकोत्तर डिप्लोमा (PGDMA) को एक और विशिष्ट सफलता तब मिली जब राष्ट्रीय प्रत्यायन बोर्ड (NBA) द्वारा इसे प्रत्यायन प्रदान किया गया। कृषि में प्रौद्योगिकी प्रबंधन में स्नातकोत्तर डिप्लोमा (PGDTMA) (एक वर्षीय) को उत्साहजनक प्रतिक्रिया प्राप्त होती रही।

उत्कृष्टता के दो केन्द्रों यथा कृषि इनोवेशन केन्द्र (CAI) तथा कृषि में ओपन एवं जीवनपर्यन्त शिक्षण के लिए केन्द्र (COLLAGE) द्वारा रिपोर्टाधीन वर्ष के दौरान बहुत अच्छी प्रगति की गई।

व्यवसाय इनक्यूबेटर केन्द्र यथा कृषि इनोवेशन केन्द्र के अंतर्गत a-IDEA द्वारा बड़ी संख्या में कृषि स्टार्ट-अप और कृषि उद्यमियों को आकर्षित करने के प्रयोजन से छः मुख्य शहरों में रोड शो करके खाद्य एवं कृषि व्यवसाय एक्सीलरेटर 2.0 जिसे एग्री-उड़ान कहा जाता है, को सफलतापूर्वक प्रारंभ किया गया। कृषि जैव-प्रौद्योगिकी स्टार्ट-अप के लिए एक बायो-इनक्यूबेटर, बायोनेस्ट (BioNest) की स्थापना करने हेतु जैव प्रौद्योगिकी विभाग, भारत सरकार के BIRAC से वित्तीय सहयोग हासिल करने में भी अकादमी सफल रहा। मॉडल डिजिटल पाठ्यक्रमों का विकास करने, सुदूर तथा ऑन-लाइन कार्यक्रमों को प्रस्तुत करने, विर्चुल कक्षाएं, MOOCs, आदि आयोजित करने हेतु COLLAge के अंतर्गत, एक स्टेट ऑफ आर्ट (उत्कृष्ट) सुविधा उत्पन्न की गई। इससे अकादमी को एक टिकाऊ रीति में राजस्व का सृजन करने में मदद मिली।

भाकूअनुप – राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी (ICAR - NAARM), हैदराबाद द्वारा अनेक सार्वजनिक एवं निजी संस्थानों के साथ बहुत अच्छी भागीदारी विकसित की गई है। अकादमी द्वारा भारत सरकार के अनेक अग्रणी कार्यक्रमों यथा स्वच्छ भारत, डिजिटल इंडिया, मेरा गांव – मेरा गौरव, फार्मर्स फर्स्ट आदि को सफलतापूर्वक लागू किया जा रहा है। मुझे यह सूचित करते हुए हर्ष का अनुभव हो रहा है कि संस्थान को “राजर्षि टंडन राजभाषा पुरस्कार” और “डिजिटल मीडिया इनीशिएटिव पुरस्कार प्राप्त हुआ है। अकादमी के उल्लेखनीय उपलब्धियों की सराहना करते हुए श्री हुक्मदेव नारायण यादव की अध्यक्षता में कृषि पर संसदीय स्थाई समिति द्वारा अकादमी का उन्नयन मानद विश्वविद्यालय के रूप में करने की सिफारिश की गई।

पूरे वर्ष परिषद से बाधारहित एवं सकारात्मक सहयोग मिलने के परिणामस्वरूप ही अकादमी के संकाय एवं स्टाफ द्वारा प्रगति करना संभव हुआ। अकादमी को निरन्तर प्रोत्साहन एवं मार्गदर्शन देने के लिए डॉ. त्रिलोचन महापात्र, सचिव (डेयर) एवं महानिदेशक (भाकूअनुप); तथा श्री छबिलेन्द्र राऊल, विशेष सचिव (भाकूअनुप) के प्रति सम्पूर्ण नार्म परिवार अपना हार्दिक आभार व्यक्त करता है। साथ ही हम अकादमी की गतिविधियों को आगे बढ़ाने में अपना पूर्ण सहयोग देने के लिए डॉ. एन.एस. राठौर, उप महानिदेशक (शिक्षा), भाकूअनुप के आभारी हैं।

मैं, अकादमी को मजबूती प्रदान करने में भाकूअनुप – राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी (ICAR - NAARM), हैदराबाद के संयुक्त निदेशक (शिक्षा), समस्त सभागाध्यक्षों, विभिन्न अनुभागों के अध्यक्ष, संयुक्त निदेशक (प्रशासन) मुख्य वित्त व लेखा अधिकारी और समस्त स्टाफ द्वारा किए गए योगदान के प्रति आभार व्यक्त करता हूँ।

इस वार्षिक प्रतिवेदन का संकलन व सम्पादन करने में किए गए अथक प्रयासों के लिए मैं, सम्पादन मण्डल के सदस्यों डॉ. रंजीत कुमार, डॉ. पी. रमेश, डॉ. डी. थम्मी राजू, डॉ. पी. कृष्णन, डॉ. एम. बालाकृष्णनन, डॉ. पी. वेंकटसन तथा डॉ. संजीव कुमार की सराहना करता हूँ। वर्ष को सफल बनाने में की गई कड़ी मेहनत और दृढ़ता के लिए समस्त नार्म परिवार प्रशंसा का पात्र है।

दिनांक : 01 जुलाई, 2018

स्थान : हैदराबाद

सिपच डी. विवास शर्मा



Director's Desk



Dr. Ch. Srinivasa Rao
Director

The ICAR-NAARM has unique place among all the institutes under the Indian Council of Agricultural Research. In the backdrop of growing need of human resource development and capacity strengthening in all the sectors of the economy, the Academy continues to build the capability of the research professionals of the National Agricultural Research, Education and Extension System (NAREES) in India. With this perspective, I am happy to present the progress and achievements of the Academy during the year 2017-18 in all spheres of research, capacity building, education and entrepreneurship development in the field of agriculture. The year has been very much rewarding for the Academy, as we raised the bars further by conducting not only large number (54) of training programmes, but also by diversifying the range of the programmes across the six strategic thematic areas of the Academy. In all, about 2500 participants from different parts of the country got benefitted through various programmes. The Academy organised two FOCARS (106th & 107th) programmes for the 84 newly recruited ARS probationers. On the pattern of FOCARS, two 1-month abridged programme for faculty of agricultural universities were also organised. Apart from regular training programmes in the form of MDP, EDP, Refresher Course, need-based programmes and off-campus training programme; through Massive Open Online Course (MOOC), the Academy has been successful in catering to the need of those professionals who are constrained to participate in on-the-campus training programmes.

The Academy is also giving due importance to research activity, the output of which are getting integrated into the capacity building programmes by adding the knowledge base. Faculty of the Academy were able to attract large amount of fund through contract and consultancy projects during this period from various public and private agencies. These projects were of topical in nature. This is evidenced from the quality research publications in high impact national and international journals, policy briefs and research reports. These have helped in improving the reach and visibility of the Academy.

The higher education programme got very good traction among the prospective students as well as employer. The 2-year full-time Post-Graduate Diploma in Management in Agriculture (PGDMA) attracted large number of students and continued its tradition of maintaining 100% placement in reputed agribusiness companies with very respectable package. The PGDMA programme got another feather in its cap by getting it accredited from National Board of Accreditation (NBA). The 1-year Post Graduate Diploma in Technology Management in Agriculture (PGDTMA) also continued to receive overwhelming response.

The two Centres of Excellence- Centre of Agri-Innovation (CAI) and Centre for Open and Lifelong Learning in Agriculture (COLLAge) made very good progress during the year. The business incubator centre viz. a-IDEA, under CAI, successfully launched the Food & Agribusiness Accelerator 2.0 called Agri-Udaan, followed by road shows in six major cities attracting large number of agri-start ups and agripreneurs enthusiasts. It also succeeded in getting the financial support from BIRAC of Department of Biotechnology, GoI to set up BioNest, a Bio-Incubator for Agri-Biotechnology Startups. Under COLLAge, an state-of-art facilities has been created to develop model digital courses, offering distance and online programmes, conduct virtual classes, MOOCs, etc. This has turned to be a sustainable revenue generation model for the Academy.

The Academy has forged very good partnership with various institutions- public and private. We have been successfully implementing various flagship programmes of Government of India, like Swachh Bharat, Digital India, Mera Gaon Mera Gaurav, Farmers' FIRST etc. I am also delighted to inform that the Academy received 'Rajashri Tandon Rajbhasha Puraskar Award' and bagged 'Digital Media Initiatives Award'. Applauding the astounding achievement of the Academy, Standing Parliamentary Committee on Agriculture, chaired by Sh. Hukmdev Narayan Yadav, recommended the Academy to be upgraded as Deemed-to-be-University.

All the progress made by the faculty and staffs of the Academy could have been possible mainly due to the unstinted and unconditional support from the Council. The entire NAARM family express sincere thanks and gratitude to Dr. Trilochan Mohapatra, the Secretary (DARE) and Director General (ICAR) and Shri Chhabilendra Roul, Special Secretary (ICAR) for continuous guidance and encouragement. We are thankful to Dr. N.S. Rathore, DDG (Edn), ICAR for providing whole hearted support in pursuing the activities of the Academy.

I acknowledge the contributions of Joint Director, all Heads of the Divisions, various sections chairman, Joint Director (Administration), Chief Finance Officer and all the staff of ICAR-NAARM in strengthening the Academy.

I place on record my appreciation for the sincere efforts put in by Drs. Ranjit Kumar, P Ramesh, D Thammi Raju, P Krishnan, M Balakrishnan, P Venkatesan and Sanjiv Kumar for compiling and editing the report. I would also like to place on record the hard work and perseverance of all the NAARM family in making the year successful.

Date: 01 July, 2018

Place: Hyderabad



(Ch. Srinivasa Rao)
DIRECTOR



कार्यकारी सारांश

अकादमी द्वारा भारत में सम्पूर्ण राष्ट्रीय कृषि अनुसंधान, शिक्षा एवं प्रसार प्रणाली (NAREES) के पेशवरों की क्षमता को बढ़ाने में केन्द्रीय भूमिका निभाई जाती है। वर्ष 2017-18 में, अकादमी द्वारा अनुसंधान योगदान, नीतिगत सहयोग और शिक्षा के साथ साथ राष्ट्रीय कृषि अनुसंधान, शिक्षा एवं प्रसार प्रणाली (NAREES) के भीतर एवं बाहर विभिन्न हितधारकों के मानव संसाधन विकास कार्य में उल्लेखनीय योगदान दिया गया। इसके अलावा, कृषि सेक्टर में उद्यमशीलता विकास के क्षेत्र में अकादमी द्वारा किए गए प्रयासों को सहर्ष स्वीकार किया गया और उनकी सराहना की गई। अकादमी ने विभिन्न कृषि विश्वविद्यालयों के प्रवेश स्तरीय सहायक प्राध्यापकों और कृषि विज्ञान केन्द्रों (KVK) के समन्वयकों के लिए क्षमता निर्माण कार्यक्रम की शुरुआत करके अपनी पहुंच और कवरेज को विस्तार दिया है।

क्षमता निर्माण करना भाकृअनुप – राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी (ICAR-NAARM), हैदराबाद का एक प्रमुख अधिदेश है। वर्ष 2017-18 में अकादमी द्वारा कुल 54 क्षमता निर्माण कार्यक्रम आयोजित किए गए। इनमें से, 36 कार्यक्रम जरूरत आधारित थे। देश के विभिन्न भागों में दस ऑफ-कैम्पस कार्यक्रम आयोजित किए गए। इन कार्यक्रमों से एक अथवा अन्य प्रकार के प्रशिक्षण कार्यक्रमों में कुल 2,495 प्रशिक्षुओं को लाभ मिला जिनमें 988 मैसिव ओपन ऑन-लाइन पाठ्यक्रम (MOOC) प्रतिभागी भी शामिल थे। अकादमी का अग्रणी कार्यक्रम जो कि कृषि अनुसंधान सेवाओं के लिए बुनियादी पाठ्यक्रम **Foundation Course for Agricultural Research Services (FOCARS)** के नाम से प्रचलित है, का आयोजन एक बार जुलाई, 2017 में किया गया और दूसरी बार जनवरी, 2018 में किया गया। इनमें कुल मिलाकर 84 नव नियुक्त एआरएस प्रोबेशनर्स शामिल थे। दो बार आयोजित कृषि विश्वविद्यालयों के संकाय के लिए बुनियादी पाठ्यक्रम (**Foundation Course for Faculty of Agricultural Universities (FOCAU)**) में कुल 102 प्रतिभागियों ने भाग लिया। इस पाठ्यक्रम की डिजाइन फोकर्स की तर्ज पर ही तैयार की गई है लेकिन इसकी अवधि एक माह है। वर्ष 2017-18 के दौरान, कार्यकारी विकास कार्यक्रम (EDP) जिसकी डिजाइन भारतीय कृषि अनुसंधान परिषद के नव नियुक्त अनुसंधान प्रबंधन पद संवर्ग के लिए तैयार की गई है, ने दस प्रतिभागियों को आकर्षित किया जबकि भारतीय कृषि अनुसंधान परिषद के आरएमपी-पूर्व संवर्ग के लिए प्रबंधन विकास कार्यक्रम (MDP) में 41 प्रतिभागियों ने भाग लिया। अकादमी द्वारा जरूरत आधारित कार्यक्रमों के माध्यम से भारतीय कृषि अनुसंधान परिषद के विभिन्न संस्थानों के कुल 113 तकनीकी एवं 240 प्रशासनिक कर्मियों को प्रशिक्षण प्रदान किया गया। कुशल सहायी स्टाफ की क्षमताओं में वृद्धि करने के उद्देश्य से भी एक प्रशिक्षण कार्यक्रम चलाया गया जिसमें 27 स्टाफ सदस्यों ने भाग लिया। अकादमी द्वारा जरूरत आधारित क्षमता संवर्धन कार्यक्रमों के माध्यम से अनेक विश्वविद्यालयों की जरूरतों को पूरा किया गया। इनमें शामिल हैं : कर्नाटक पशु चिकित्सा, मात्स्यिकी एवं पशु विज्ञान विश्वविद्यालय, बीदर; भाकृअनुप – राष्ट्रीय डेरी अनुसंधान संस्थान (ICAR- NDRI), करनाल; वसंतराव नाइक मराठवाडा कृषि विद्यापीठ, परभनी; बागवानी विज्ञान विश्वविद्यालय, शिवमोगा; दंतेवाडे राष्ट्रीय कॉलेज, नागपुर (शिवाजी एजुकेशनल ट्रस्ट, अमरावती); श्री वेंकटेश्वर पशु चिकित्सा विश्वविद्यालय, तिरुपति; जवाहर लाल नेहरू कृषि विश्वविद्यालय, जबलपुर; तथा भाकृअनुप – केन्द्रीय मात्स्यिकी शिक्षा संस्थान (ICAR- CIFE), मुम्बई। रिपोर्टीधीन वर्ष 2017-18 में भाकृअनुप – राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी (ICAR-NAARM), हैदराबाद विभिन्न प्रायोजित कार्यक्रमों का आयोजन करके लगभग रुपये 68 लाख का राजस्व सृजन करने में सफल रहा।

शिक्षा, अकादमी के एक प्रमुख अधिदेशों में शामिल है जिसे दो शैक्षणिक कार्यक्रम प्रस्तुत करके पूरा किया गया। अकादमी द्वारा पूर्णतया आवासीय प्रबंधन (कृषि) में दो वर्षीय स्नातकोत्तर डिप्लोमा – पीजीडीएमए की सुविधा प्रदान की जाती है जिसमें 399 आवेदकों को अपनी ओर आकर्षित किया। अंततः प्रवेश के लिए 34 आवेदकों का चयन किया गया। उत्तीर्ण बैच के छात्र विभिन्न कम्पनियों में औसतन प्रति वर्ष रुपये 7.25 लाख का पैकेज हासिल करने में सफल रहे। वर्ष 2017-18 के दौरान, कृषि के प्रौद्योगिकी प्रबंधन में स्नातकोत्तर डिप्लोमा (एक वर्षीय) (PGDTM-A) में 65

छात्रों ने प्रवेश लिया। इनमें से कई सेवारत अभ्यर्थी थे। शैक्षणिक कार्यक्रमों से अकादमी को वर्ष 2017-18 में लगभग रुपये 2.1 करोड़ का राजस्व सृजन करने में मदद मिली।

अनुसंधान के मोर्चे पर, भाकृअनुप – राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी (ICAR-NAARM), हैदराबाद द्वारा कुल 34 अनुसंधान परियोजनाएं चलाई गईं जिनमें घरेलू तथा बाह्य वित्त पोषित अनुबंध अनुसंधान परियोजना प्रत्येक की संख्या 16-16 थी। संस्थान, दो परामर्शी परियोजनाओं को हासिल करने में भी सफल रहा। प्रायोजित परियोजनाओं में से, सात परियोजनाएं विभिन्न भाकृअनुप स्कीमों से सम्बद्ध थीं। अन्य परियोजनाओं में जैव प्रौद्योगिकी विभाग (DBT); विज्ञान व प्रौद्योगिकी विभाग (DST); तम्बाकू बोर्ड, तेलंगाना सरकार; आईएफपीआरआई; मायको मोनसैन्टो बायोटेक (इंडिया) आदि शामिल थे। अनुसंधान परियोजनाएं, अकादमी के सभी छः संभागीय विषयों पर केन्द्रित हैं। परियोजनाओं को छः विषयी क्षेत्रों में वर्गीकृत किया गया है जैसे कि सम्यक प्रगति के लिए कृषि व्यवसाय प्रबंधन; समृद्ध शिक्षण गुणवत्ता एवं वातावरण के लिए शिक्षा प्रणालियां प्रबंधन; नेतृत्वशीलता तथा शासन प्रणाली के लिए क्षमताओं में सुधार करना; बाजार चालित वातावरण में प्रसार प्रणालियां प्रबंधन; नवोन्मेष एवं शासन प्रणाली को बढ़ावा देने के लिए सूचना एवं संचार प्रबंधन; तथा नवोन्मेष एवं टिकाऊ विकास के लिए विज्ञान एवं प्रौद्योगिकी को गतिशील बनाना।

वर्ष 2017-18 के दौरान अकादमी के संकाय सदस्यों द्वारा विभिन्न प्रारूप में कुल 245 अनुसंधान प्रकाशनों में अपना योगदान दिया गया। सहकर्मी समीक्षित अनुसंधान पेपर्स में 110 पेपर शामिल थे जो कि वर्ष 2017-18 में संस्थान संकाय के कुल प्रकाशनों का 43 प्रतिशत से भी अधिक है। प्रकाशनों के अन्य प्रारूपों में शामिल था : पुस्तकें, पुस्तक अध्याय, लोकप्रिय लेख, प्रशिक्षण मैनुअल, सेमिनार का कार्यवृत्त तथा अनुसंधान परियोजनाएं आदि।

भारतीय कृषि अनुसंधान परिषद द्वारा प्रारंभ की गई पहल फार्मर्स फर्स्ट (Farmer FIRST) का भाकृअनुप – राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी (ICAR-NAARM), हैदराबाद एक कंसोर्शिया भागीदार है। रिपोर्टींगीन अवधि में संस्थान, क्षमता निर्माण गतिविधियों और अनेक कार्यशालाओं का आयोजन करने में सक्रिय बना रहा। किसानों से सीधी प्रतिक्रिया जानने के लिए पहली बार एक अनूठी कार्यशाला ‘चलों किसानों की सुनें (Let's Listen to Farmers)’ का आयोजन किया गया। इस कार्यशाला में 25 राज्यों से लगभग 143 किसान सहित कुल 200 प्रतिनिधियों ने भाग लिया।

अकादमी के संकाय सदस्यों की जानकारी को अद्यतन बनाये रखने और शिक्षा क्षेत्र में नवीन घटनाओं के साथ उन्हें अद्यतन बनाये रखने के प्रयोजन से भाकृअनुप – राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी (ICAR-NAARM), हैदराबाद के संकाय सदस्यों को देशभर में आयोजित विभिन्न क्षमता निर्माण कार्यक्रमों में नियमित रूप से भेजा जा रहा है। वर्ष 2017-18 में, भाकृअनुप – राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी (ICAR-NAARM), हैदराबाद के संकाय सदस्यों ने कुल 18 विभिन्न क्षमता निर्माण कार्यक्रमों में भाग लिया।

अकादमी में स्थित कृषि इनोवेशन केन्द्र (Centre for Agri-Innovation) द्वारा शिक्षण सामग्री के माड्यूल्स, दिशानिर्देशों और अन्य प्रारूपों का विकास करके कृषि व्यवसाय इनक्यूबेटर्स को हैंड-होलिंग सहयोग प्रदान किया जाता है। आज तक, कुल 36 इनक्यूबेटीज को शामिल किया गया है और इनमें से 13 को प्रशिक्षित किया जा चुका है। रिपोर्टींगीन वर्ष के दौरान, विभिन्न स्थानों पर अलग-अलग विविध कार्यक्रम आयोजित किए गए जिनमें शामिल हैं : एग्री उद्यान, 6 रोड शो, 3 जागरूकता कार्यशाला और अन्य। अकादमी में एक स्टेट ऑफ दि आर्ट प्रौद्योगिकी संवर्दि

ति शिक्षण संसाधन केन्द्र (TELAGe) की स्थापना की गई ताकि डिजिटल पाठ्यक्रमों के मॉडल का विकास किया जा सके, सुदूर एवं ऑन-लाइन कार्यक्रम चलाए जा सकें, विच्युल कक्षाएं आयोजित की जा सकें और घरेलू कार्यक्रमों को सहयोग किया जा सके। अकादमी में मैसिव ओपन ऑन-लाइन पाठ्यक्रम (MOOC) पाठ्यक्रम चलाए जाते हैं। साथ ही संस्थान द्वारा अन्य एजेंसियों को डिजिटल सामग्री का विकास करने में परामर्शी मोड में सहयोग प्रदान किया जाता है। मैनेज (MANAGE), NIRD & PR जैसे संस्थानों के लिए क्रियान्वित कार्यक्रमों से लगभग रुपये 10 लाख का राजस्व सृजित हुआ।

वर्ष 2017-18 के दौरान भाकृअनुप – राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी (ICAR-NAARM), हैदराबाद को अनेक विशिष्टज्ञों के आगमन का स्वागत करने का सुअवसर मिला। इनमें से कुछ का उल्लेख इस प्रकार है : ग्यारह माननीय सांसदों के साथ माननीय श्री हुक्मदेव नारायण यादव की अध्यक्षता में कृषि पर संसदीय स्थाई समिति; श्री सुदर्शन भगत, माननीय कृषि एवं किसान कल्याण राज्य मंत्री; माननीय डॉ. इयासुअब्राहा, केन्द्रीय मंत्री एवं माननीय श्री टेस्फाये मेन्गिस्टेडोरी, राज्य मंत्री। रिपोर्टाधीन वर्ष में श्री गजेन्द्र सिंह शेखावत, माननीय कृषि एवं किसान कल्याण मंत्री ने भाकृअनुप – राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी (ICAR-NAARM), हैदराबाद का दौरा किया और वहां दिनांक 22-23 दिसम्बर, 2017 को “चलो किसानों को सुनें” विषय पर आयोजित राष्ट्रीय कार्यशाला में 25 राज्यों के लगभग 150 किसानों के साथ परस्पर बातचीत की।

वर्ष 2017-18 के दौरान भाकृअनुप – राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी (ICAR-NAARM), हैदराबाद में सातवें पंचवर्षीय समीक्षा दल (QRT) और सातवीं अनुसंधान सलाहकार समिति (RAC) की बैठकें भी आयोजित की गईं। भाकृअनुप – राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी (ICAR-NAARM), हैदराबाद द्वारा अनेक आयोजनों की मेजबानी की गई जैसे कि एग्रीकल्चरल इकोनॉमिक्स रिसर्च एसोसिएशन (AERA) का रजत जयंती सम्मेलन; कृषि वैज्ञानिक चयन मण्डल (ASRB) का स्थापना दिवस-2017; तथा भाकृअनुप अंतर-जोनल खेलकूद प्रतियोगिता-2017 आदि। फोकार्स के वर्तमान मॉडल से संबंधित मुद्दों पर चर्चा करने के उद्देश्य से भाकृअनुप – राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी (ICAR-NAARM), हैदराबाद द्वारा “फोकार्स का पुनः अवलोकन : प्रशिक्षित वैज्ञानिकों के विचार एवं प्रतिक्रिया” विषय पर एक राष्ट्रीय कार्यशाला आयोजित की गई जिसमें फोकार्स के 94वें से 103वें बैच से जुड़े वैज्ञानिकों को आमंत्रित किया गया।

रिपोर्टाधीन वर्ष अन्य संगठनों/संस्थानों के साथ समझौता ज्ञापन (MoU) के रूप में अनेक पेशेवर सहयोग बनाने का साक्षी बना। संस्थानों में शामिल थे : इन्दिरा गांधी राष्ट्रीय वन अकादमी (IGNFA), देहरादून; प्रशासनिक स्टाफ कॉलेज ऑफ इंडिया (ASCI), हैदराबाद; टाटा इंस्टिट्यूट ऑफ सोशल साइन्सज (TISS), हैदराबाद और कई सार्वजनिक एवं निजी संगठन। इन समझौता ज्ञापनों से परस्पर शिक्षण के लिए सहयोग बढ़ने और प्रशिक्षण गतिविधियों में मजबूती आने का विश्वास है।

भाकृअनुप – राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी (ICAR-NAARM), हैदराबाद द्वारा अनेक मोर्चों पर उत्कृष्टता हासिल की गई जो कि इसे मिले पुरस्कारों में प्रतिबिम्बित होता है। इस वर्ष संस्थान को वर्ष 2015-16 के लिए राजर्षि टंडन राजभाषा पुरस्कार मिला। अकादमी को पब्लिक रिलेशन सोसायटी ऑफ इंडिया (PRSI) द्वारा डिजिटल मीडिया पहल पुरस्कार भी प्रदान किया गया।



Executive Summary

The Academy has played a pivotal role in enhancing the capacity of the professionals of the entire National Agricultural Research, Education and Extension System (NAREES) in India. In the year 2017-18, the Academy has contributed significantly in human resource development of different stakeholders within as well as outside NAREES along with research contribution, policy support and education. Apart from this, the efforts put forward by the Academy in the field of entrepreneurship development in agriculture sector has been well received and applauded. The academy further expanded its reach and coverage in offering capacity building programmes for entry level Assistant Professors of different agricultural universities and Programme Co-ordinators of Krishi Vigyan Kendras (KVKs).

Capacity building is one of the important mandate of the Academy. There were 54 capacity building programmes organized in the year 2017-18. Out of these, 36 were need based programmes. There were 10 off-campus programmes which were organized in various parts of the country. These programmes benefitted 2,495 trainees in one or other form of the training programmes, which includes 988 Massive Open Online Course (MOOC) participants. The flagship programme of the Academy popularly known as **Foundation Course for Agricultural Research Services (FOCARS)** was organized twice- one in July 2017 and other in January 2018 and had 84 newly appointed ARS scientist probationers in all. One hundred and two participants have undergone the two **Foundation Course for Faculty of Agricultural Universities (FOCFAU)** which is designed in the same line of FOCARS but for one-month duration. The Executive Development Programme (EDP), which is designed for the newly recruited Research Management Position cadre of ICAR attracted 10 participants, while the Management Development Programme (MDP), which is meant for the pre-RMP cadre of ICAR had 41 participants during 2017-18. The Academy trained 113 technical and 240 administrative personnel of various ICAR institutes by way of need based programmes. In order to enhance the capabilities of the skilled support staff, a training programme was also organized which was attended by 27 staff. It catered the needs of many universities by imparting need based competency enhancement programmes. These are Karnataka Veterinary, Fishery and Animal Sciences University, Bidar; ICAR – National Dairy Research Institute, Karnal; Vasant Rao Naik Marathwada Krishi Vidyapeeth, Parbhani; University of Horticultural Sciences, Shivamogga and Dhanwate National College, Nagpur (Shivaji Educational Trust, Amaravathi), Sri Venkateswara Veterinary University, Tirupati, Jawaharlal Nehru Agricultural University, Jabalpur and CIFE, Mumbai. The Academy was able to generate approx. ₹68 lakh in 2017-18 by organizing various sponsored programmes.

Education is also one of the mandate of the Academy which is fulfilled by offering two academic programmes. One is Post Graduate Diploma in Management (Agriculture), a two-year fully residential programme which attracted 399 applications. Finally, 34 were selected for admission. The pass out batch students were able to grab average Cost to Company (CTC) package of ₹7.25 lakh per annum. The one-year Post Graduate Diploma in Technology Management in Agriculture (PGDTM-A) had 65 students admitted in 2017-18 many of whom are in-service candidates. Educational programmes helped the Academy to generate approximately ₹2.1 crore in 2017-18.

In the research front, the Academy had 34 research projects, in which the in-house and externally funded contract research projects were 16 in number each. The Academy was able to take two consultancy projects. Out of the sponsored projects, seven were from different ICAR schemes. Others include DBT, DST, Tobacco Board, Telangana Govt., IFPRI, Mahyco Monsanto Biotech (India) etc. The research projects span over all the six divisional themes of the Academy. The projects have been categorized into six thematic areas viz. Agribusiness Management for Inclusive Growth, Education Systems Management for enriched learning quality & environment, Enhancing Capacities for Leadership and Governance, Extension Systems Management in a Market-driven Environment, Information and Communication Management for Promoting Innovation and Governance, Mobilizing Science and Technology for Innovation and Sustainable Development.

Faculty members contributed 245 publications in 2017-18 in various forms. Peer reviewed research papers include 110 which is more than 43 percent of the total publications by NAARM faculty in 2016-17. Other forms of publication include books, book chapters, popular articles, training manuals, seminar proceedings, research projects etc.

NAARM is one of the consortia partner of 'Farmers FIRST' initiative launched by ICAR. The Academy was involved in the capacity building activity and organized several workshops. A unique workshop and first of its kind- '**Let's listen to Farmers**' was also organized to collect feedback directly from the farmers. Two hundred delegates including about 143 farmers from 25 states participated in this workshop.

In order to upgrade the knowledge and remain updated with the happenings in the academic arena, NAARM faculty are regularly being sent for attending various capacity building programmes across the country. In the year 2017-18, NAARM faculty attended 18 different capacity building programmes.

The Academy houses **Centre for Agri-Innovation (CAI)**, which provides hand-holding support to the agri-business incubators by inter alia developing

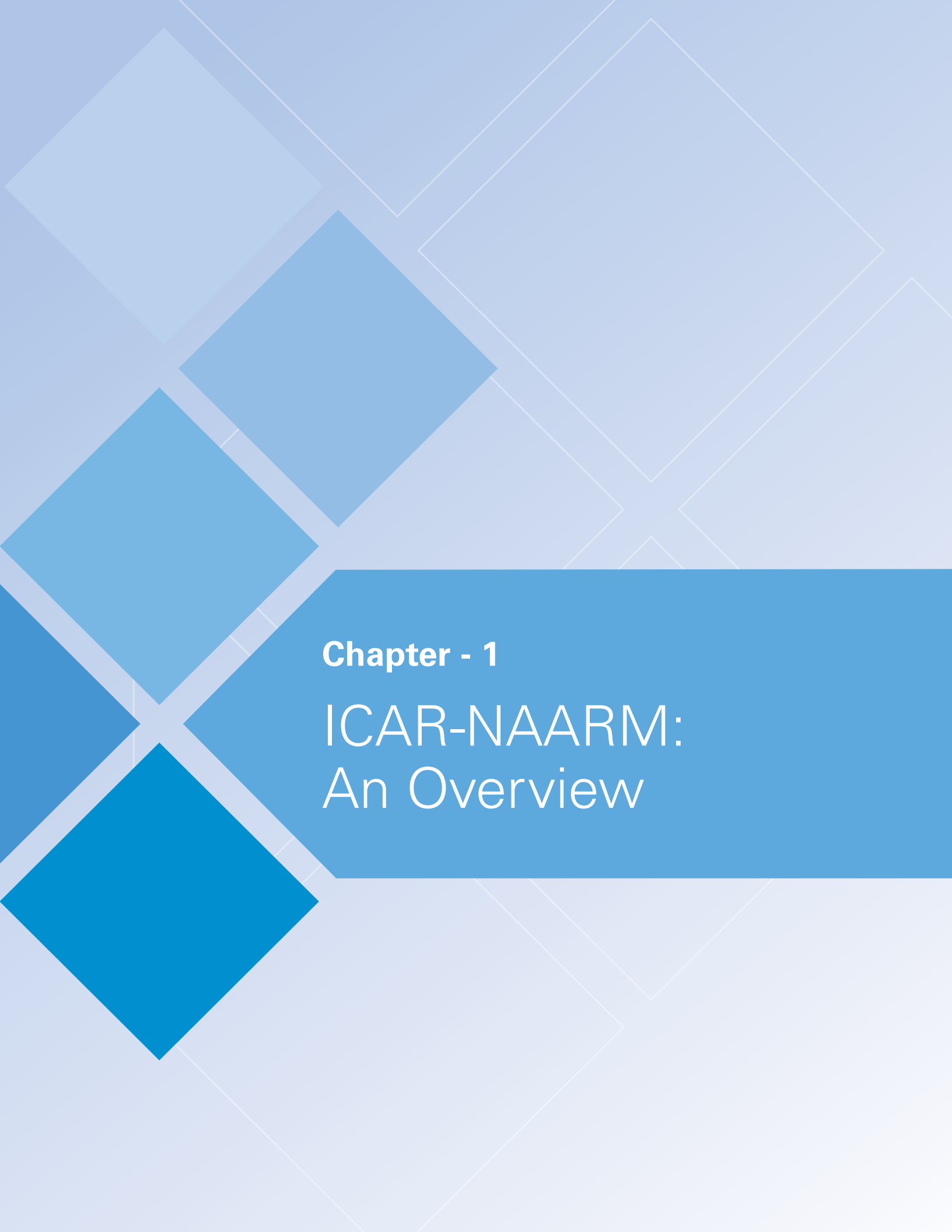
modules, guidelines and other forms of learning material. Till date 36 incubatees have been inducted and out of that, 13 have already graduated. During the year, various events were organized in different places which includes AGRI UDAAN, 6 road shows, 3 sensitization workshop and many more. A state of art Technology Enhanced Learning in Agricultural Education Lab (TELAge) was established in the Academy to develop model digital courses, offer distance and online programmes, conduct virtual classes, and support some of the in-house programmes. The facility is used to offer MOOCs. It has been also used to provide consultancy-mode support to other agencies in developing and hosting the digital content. Programmes executed for institutes like MANAGE, NIRD & PR has generated a revenue of about ₹10 lakh.

The Academy hosted various dignitaries visiting in 2017-18. Naming a few are Parliamentary Standing Committee on Agriculture headed by Hon. Sh. Hukumdev Narayan Yadav, along with other eleven members of Parliament; Shri. Sudarshan Bhagat, Hon'ble MoS, Ministry of Agriculture and Farmers' Welfare; Ethiopian Agriculture Ministers- H.E. Dr Eyasu Abraha, Cabinet Minister and H.E. Mr Tesfaye Mengiste Dori; State Minister. Shri. G.S. Shekhawat, Hon. MoS, Ministry of Agriculture and Farmers Welfare visited the Academy and interacted with the farmers during a national workshop "Let's listen to farmers".

Seventh Research Advisory Committee (RAC) meetings was organized in the year 2017-18. The Academy hosted many other events like Silver Jubilee Conference of the Agricultural Economics Research Association (AERA), ASRB Foundation Day-2017, ICAR Inter-Zonal Sports Meet-2017 etc. In order to discuss various issues pertaining to the present model of FOCARS, the Academy organized National Workshop on "Revisiting FOCARS: Reflections and Feedback of Trained Scientists" by inviting scientists belonging to 94th to 103rd batches of FOCARS.

The year witnessed various professional collaborations in the form of MoUs signed with other organizations/institutes like Indira Gandhi National Forest Academy (IGNFA), Dehradun and Administrative Staff College of India (ASCI), Hyderabad, Tata Institute of Social Sciences (TISS), Hyderabad and many public and private organizations. The MoUs are believed to enhance collaboration for mutual learning and strengthen the training activities.

The Academy achieved excellence in many fronts which is reflected by the awards received. The Academy was awarded with Rajarshi Tandon Rajbhasha Puraskar for the year 2015-16. The Academy also bagged Digital Media Initiatives award of Public Relations Society of India (PRSI).



Chapter - 1

ICAR-NAARM: An Overview



ICAR-NAARM: An Overview



Vision

A global knowledge institution enabling National Agricultural Research and Education Systems adapt to change through continuous innovation.



Mission

To enhance leadership, governance and innovation capacities of National Agricultural Research and Education System (NARES) through capacity strengthening, education, research, consultancy and policy support.

Since its establishment in 1976, the National Academy of Agricultural Research Management (NAARM) has played a key role in enhancing the capacities of individuals and institutions of National Agricultural Research and Education System (NARES) in agricultural research, education and technology management. Effective and efficient management of the vast pool of manpower of NARES poses a formidable challenge. Over the last four decades, the Academy has been contributing significantly to the development of competencies and capacities of scientists and research leaders in the Indian Council of Agricultural Research (ICAR) and Agricultural Universities (AUs). The overarching objective of these efforts is to build an ecosystem which is adaptive and responsive to the needs of its stakeholders. Through innovative, need-based human resources development programmes, and demand-driven research initiatives in the specialized areas of agricultural research and education management, the Academy has helped the systems in evolving appropriate policies to improve the efficiency and effectiveness of research and education in agriculture sector. To fulfil this, the Academy has expanded its activities in an effort to reach all stakeholders, even beyond NARES including scientists, policy makers, farmers, extension personnel, entrepreneurs and R&D managers in agribusiness sector. It has also facilitated several new policies, institutional mechanisms, and initiatives leading to enhanced capacities in the NARES. The Academy strives to generate a sense of fraternity and inculcate a scientific work culture amongst the agricultural scientists and faculty across the country.

Additionally, the Academy has also widened the scope of its activities and initiated academic programmes in agricultural management and technology management to develop a new generation of young leaders and managers who can contribute to sustainable agricultural development at the national level. The Academy's higher education programs viz. Post Graduate Diploma in Management- Agriculture (PGDMA) and Post Graduate Diploma in Technology Management in Agriculture (PGDTMA) are gaining huge popularity among its clientele. Recognising the process standardization in agribusiness management education, the AICTE-approved PGDMA program also got accredited by the National Board of Accreditation (NBA).

Mandate

- Enhance individual and institutional competencies in managing innovation through capacity development, research and policy advocacy.
- Serve as a think tank for National Agricultural Research and Education System and facilitate strategic management of human capital.





The Technology Business Incubator of the Academy viz. a-IDEA (Association for Innovation Development of Entrepreneurship in Agriculture) got national recognition due to its constant support to Agri-Startups. Being a unique institution of its kind, the Academy offers complete value chain in agricultural management and aims to emerge as an institution par excellence to facilitate and support a culture of dynamic management in agricultural research and education.

1.1 Strategic framework

In tune with its Vision, Mission and Mandate, the Academy has developed a six-tier strategic framework during 12th Plan to steer different activities to achieve the set goals and targets. Those are:

1. Agribusiness Management for Inclusive Growth

- Enhance capacities for food and agribusiness management education and research in NARES
- Enhance capacities for entrepreneurship development in NARES and strategic management of agribusinesses

2. Education Systems Management for Enriched Learning Quality & Environment

- Enhance capacities for agricultural education policy, planning and evaluation in institutions of NARES
- Enhance capacities for faculty excellence and technology-enhanced learning for increasing learning effectiveness and continuous learning by deploying emerging media to create new and vibrant education environments

3. Enhancing Capacities for Leadership and Governance

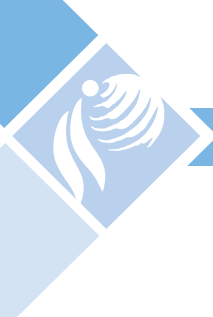
- Institutionalize a framework for leadership development at all levels of NARES – early, mid- career, senior professionals and manager across all functions (research, education, extension)
- Institutionalize good governance in NARES by building capacities for effective management of institutions, performance assessment, accountability, and developing strong organizational value systems and work culture that promote innovation

4. Extension Systems Management in a Market-driven Environment

- Enhance operational, adaptive and generative capacity of front-line agricultural extension systems to address emerging challenges
- Enhance capacities for use of IoTs to provide customized knowledge, skills and solutions to farmers, farmer groups, rural communities, and for social networking

5. Information and Communication Management for Promoting Innovation and Governance

- Enhance capacities for integrating ICTs and Internet of Things (IoT) in institutional management and governance
- Enhance capacities for managing data, information and knowledge in agriculture, scientific communication and institutional governance



6. Mobilizing Science and Technology for Innovation and Sustainable Development

- Enhance capacities for research policy, strategy, priority setting, planning, management, monitoring and evaluation for a more pluralistic NARES, in the emerging contexts of food security, climate change and globalization
- Enhance capacities for technology foresight and strategic management of intellectual property and commercialization of technologies

1.2 Organization and Management

The academic, research and capacity building programmes of the Academy along the above strategic framework is carried out through six divisions:

- Agribusiness Management (ABM)
- Education Systems Management (ESM)
- Extension Systems Management (XSM)
- Human Resources Management (HRM)
- Information and Communication Management (ICM)
- Research Systems Management (RSM)

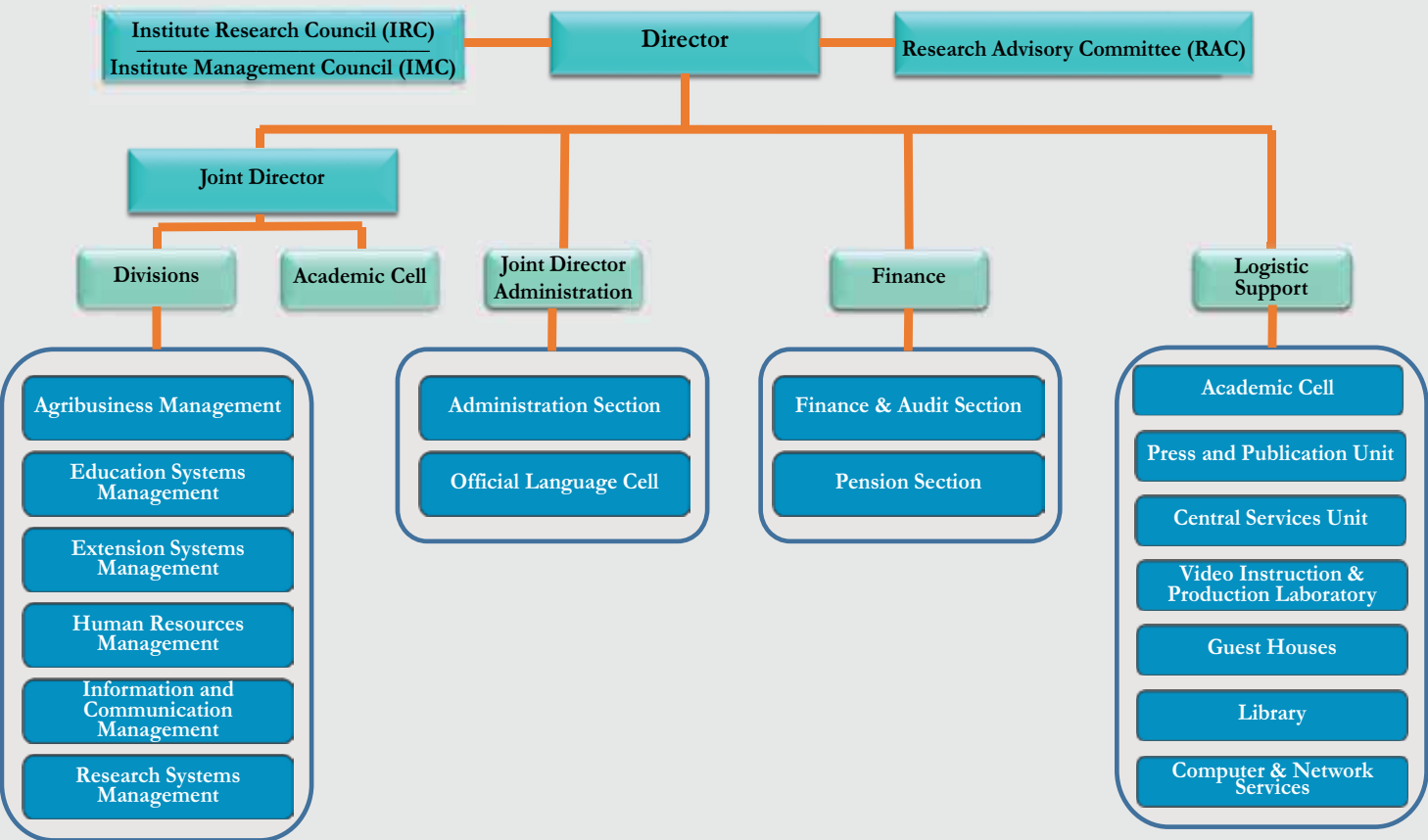


Figure 1.1: Organogram of the Academy



1.2.1 Research Advisory Committee (RAC) of the Academy

The Indian Council of Agricultural Research (ICAR) constituted the Research Advisory Committee (RAC) for three years from 24.11.2016 to 23.11.2019 as given in Table 1.1.

Table 1.1. Research Advisory Committee of the Academy till 23 Nov 2019

Padma Shri Prof. S.K. Sopory Ex-Vice Chancellor, Jawaharlal Nehru University, New Delhi	Chairman
Dr. K. C. Bansal Ex- Director, ICAR-NBPGR, New Delhi	Member
Dr. P.K. Joshi Director for South Asia, International Food Policy Research Institute, New Delhi	Member
Dr. C. Shambu Prasad Professor, Institute of Rural Management- Anand	Member
Dr. Ishwar Murthy Professor, Decision Science & Information Systems, Bengaluru	Member
Dr. R.K. Samanta Ex- Director, ICAR-NAARM, Hyderabad	Member
Dr. N.S. Rathore DDG (Agri. Education), ICAR, New Delhi	Member
Dr. Ch. Srinivasa Rao Director, ICAR-NAARM, Hyderabad	Member
Shri Venkateshwarlu Vasanthapu Rajendranagar, Mahabubnagar	Member
Shri M. Dharma Rao Ex- MLA, Hanumakonda Dist	Member
Dr. S.K. Soam Head, ICM Division, ICAR-NAARM, Hyderabad	Member Secretary



Members of Research Advisory Committee (RAC) with NAARM faculty



1.2.2 Institute Management Committee (IMC)

The following officials were nominated as the members of the Institute Management Committee of NAARM for a period of 3 years with effect from 1st September, 2016..

Table 1.2 Institute Management Committee of NAARM

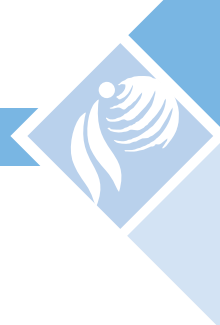
Director ICAR-NAARM, Hyderabad	Chairman
Director of Agriculture Department of Agriculture, Government of Telangana, Basheerbagh, Hyderabad-500001	Member
Director of Agriculture Department of Agriculture, Government of Andhra Pradesh, Secretariat, Basheerbagh, Hyderabad-500001	Member
Director Prof. Jayashankar Telangana State Agricultural University, Rajendranagar, Hyderabad-500030	Member
Shri. Venkateshwarlu Vasanthapu H.No. 1-3-158/401, BSN Reddy Complex, Rajendranagar, Mahabubnagar, Telangana-509001	Member
Shri. M. Dharma Rao H.No. 2-4-549, Hastinapur Colony, Hanumakonda district, Telangana-506001	Member
Joint Director (Admin.) ICAR-NAARM, Hyderabad	Member Secretary

1.3 Linkages/Networking/Collaborations

The mandate of the Academy demands high co-dependency on linkages and collaborations with various national as well as international organization. The Academy has a strong network with all the 108 ICAR Research Institutes, 3 Central Agricultural Universities and 71 Agricultural Universities (AUs), and plays a crucial role through capacity building of these institutions, agribusiness industry, scientists, and academia. Its network spans across various other National as well as International Institutions.

1.3.1 National Institutes/ Organizations

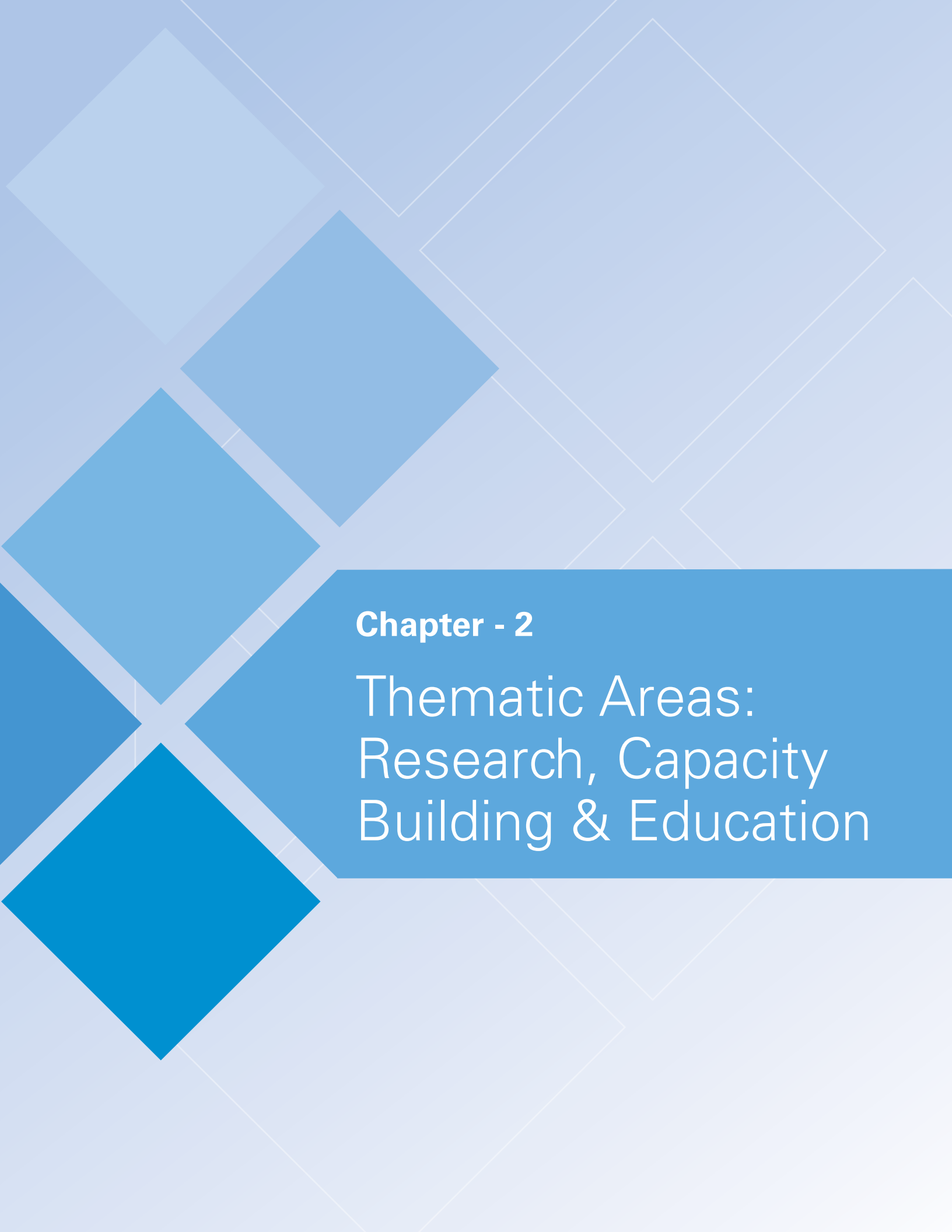
1. All the ICAR institutes, and Agricultural Universities.
2. Various departments of the Government of India like, NITI (National Institution for Transforming India) Aayog, Department of Science and Technology (DST), Department of Biotechnology (DBT), National Science & Technology Entrepreneurship Development Board (NSTEB).
3. University of Hyderabad, Osmania University, Jawaharlal Nehru Technological University (JNTU), Indian Institute of Information Technology (IIIT), Council of Scientific & Industrial Research (CSIR), Indian Council of Medical Research (ICMR) and various institutions located at Hyderabad.



4. MoEF & CC, NBA, PPV&FRA, NFDB and State Biodiversity Boards.
5. Management Institutions located at Hyderabad *viz.* the Administrative Staff College of India (ASCI), Indian School of Business (ISB), National Institute of Agricultural Extension Management (MANAGE), National Institute of Rural Development and Panchayat Raj (NIRD & PR), and Institute of Public Enterprise (IPE); and across the country such as Indian Institute of Management (IIM).
6. State Departments of Agriculture of various States (Telangana, Andhra Pradesh).
7. Private Sector Organizations (Agri Input Companies, Service Companies, *etc.*).
8. Non-Government Organizations (NGOs).
9. Other organizations *viz.* National Innovation Foundation (NIF), Federation of Indian Chamber of Commerce and Industry (FICCI), Centre for Innovation Incubation and Entrepreneurship (CIIE) *etc.*
10. Krishi Vigyan Kendras (KVKs).

1.3.2. International Organizations

1. CGIAR Institutions (CIMMYT, ICRISAT, IFPRI, *etc.*)
2. World Bank, FAO other United Nations Organizations.
3. Department of International Development (DFID, UK).
4. SAARC Agricultural Centre (SAC).
5. NARS of South Asian Countries such as Bangladesh Agricultural Research Council (BARC), Nepal Agricultural Research Council (NARC), Pakistan Agricultural Research Council (PARC), and other NARS Institutions in Afghanistan, Sri Lanka and other ASEAN countries.
6. NARS in other African and South East Asian countries such as Kenya, Mali, Tanzania, Nigeria, Malawi, Liberia, and Philippines.
7. Leading Land Grant and State Universities of USA and other Universities in Europe, Australia and other developing and developed countries.



Chapter - 2

Thematic Areas: Research, Capacity Building & Education



Thematic Areas: Research, Capacity Building & Education

Achievements

Achievements under different thematic areas are presented against FOUR aspects.



Research



Capacity Building



Workshop/
Conference



Education

SIX Thematic Areas

1. Agribusiness Management for Inclusive Growth
2. Education Systems Management for Enriched Learning Quality & Environment
3. Enhancing Capacities for Leadership and Governance
4. Extension Systems Management in a Market-driven Environment
5. Information and Communication Management for Promoting Innovation and Governance
6. Mobilizing Science and Technology for Innovation and Sustainable Development

Keeping in view the targets envisaged in the 2050 Vision documents of the Academy and ICAR and the premises of keeping innovation at the centre of the debate and discussion, the Academy recognizes the need for developing knowledge-driven agriculture with new research governance models. Recognizing the enabling nature of role of emerging science and technologies across diverse sectors, it voices for their integration into the agricultural value chain. Further, the evolving changes in agri-education sector shifting focus from ‘teaching to learning outcome’ signifies the requirement to develop newer institutional mechanisms and capacities of the system. The Academy also identifies the crucial need for a continuous stakeholder engagement to usher effective and efficient performance in the NARES with more transparency and accountability systems in place. Therefore, the entire activities of research, capacity building and education have been aligned along six-tier strategy approach. Through a structured deployment of this strategy, it is envisaged that the Academy can play an effective role in the planned transformation process of the NARES.

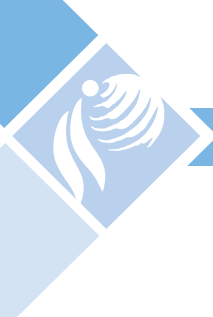
The subsequent sections present the research projects, capacity building programmes and the conferences/workshops conducted by the Academy under the six strategic areas which are already mentioned in the previous chapter. The codes given in the side pane can be used to get the relevant information of the project/capacity building programme/workshop, which are placed in the end of the chapter

2.1 Agribusiness Management for Inclusive Growth

With the shift in agricultural development strategies from a pure production-oriented approach to a broader systems perspective, the importance of the agribusiness sector has grown significantly. Seen as an engine for growth, agribusiness and its related industries are receiving increased attention in policies and strategies that aim to promote investments in agro-enterprises and agro-based value chain development. This has prompted a need for deeper understanding of agribusinesses, agro-industries and agri-food value chains. The theme on Agribusiness Management for Inclusive Growth enables the Academy to harness its unique strength of blending research, training, and education and consultation activities for developing future leader in the field of agribusiness. It broadly aims to:

- Enhance capacities for food and agribusiness management education and research in NARES
- Enhance capacities for entrepreneurship development in NARES and strategic management of agribusinesses

Rapid urbanization, rising disposable income of households, expanding and diversifying food demand and globalization of agricultural economy has given impetus to research and capacity building in the area of Agribusiness Management. In order to meet the broader objectives of the theme, the Academy focuses on the following thematic area



for the research, capacity building and education:

- 2.1.1. Agribusiness environment:** with major focus on value chain analysis for agri-commodities and services; consumers demands; price behaviour & forecast analysis
- 2.1.2. Public policy and impact:** with focus on market reforms like e-NAM, GST, etc; financial inclusion and its spillover; spread and convergence of crop insurance
- 2.1.3. Markets and institutional innovation:** market integration of commodities; producers companies and FPOs; inputs and service delivery system; diversification and intensification of agriculture for high value commodities; business analytics
- 2.1.4. Nurturing future leaders in agribusiness:** through imparting quality education in agribusiness management



SIL/2016/08/118
SOL/2015/05/108
CCP/2017/16/136
SIL/2017/10/130

2.1.1 Agribusiness Environment

The agribusiness sector, which comprises the business activities performed from farm to fork, is a major generator of employment and income worldwide. In contrast to the dynamism evident in other sectors, investment in Indian agriculture and agribusiness—public and private, domestic and foreign—has been low and, until very recently, has shown little growth. Private agribusiness investment depends on state and central regulatory policies and business environment—including marketing, interstate movement, storage, and taxation policies. Prospects for agricultural investment hinge largely on the environment for private investment by farmers and agribusinesses. Key factors likely to shape the climate for private investment by farmers and agribusinesses include the extent to which domestic demand for agricultural products makes investment potentially profitable; a range of government regulatory, credit, tax, and other policies that affect incentives for domestic and foreign investment in Indian agriculture; and infrastructure constraints.

Strategic and Financial Analysis of Livestock Feed Industry in India

Table 2.1: Sample feed manufacturers’ profile

Annual Turnover	No.	Location of Manufacturing unit
< 1 Lakh tonnes	2	Uttar Pradesh and Punjab
1-10 Lakh tonnes	2	Gujarat and West Bengal
> 10 Lakh tonnes	2	Kerala and Maharashtra

The study was undertaken with the objectives of studying the current status of livestock feed industry in India; analysis of the financial performance of livestock feed manufacturing units; and identifying the problems and prospects of livestock feed industry in India.

The financial analysis shows the spread of the feed manufacturers in terms of annual turnover as well as geographically (Table 2.1). Though the average profit margin ratio is lesser than best agribusiness companies, the return on equity ratio is higher indicating the desire to retain the shareholders in the livestock feed manufacturing business in the short run. The asset turnover ratio indicates better efficiency of livestock feed mills. The inventory turnover ratio indicates high movement of the products which is due to higher demand in the market. shows the various financial ratios of livestock feed manufacturing units (Figure 2.1):



Figure 2.1: Average key financial ratios of livestock manufacturing units

Fluctuation in price of the raw materials is found to be the biggest constraint followed by the new tax system of GST. Increasing demand is considered to be the biggest driver for feed industry. The major findings out of the study can be summarized as below:

1. There exists deficit in the availability and requirement of all types of feed.
2. Region-wise variation exists in the composition of feed manufactured by the feed manufacturers.
3. The financial analysis clearly reveals the encouraging sign for the industry to flourish.
4. The economies of scale favours the financial ratios, particularly the return on equity and the profit margin of the feed manufacturing units.
5. The fluctuation in the price of raw materials is found to be the major constraint for the feed manufacturers.

Socio-economic Analysis of Finger Millet (ragi) in India

The study was undertaken to estimate the growth rate in area, production and productivity of ragi crop and, its trend and instability in the study area; to estimate market demand and consumer preference for ragi; and to assess the value chain of ragi and analyze its market competitiveness and market integration. During the period under report, the traits preferred by the ragi growers; shift in consumption pattern of ragi and its demand projections for the future were studied.

Table 2.2: Traits considered in the conjoint analysis

S. No	Traits/ Attributes	Levels of Attributes
1	Duration*	1- Short and 2- Long
2	Yield [#]	1- Low and 2- High
3	Colour of the grains	1- White and 2- Red
4	Drought resistance	1- Susceptible and 2- Tolerant
5	Plant height	1- Dwarf and 2- Tall

* Short: Less than 140 days; Long: More than 140 days

[#] Low: Less than 2 tonnes; High: More than 2 tonnes



Use of tractor for threshing of finger millet in Tamil Nadu



Straw of ragi crop stored for fodder purposes in Tamil Nadu

The traits preferred by the growers were provided by the breeders and physiologists involved in this study. These traits were narrowed down based on farmers' revealed preference for preferable ragi seeds. Subsequently, five traits were retained for further analysis. They were duration of the crop, yield, colour of the grains, drought resistance and plant height and the level of these traits / attributes. The results reveals that almost all ragi growers in Tamil Nadu prefer long duration ragi crop over short duration crop, because short duration crop comes to harvest during rainy reason and hence, have a higher chance of crop damage. Preference for taller plant by them was because of more fodder yield, which could be used for feeding their farm animals. The results also reveal that drought resistance characteristics and red colour grains were the other important traits preferred by them. Farmers prefer long duration crop over short duration for the reason of avoiding crop getting damaged due to monsoon during the time of harvesting (Table 2.2).

The demand projections for ragi for rural and urban India are obtained for the year 2020, 2025 and 2030 by using the following formula for the three time periods considered:

$$Q_{2020} = Q_{2011} (1+r)^9$$

Where, Q_{2011} is the monthly quantity of ragi consumed in 2011-12 by the concerned group of population (rural/urban);

Q_{2020} is the projected quantity of ragi demanded for the group (rural/urban) for the year 2020;

and 'r' is the sum of population growth rate and growth rate of per capita ragi consumption

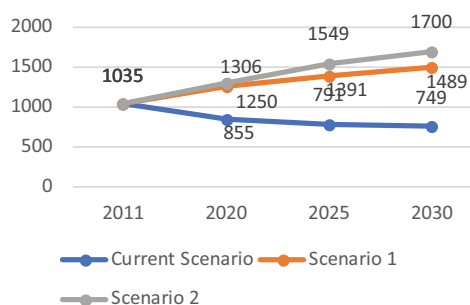


Figure 2.2: Demand of finger millet in India ('000 tonnes)

Table 2.3: Demand Forecast of Ragi in Major States and India

Year	States ('000 tonnes)				All India
	Karnataka	Tamil Nadu	Andhra Pradesh	Maharashtra	
2011-12 (Actual)	740	78	110	38	1035
2020	483	37	86	25.3	855
2025	386	32	75	20.5	791
2030	313	31	65	17	749

Considering the past trend of ragi consumption in selcted states, the future demand of ragi is expected to decline in all these states (Table 2.3). However, it is further felt that there is a consistent improvement in the awareness levels about the nutritive values of this cereal among the Indian population. Hence, a scenario analysis was attempted to project the demand by the total population in the country, by assuming 0.5% and 1% growth in the per capita consumption of ragi. Accordingly, the demand of ragi in 2020, 2025 and 2030 under the two scenarios for the country was depicted in Figure2.2.



Estimating Consumption of Different Food Commodities in Telangana State

The consumption of different food items in Telangana State was estimated by surveying about 6200 households across all 31 districts of the state covering both rural as well as urban households. The data collection was accomplished by using GPS enabled tablet based survey system.

The average monthly per capita consumption (MPCC) of cereals is 12.6 kg in the state (Figure 2.4). Rural consumption of cereals is significantly higher than that of urban consumption. Among cereals, rice is the most consumed comprising about 77 percent of the total cereals consumption.

The monthly per capita consumption of pulses in the state is 1.74 kg and it is almost equal in both rural and urban areas. Arhar/tur occupies 41 percent share in quantity among all pulses consumed in Telangana, followed by moong and urad both with 18% consumption share among pulses.

Telangana is one of the major spices consumption state in the country. The average monthly per capita spices consumption in the state is 636 g. The urban people are consuming relatively more spices than their rural counterpart. Ginger (146 g per capita per month), tamarind (141 g per capita per month) and dry red chilli (98 g per capita per month) are the major spices consumed in Telangana.

Telangana has high non-vegetarian eating population. The consumption of non-vegetarian items in the form of meat/fish/egg (MFE) is about 1.38 kg per month. MFE is consumed more in rural areas. Chicken meat comprises major portion of MFE consumption (550 g per capita per month) in rural and urban areas. The survey revealed that about 95 percent households are consuming non-vegetarian food items in the state. The monthly per capita consumption of MFE in Jayashankar Bhupalpally district is the highest with 1.91 kg while Vikarabad district has the lowest consumption (1.06 kg per capita per month).

About 4.6 litres of liquid milk per capita per month is consumed in the state. Rajanna Siricilla district has the highest MPCC (6.27 litres) while Adilabad has the lowest level of consumption (2.34 litres). The urban areas have higher level of MPCC (5.09 litres) than the rural areas (3.86 litres). The milk products, which include curd, ghee, paneer, cheese, butter, etc. is consumed at the rate of 920 g per capita per month in Telangana. Similarly, the consumption of edible oil is at 1.22 litres per capita per month in the state. In Adilabad and Kumarambheem Asifabad district, the per capita consumption of edible oil is the highest with 1.4 litres, while Vikarabad district has the lowest per capita consumption (0.99 litre).



Project planning meeting with officials of Telangana state



Survey of rural women for food consumption pattern using tablet by the field investigators in Telangana State

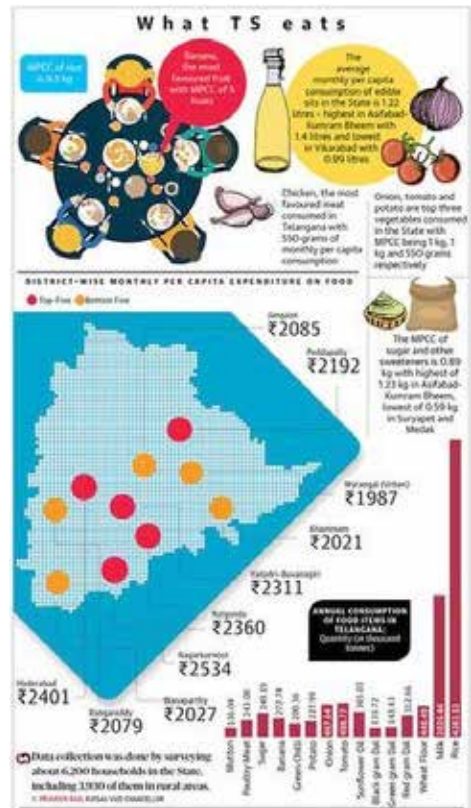


Figure 2.3. Media coverage of the study (<http://www.thehindu.com/news/cities/Hyderabad/production-of-veggies-fruits-short-of-demand/article23674326.ece>)



Figure 2.5: District wise per capita expenditure on food in Telangana state

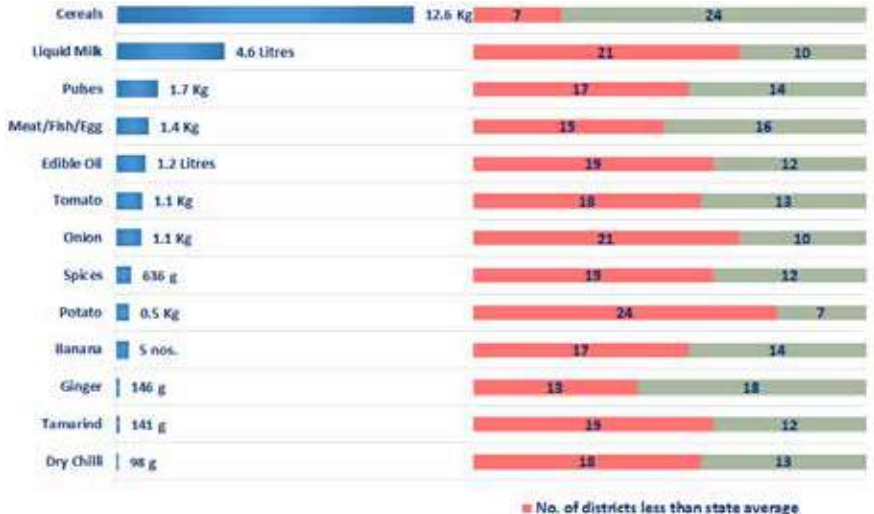


Figure 2.4: Monthly per capita consumption in Telangana state and number of districts less than and greater than state average consumption

Among vegetables, onion and tomato occupies major share with about 1 kg MPCC. Potato is also another most consumed vegetables with a MPCC of 510 g. Banana is the most favorite among fruits in the state with a MPCC of 5 fruits. The consumption ranges from 3.3 in Vikarabad district to 6.97 in Nirmal district. The average per capita monthly expenditure on food is ₹2156, (Figure 2.5) out of which, cereals occupies the highest share (19 %) followed by milk and milk products (15%). The study has significant implication on the entire food economy of the state. The information emerged from the study gives an important indication of nutritional status across the districts, which can be helpful in planning for nutritional security in the state. Further, the study would also help the state in designing crop colony strategies at district level.

2.1.2 Public Policy and Impact

Financial Inclusion of Small Holders and Landless Households in Madhya Pradesh

The research study has been carried out with the objectives of assessing the extent and level of financial inclusions for smallholders and landless households, estimating the plausible income gains to the smallholders and landless households with better financial inclusion, finding the determinants of financial inclusion in the study area, and suggesting strategies for enhancing the financial inclusion of smallholders and landless households. Two districts of Madhya Pradesh has been selected for the study- Morena and Jabalpur. Jabalpur falls under the category of high financial inclusion and Morena in low financial Inclusion. Data from 104 respondents has been collected from Jabalpur district. The preliminary result of the study is given below.



Table 2.4: Use of transaction of banking services among farmers in Jabalpur district, Madhya Pradesh
(Figures in percentage)

Land Holding Category	State Bank of India	Other Nationalized Bank	Grameen Bank (RRB)	Co-operative Bank	Post Office	SHG
Landless	29	79	4		29	
Marginal	23	75	9	16	16	5
Small	53	59	3	28	9	
Overall	34	71	6	15	17	2

The majority of the farmers in all the land holding category are associated with the nationalized bank with 79 percent in landless, 75 percent in the marginal and 59 percent in small category. State Bank of India is next popular bank in the sample households and around 34 percent of the all the respondents are associated with State Bank of India in all the landholding classes of the farmers. Cooperative banks provide free credit for the purchase of the agricultural inputs in the rural areas. Small farmers (around 28 percent) are associated with the cooperative banks and take the benefits of the free credit. 16 percent of small farmers are associated with the cooperative banks. (Table 2.4) They are not able to get the benefits or excluded from the ambit of the cooperatives in rural area of the Jabalpur district of the Madhya Pradesh.



Survey of smallholders in Jabalpur district, MP

Transaction of financial services refers to use of cash or other means such as cheque, debit cards or credit cards to make or receive payments. At basic level it means obtaining cash for personal use, the withdrawals and converting cheque to cash at broader perspective. Transaction also include payments and receipts through financial network both at domestic and international level. Among the farmers' group, more use of branch banking is found among marginal (35%) followed by small farmers (28%). ATM and cheque are more popular among small farmers and 18 percent and 26 percent of small farmers used Cheque and ATM services respectively. 56 percent of the farmers used the ATM services among the sample villages in the Jabalpur district. Mobile banking and internet banking are also becoming popular after opening of small payment bank like Paytm, Phonepay etc. Among all the farmers' group 29 percent of the farmers used each of the mobile banking and internet banking for the transfer and receipts of money (Figure 2.6).

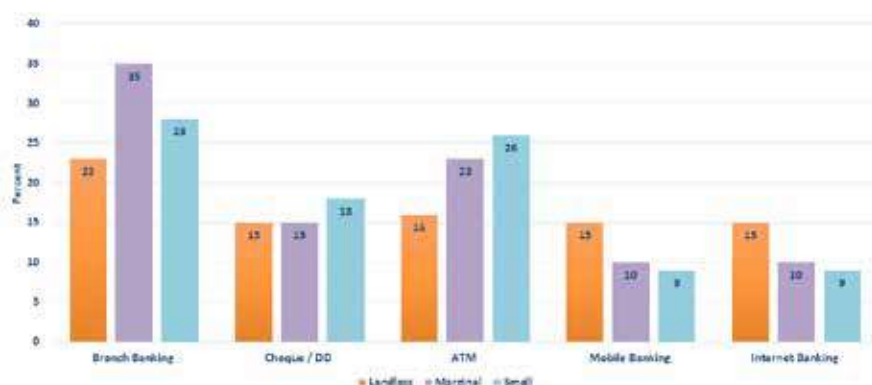
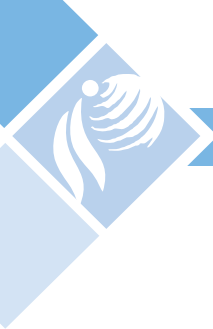


Figure 2.6: Use of banking services by land-size category



NBP/2017/11/20

Developing Business Proposal for Producer Companies

Producer Companies (PCs) and start-ups are formed exclusively by primary producers/innovators without much corporate linkage needs to be supported through hand holding at least in the initial stages. The success of PCs and Start-ups lies in providing goods and services in innovative and efficient manner. The training programme aimed to build an integrated view and enhance the knowledge base to cater to the agri-business management challenges in the current scenario. On completion of the training programme, the participants gained the skills necessary for critically assessing both established and emerging approaches to business plan analysis and developing their own approaches. The objectives of the programme was to provide an overview of the key concepts of business proposals with respect to agricultural technologies focusing more on PCs and Start-up scale. Total 12 participants attended the programme.

2.1.3 Markets and Institutional Innovation

Structural Changes in Agricultural Marketing in India

SIL/2017/11/131
CCP/2017/15/135

The Electronic National Agriculture Market (e-NAM) initiative is considered to be the game-changing policy initiative in agricultural marketing in India. The e-NAM seeks to leverage the physical infrastructure of APMC mandis through an online trading portal, enabling buyers situated even outside the state to participate in trading at the local level. According to Ministry of Agriculture & Farmers Welfare, as on 31st March 2018, 585 Mandis across 16 States are live on e-NAM. The present study has been undertaken to examine the readiness of the APMC Mandis for implementation of e-NAM, awareness of different stakeholders about e-NAM, level of farmers' participation and estimate the benefits from this structural change in selected market. The survey was conducted during April- May 2017 in 4-5 major APMC mandis in two states- Madhya Pradesh and Telangana. Total 96 farmers, 23 traders and mandi official in 4 APMC mandis- Bhopal, Dewas, Indore and Sehore in Madhya Pradesh state were interviewed. Similarly, 94 farmers, 60 traders and 7 mandi officials in Hyderabad, Karimnagar, Thirulmalgiri, Nizamabad, Warangal, Suryapet and Badepally mandis of Telangana state were interviewed. Among these mandis, Nizamabad market committee had already implemented e-tendering system with the collaboration of NCDEX before launch of e-NAM. According to mandi officials in Telangana state, almost all commodities are traded on e-NAM platform, except in Malakpet (Hyderabad) where only chillies are traded.



Farmers with different grain quality and lot size in a mandi of Telangana state

Awareness of farmers: In Telangana, except in Nizamabad market, less than 50 % farmers are aware about the e-NAM in all the selected markets. In Nizamabad, more than 75% farmers know about it because of already existing e-tendering process was implemented in association with NCDEX. In Madhya Pradesh, none of the farmers in the selected markets know about e-NAM.

Implementation of e-NAM: The implementation of e-NAM framework is reviewed based on 12 parameters like gate entry, e-auction, assaying, sale



receipts, etc. which are given in the table 2.5. In Telangana mandis, real time information about the arrival of commodities was not available on the e-NAM platform. Therefore, there was no participation of buyers from other market. Assaying and standardization facility was also not available in the markets, which was creating major hurdles in implementation of e-NAM. Except in Nizamabad direct purchase center (DPC), the payment to the farmers was done through commission agent by cheque/RTGS/cash in all other markets. Only part payment to the farmer was done in cash to meet the labour and transport expenditure. In the selected markets, the electronic weighment was not integrated into the portal except in Nizamabad DPC. In Nizamabad DPC, mandi officials were announcing the winner bid lists. In other markets, manual receipts are given to the farmers instead of generating sale receipts.



Discussion with farmers and traders of groundnut in an APMC mandi of Telangana state

Table 2.5: Implementation status of e-NAM in selected APMC mandi as on survey date

Particulars	Telangana						Madhya Pradesh			
	KMGR	TMLG	NZMD	WRGL	SRPT	BDPL	BPL	INDR	SHRE	DWS
Gate entry/ Lot generation	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Generation of Unique ID	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Sample/ Heap	x	x	x	x	x	x	x	x	x	x
*e-Auction	x	x	✓	x	x	x	x	x	x	x
Assaying	x	x	x	x	x	x	x	x	x	x
Best price-SMS sent to farmers	x	x	x	x	x	x	x	x	x	x
Generation of sale receipt	x	x	✓	x	x	x	x	✓	✓	✓
Online payments to the farmers	x	x	✓	x	x	x	x	✓	✓	✓
Immediate cash payment	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Permit/Gate exist	✓	✓	✓	✓	✓	✓	x	✓	x	x

MLPT: Malakpet; KMGR: Karimnagar; TMLG: Thirumulgiri; NZMD: Nizamabad; WRGL: Warangal; SRPT: Suryapet; BDPL: Badepally; BPL: Bhopal; INDR: Indore; SHRE: Sehore; DWS: Dewas; KHRI: Kheri; STPR: Sitapur; SLTPR: Sultanpur

In the mandis of Madhya Pradesh, following observations were found:

1. e-Auctioning is not yet implemented but the trade data is manually entered into the system after the trade.
2. There are no commission agents in Madhya Pradesh.
3. Payment process in all the markets is not yet integrated with e-NAM.
4. Assaying and SMS services are not implemented in any of the selected markets.
5. Sales receipt and farmer's receipt is generated in all the mandis except in Bhopal.

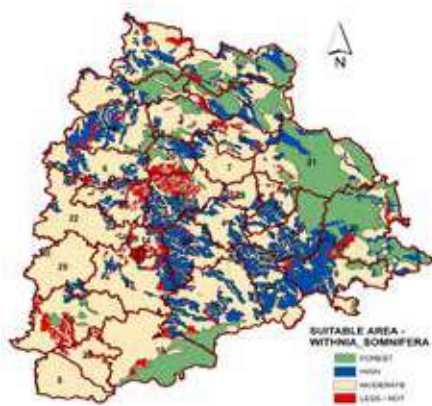
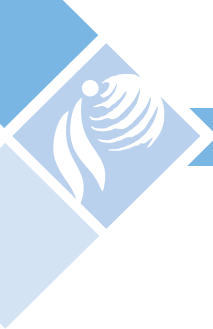


Figure 2.7: Suitable area for ashwagandha in Telangana state

Potential for Medicinal Plants Cultivation in Telangana

The aim of the project is to identify the areas suitable for cultivation of medicinal plants in Telangana. Crops which have good market potential were selected for the study, namely, Aswagandha, Sandal Wood, Aloe vera, Red Sanders, Mucuna, Tulsi, Vacha and Guggul. Various factors like weather factors and soil factors were used for identifying the suitable tracts. The levels of the variables affecting various medicinal crops were determined through discussions with experts. Geo-Spatial analysis was carried out for the entire state for identifying the area suitable for various medicinal plants using GIS spatial data viz., Telangana district map, mandal map, soil map; and non-spatial data viz., monthly average temperature, rainfall, pH (potential of hydrogen) using Arc-GIS software with the help of Map Overlay analysis. Map Overlay analysis is a group of methodologies applied in optimal site selection or suitability modelling. It is a technique for applying a common scale of values to diverse and dissimilar inputs to create an integrated analysis for classifying the suitability area into different classes. Using GIS, the area suitable for cultivation of the following eight medicinal plants were extracted and mapped for all mandals in 31 newly formed districts of Telangana (Figure 2.7). Using these maps, the mandals suitable for cultivation of the medicinal plants can be located for promotion of these crops. The problems faced by medicinal plant growers in Telangana was collected through primary survey. Since a majority of the farmers surveyed were first time growers of medicinal plants, they did not face any issues. Some of the important issues faced by the medicinal plants growers are lack of market and market information, lack of cheaper transportation, non-availability of labour, erratic monsoon and non-availability of crop loan for growing medicinal crops.

Advanced Analytics in Developing Agricultural Market Intelligence

The 21-days Winter School programme was sponsored by ICAR in which 19 scientists of ICAR institutes and faculty of Agricultural Universities participated. The objectives of the programme were to acquaint the participants with advancements in the field of market intelligence; to impart skills in designing survey instruments and conducting surveys; to make the participants having practical exercises in handling and analyzing the data using advanced statistical software, and to equip the trainees with advanced statistical methods used in marketing intelligence. The programme covered Advances in data collection, Price forecasting, Modelling volatility, Big Data Analytics, Demand and supply dynamics, Marketing information system, Determinants of prices, Social marketing tools, Marketing integration, Price discovery and determination, and ICT in agricultural marketing. Group assignments on various topics related to market intelligence were assigned to participants in a group of 3-4 members. They made their presentations before faculty and other participants and submitted write up. Cross-learning among the participants were facilitated through having a regular slot every week where trainees who gained expertise in using statistical software conducted hands-on sessions for other participants. Experts from successful market intelligence cell demonstrated how market intelligence cell was functioning and what are the issues confronted by them.



NBP/2017/17/32



2.1.4 Nurturing Future Leaders in Agribusiness

Post Graduate Diploma in Management-Agriculture (PGDMA)

Quality education in agribusiness is one of the most critical areas wherein institutions and universities attempt to link the academic innovations to the practices. In recent years, several organisations come up with agri-business management programme, though it is always point of mute concern among industrialists that there is serious lack of quality and employable graduates in this sector. The NAARM started 2- year full-time AICTE-approved agribusiness management education programme called PGDMA in the year 2009. The aim of the program is to help the students build solid academic background, strong characters and leadership skills to meet the demand for employers ranging from the agricultural business, financial institutes to the public services. The 30 students of 8th Batch of PGDMA (2016-18) successfully completed their final project and were awarded with provisional course completion certificate. During the academic year (2017-18), 15 agribusiness companies from different sectors have participated in the placement process. The academy's PGDMA program could attract 7 new companies during this season for final placements. The CTC (Cost to Company) offered by the companies ranged between ₹6.0 Lakhs and ₹11 Lakhs per annum with average CTC of ₹7.25 Lakhs per annum. The placement process witnessed participation of companies cutting across the major sectors of agribusiness: Input (Seeds, Fertilizers and Pesticides), Micro-finance, Commodities Exchange, Food Retail, Crop Advisory, Market Research and Consultancy Services.



ACA/2016/01/55

ACA/2017/03/57



Figure 2.8: Prestigious recruiters for 8th Batch of PGDMA

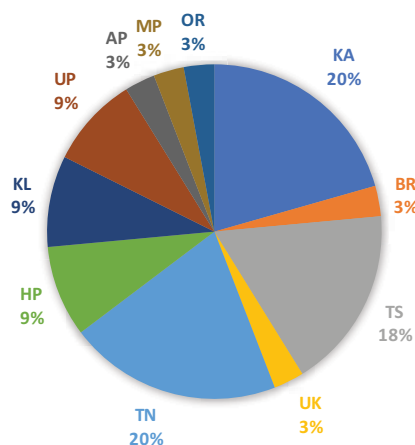


Figure 2.9: States representation in the 9th batch of PGDMA

Thirty four students took admission in 9th batch of PGDMA, out of which 19 are females and 15 are males. These students, representing 11 domicile states, have graduated from 17 different Agricultural Universities with different backgrounds of agriculture, horticulture, fisheries, food technology, biotechnology, agricultural engineering, etc. The academic session started on 20th June 2017.

Admission process of 10th batch of PGDMA also started in December 2017 by inviting applications from the graduating students from any agricultural universities.

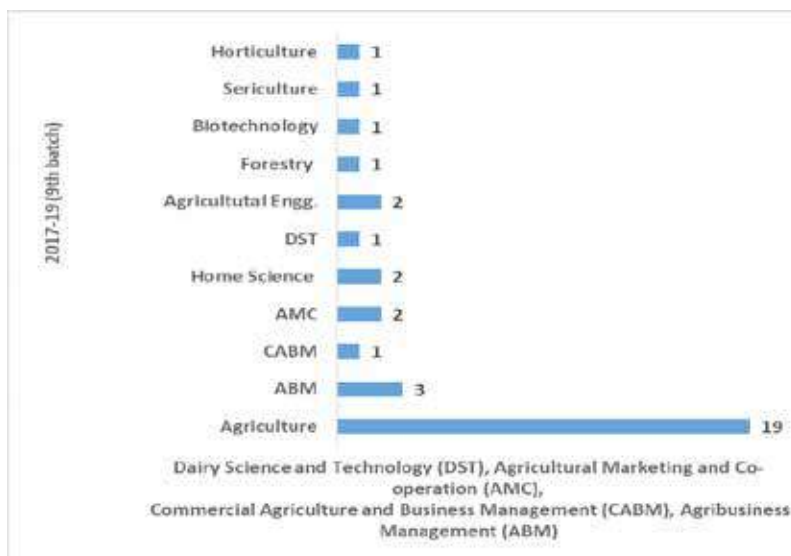


Figure 2.10: Distribution of students across different disciplines

SANKALP: Annual Business Competition Event



Figure 2.11: Sankalp '17 events

The SANKALP, an annual business competition event, is organised by the NAARM-PG Alumni Association (NPAA), which aims at providing a platform to young management graduates to Collaborate, Compete and Connect. The SANKALP '17 was organised during August 28-29, 2017, which was second in series. About 200 students from 21 management and engineering institutes across the country participated in the event. The event received very good response from corporate sector for sponsoring. The total prize money was Rs. 1,00,000/- across 8 events organised.



Winners receiving prize from Director, NAARM



2.2 Education Systems Management for Enriched Learning Quality & Environment

This theme is one of the core themes of the Academy, which essentially aims to:

1. Enhance capacities for agricultural education policy, planning and evaluation in institutions of NARES
2. Enhance capacities for faculty excellence and technology-enhanced learning for increasing learning effectiveness and continuous learning by deploying emerging media to create new and vibrant education environments

In order to advance the objectives of this broad theme, the Academy focused on the following broad areas for its research, academic and capacity building activities.

2.2.1 Lifelong Learning for Professional Development

2.2.2 Education beyond Higher Educational Institutions

2.2.3 Technology Enhanced Learning

2.2.4 Policy Advocacy on Education Systems Management

2.2.1 Lifelong Learning for Professional Development

Professional development and competency enhancement is a continuous phenomenon for any organization for achieving the par excellence in the respective domains. The major thrust of the academy is providing the opportunities for lifelong learning for National Agricultural Research and Education System. In this endeavor, multipronged strategies are adopted. Processes and methodologies are envisioned to enhance the quality of agricultural education in the country. A unique initiative in this direction is competency enhancement of faculty through Foundation Courses and need-based programmes. The outcomes, feedback from participants backed up by research studies formed the basis for designing the courses for lifelong learning.

Assessment of Technological Interventions in Teaching-Learning Process

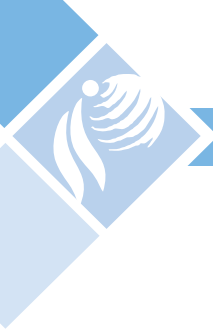
Enhancement in the outcomes of learning is a challenge. The Academy continuously draw support by conducting research for building its capacity building programmes; while aligning to the mandate. One such study on **Assessment of Technological Interventions in Teaching-Learning Process**, funded by the institutional research budget was taken up during the reporting year on a pilot basis. The objectives of the study include 1) to study faculty perceptions on technology experiences and expectations through Educational Technology (ET), 2) to study the student's overall attributes towards technology adoption in learning process, 3) to identify the constraints and to propose suitable strategies for effective functioning of ET cells. The data were collected from Tamil Nadu Agricultural University from three levels of respondents – students, faculty and administrators.

SIX Thematic Areas

1. Agribusiness Management for Inclusive Growth
2. Education Systems Management for Enriched Learning Quality & Environment
3. Enhancing Capacities for Leadership and Governance
4. Extension Systems Management in a Market-driven Environment
5. Information and Communication Management for Promoting Innovation and Governance
6. Mobilizing Science and Technology for Innovation and Sustainable Development



Pilot Study



The study reported utilization pattern of ET resources by the students. Prominent among them are use of e-learning platform (45%), access to online journals (20%), multimedia software (17%), computer lab with all gadgets (13%) and statistical software (5%). The perception of students on the benefits of technology-enhanced courses include peer communication (87.7%), access to grades (84.8%), more convenient to use (83.3%), increases students' awareness on ET (79%) and access to course material (76.8%) etc.

The perceptions of the faculty towards educational technology indicated that it helped them to complete their academic tasks with ease and quick which resulted in more productive time for teaching purpose, helped them to clearly define the purpose, engage students and provide a deeper understanding of lecture material. Faculty also perceived that actual learning and activity-based interaction from educational technology besides easy integration of technology in teaching. However, the faculty further needed training on m-learning (development of mobile app), e-learning, multimedia tools, conducting online quiz and evaluation etc.

The administrators reported challenges in technology adoption for teaching-learning process which include faster obsolescence of technology (76%), lack of manpower support in instructional lab (72%), less time to develop lessons (69%) students scheduling for use of technology (68%) and constant change in the teaching style due to new technology development (56%) etc. The administrators suggested more budget allocation for equipment and other facilities besides regular faculty capacity building programme on ET for quality education in agriculture.

Foundation Course for Faculty of Agricultural Universities (FOCFAU)



FOF/2017/01/02
FOF/2018/02/41
OFC/2017/01/01
OFC/2017/02/03
NBP/2018/29/47
NBP/2018/33/51

The Academy initiated foundation courses for the newly recruited Assistant Professors / Scientists of State Agricultural Universities since 2014. During the current year three such programmes were organized; out of which two programmes for Acharya N G Ranga Agricultural University, Andhra Pradesh exclusively. The objectives of the one month long programme were 1) to orient the participants to the national and international agricultural scenario, 2) to enhance knowledge, skills and attitude (KSA) of the participants in various aspects of education, teaching-learning process and technology in education, 3) to expose the participants to research and technology transfer processes and 4) to sensitize the participants to organizational capacity related to human capital, administrative and financial management.

The major components of the programme include foundations of Teaching-Learning, Teaching Management, Communication and Linguistic Skills, Human Resource Management, Information Communication and Extension Management, Research Methodology & Project Proposal development etc. The practical approach was predominant during the programme to improve the teaching, research and extension skills through micro teaching, human resource management, digital presentation and data analysis exercises. In-house practical sessions reinforced through field visits to reputed National



and International institutions, helped the participants to reflect the experiences and consolidate their learnings at the Academy.

The 4th Foundation programme was attended by 36 participants representing 7 states and 7 Agricultural Universities, while 5th and 6th programmes were attended by 97 newly recruited Assistant Professors/ Scientists/ SMSs of ANGARU, Guntur of Andhra Pradesh. The trainees represented agriculture, fisheries and basic science faculty belonging to different disciplines and varied experience. Eminent personalities of repute viz Dr. Harsh Kumar, C. Sridhar, R.S Paroda, Mangala Rai etc. motivated trainees. Drs. K. Chanth Kumar, Dr. G.S Haritha, Dr. Pradeep Manyam were adjudged the 'Best Trainee' award for his/ her outstanding performance during 4th, 5th and 6th programmes respectively. A revenue of Rs. 49.59 Lakhs/- generated through these programmes.

Following are the details of the programmes

1. 4th Foundation Course for Faculty of Agricultural Universities (FOCFAU) from 2-31, May 2017
2. 5th Foundation Course for Newly Recruited Assistant Professor, Scientists of ANGRAU from 17-1-2018 to 15-2-2018
3. 6th Foundation Course for Newly Recruited Assistant Professor, Scientists of ANGRAU from 12-3-2018 to 10-4-2018

Need-based Competency Enhancement Programmes

The Agricultural Universities realized the importance of competency enhancement of their faculty and NAARM has been recognized as the premier institute in the area of education systems management. During the year the Academy catered the needs of five universities in imparting need based competency enhancement programmes. Karnataka Veterinary, Fishery, and Animal Sciences University, Bidar; ICAR – National Dairy Research Institute, Karnal; Vasanthrao Naik Marathwada Krishi Vidyapeeth, Parbhani; University of Horticultural Sciences, Shimoga and Dhanwate National College, Nagpur (Shivaji Educational Trust, Amaravathi) were benefitted by these programmes. Out of these five programmes, four were off campus and one on campus. The academy has been extending support to higher educational institutions beyond National Agricultural Research and Education System also.

The broad objectives of these programme are to reflect on and renew the knowledge and skills of the faculty of the University for improving the quality of higher education and human resources management and to impart knowledge and skill in writing winning research proposals and impart knowledge on the administrative and financial matters.

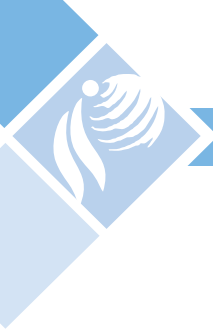
1. Administrative Training for SC/ST Teachers of Veterinary Colleges, Shimoga and Hassan, Karnataka from 24-26 April, 2017 attended by 17 faculty.
2. Competency Enhancement of Faculty and Scientists of ICAR-NDRI from 25-30, May 2017, attended by 32 faculty/ scientists.



**Dr. B Janardhan Reddy, GHMC
Commissioner, Hyderabad addressing the
participants during Valedictory Programme
of 4th FOCFAU**



**Certificate distribution during
administrative training for faculty of
Veterinary science**



3. Competency Enhancement for Efficiency and Effectiveness at Workplace for Faculty and Scientists of Vasantao Naik Marathwada Krishi Vidyapeeth, Parbhani organized from 4-8, September 2017, attended by 36 faculty members.
4. Capacity Building Programme on Enhancing Efficiency and Effectiveness at Workplace for Faculty & Scientists of UAHS, Shimoga from 29-1-2018 to 2-2-2018, attended by 30 Assistant Professors.
5. Faculty Development Programme on Enhancing Efficiency and Effectiveness at Workplace at Dhanwate National College, Nagpur from 6-2-2018 to 10-2-2018 attended by 35 faculty members of management/science / commerce education.

The academy has generated a revenue of ₹ 9.55 lakhs through these programmes. The Academy also plays an important role in supporting other organizations in conducting similar programmes by faculty participation. The higher education institutions served by NAARM during the year are – University of Agricultural Sciences, Dharwad; Tamil Nadu Veterinary and Animal Sciences University, Chennai; Karnataka Veterinary, Fisheries and Animal Sciences Universities, Bidar; Maulana Azad National Urdu University, Hyderabad; Professor Jayashankar Telangana State University, Hyderabad; Banarus Hindu University, Varanasi; Sri Venkateswara Veterinary University, Tirupati; Birsa Agricultural University, Ranchi etc.

2.2.2 Education beyond Higher Education Institutions

The endeavor to enhance the quality of agricultural education, and providing access and equity to all stakeholders, the Academy took the path of reaching out the unreached with the help of technologies; thus contributing to redefining of educational process.

Asynchronous Learning in Agricultural Higher Education through MOOCs: A Case Study

Dynamics of traditional learning and asynchronous learning always vary. Despite technological explosion and support by public and private educational entities, the synchronous and asynchronous learning yet to take off significantly. The debate on traditional vs online education is always live.

A research project **Asynchronous Learning in Agricultural Higher Education through MOOCs: A Case Study** was undertaken to broadly understand how asynchronous learning takes place through Massive Open Online Courses. The objectives of the project were to investigate perceptions and expectations related to MOOCs from the perspective of Agricultural Education, to explore learning analytics of MOOCs, to analyse critical success factors and to identify suitable courses for MOOCs in Agri. Education and suggest a framework for development of such courses.

The learner's educational background, institutional representativeness, stage of career, content delivery, expectations from MOOC was ascertained through an online questionnaire from the 1003 registered users before the launch of MOOC course "Competency enhancement for effective teaching".



SIL/2016/05/115



Subsequently, viewing behavior, learning curves, spatial and temporal access of content, use of internet browsers / IP address, quality attributes of resources, goals of learning through MOOCs, motivational factors for online education (learning), sources of information, comparative success rates (Table 2.6), learners engagement, learning strategies, withdrawals / dropouts from MOOCs, effectiveness MOOCs, extent of change over a period of time (longitudinal study) (Table 2.7) and courses suitable for agricultural education etc. were ascertained from 982 participants of MOOC on Teaching Management.

Majority of the online learners adopted learning strategies such as watching videos (85.78%), reading course materials (83.35%), assignments (64.71%) and combination thereof (Figure 2.12).

The success of this course was mainly attributed to relevance, personal interest, desire for expansion of skill sets, confidence etc. (Figure 2.13).

The issue of quality enhancement of agricultural education has been addressed under this project through Massive Open Online Courses (MOOC). The MOOCs reached beyond higher agricultural institutions to the STEM (Science, Technology, Engineering and Mathematics) organizations. About 1985 faculty and aspiring faculty benefitted from MOOC over the last two years and more than 250 Institutions participated (majority from SAUs) in MOOCs and Rs 6.61 lakhs earned through certification.

Table 2.6: Trainees engagement and success rates

Item	MOOC 1 (n=1003)	MOOC 2 (n=982)
Discussion Threads	305	1290
Assignments	1456	1648
Feedback	386	448
Mock Exam (Online)	341	307
Final Exam (Online)	473	506
CoP (Community of Practice)	13	130
Challenges/Problems in Teaching	0	180
Total events	2974	4509
Engagement in events	2.97	4.59
Marks in Final Examination	68.0 (n=473)	67.0 (n=506)
Completion Rates	350 (34.89%)	421 (42.87%)
Participation Rate	179 (17.85%)	95 (9.47%)
Success Rate	52.74%	52.55%

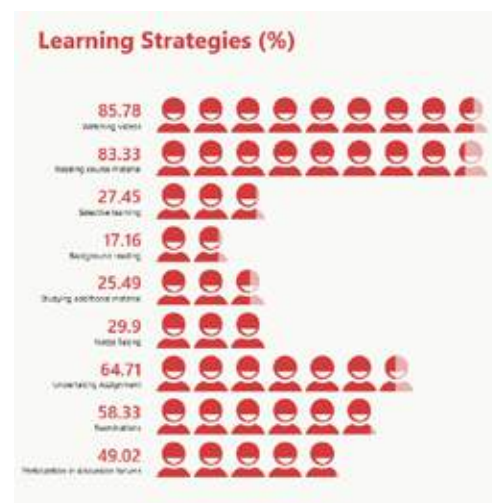
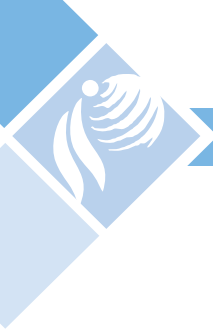


Figure 2.12: Learning strategies



Figure 2.13: Motivational factors for online learning

**Table 2.7: Extent of Change through Online Education (MOOCs)**

Age (years)	Upto 30%	30 to 60%	More than 60%	Total %
Less than 30	10.40	8.91	9.41	28.71
31-40	15.35	16.34	18.81	50.50
41-50	4.95	3.96	6.44	15.35
More than 50	2.48	0.99	1.98	5.45
Total	33.17	30.20	36.63	100.00



MOO/2017/01/29

Massive Open Online Course on Technology Management

Majority of the educational stakeholders are devoid of opportunities to enhance their pedagogical skills owing to many reasons. The situation in professional education is different as the faculty are not exposed to pedagogy, teaching management etc. A **Massive Open Online Course on Teaching Management (MOOC)** has been initiated by the Academy during the year and the Academy first to introduce such Massive Open Online Courses across National Agricultural Research and Education System. During the year a MOOC on Teaching Management was offered from 1-30, November, 2017. The objective of this online course is to make participants understand various dimensions of teaching management and develop effective teaching styles for quality education. The target learners are the young faculty, aspiring faculty (PG and Ph D students) and other interested learners. All learners irrespective of disciplines, institutions are provided an opportunity. About 982 learners participated in the programme representing many SAUs; out of which 529 learners completed the course successfully. The content was delivered at <http://elearning.naarm.org.in/> through videos, ppts and reading materials as follows.

- Videos - 17 (5h 33m)
- PPTs - 15 (241 nos)
- PDFs- 9 (135 pages)

This MOOC is 3rd in series and achieved a success rate of 52.55%. The innovations introduced in this course are Online Certification (first time), QR Code, use of Google Analytics and implementation of digital transactions (99%) in collection of certification fee.



MOOC (online) participants interacting with Programme Director at Shivamogga

2.2.3 Technology Enhanced Learning

Enriched learning quality in Agricultural Education necessitates integration of technology in teaching-learning process. Several technological advancements are bringing sweeping changes in the society and education sector is no exception. In this direction, the Academy focused on building capacities of NARES on technological integration with sound research base. The Academy's mission on technology enhanced learning stems from earlier experience on eLearning.



Technology Enhanced Learning in Agricultural Education

This project was sanctioned by ICAR under Niche Area of Excellence. The project is envisaged to develop suitable digital content development protocols from production (by a domain specialist who can be a teacher, scientist or subject expert) to consumption (by a learner who can be a student, farmer, rural youth/entrepreneur). The protocols include

- practice of technology-enabled pedagogy that contours along the various natural styles of teaching
- methods of content capture
- processes of refinement and storage in appropriate digital formats
- mode of delivery to the end-user either offline/online
- integration of interactivity to the digital content
- repurposing methods of the granular content for purposes like distance, vocational, open access, massive open online courses



SOL/2014/09/100



TELAge Lab

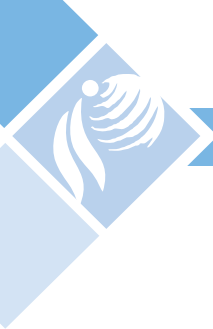
TEL- A State of Art Resource Centre

A state of art Technology Enhanced Learning Resource Centre is established which is scalable, easily implementable, cost-effective and with less manpower-intensive. The protocols were constantly refined and upgraded to enhance better quality in content.

Besides these, similar facilities were established in identified consortia centres (TANUVAS, PJTSAU, and DRNVJIRD) and sensitized about digital content production practices from time to time. The practices and knowledge generated were shared through capacity building programmes and policy workshops. Over 2000 personnel were exposed to the TEL protocols developed.



Figure 2.14: Screen Shot of MOOC on Teaching Management



The protocols were used to develop model digital courses, offer distance and online programmes, conduct virtual classes, and support some of the in-house programmes. A total of 6 digital courses were developed. A distance education programme was offered completely in online mode with 18 participants. A total of about 100 streaming hours were provided for virtual classes to support contact sessions of another distance education programme.

Hosting of Massive Open Online Courses

The facility was innovatively used to offer Massive Open Online Courses through open source e-learning methodology and the content of 3 MOOCs produced (Figure 2.14).

eLearning portal (<http://elearning.naarm.org.in/>)

A portal is established which acts as a single window gateway to access educational content of various purposes-

- Academic (teachers, students) – Traditional courses in residential mode and distance education courses
- Vocational (Subject Matter Specialists and farmers/rural youth/ rural entrepreneurs),
- MOOCs (Content developers and any learner),
- Open courses of general interest



Figure 2.15: Average Hits

As seen in figure 2.15, the website activity is more in the months of November to January. The activity touches a maximum of over 60000 per month during December. This is because of offer of MOOC during this period. There is online activity during other months of a year also with the overall average hit rate of portal 21015 per month.

Figure 2.15 shows the overall activity history of e-learning portal since its launch. The three peaks seen in the figure depict the MOOC activity during those periods. It is significant to note that there is a steady increase in MOOC activity peaks over three years. The MOOC activity grew at a rate of 217 percent over three years. It is a clear indicator that MOOC has a positive influence among the learners.



Revenue Model

The methodology is blossomed into a revenue generation activity also by providing the consultancy-mode support to other agencies in developing and hosting the digital content. Programmes executed for institutes like MANAGE, NIRD & PR which generated a revenue of about 10 lakh INR.

The project outcomes bear a key significance for Agricultural Education and Knowledge as the emphasis is laid on programmes like NAHEP of ICAR and SWAYAM of MHRD.

The project outputs for the year 2017-18 are summarized as below

Type of Output	Quantified Output
i. Digital Video content hours	314
ii. Number of Video Modules	162
iii. Number of Online Courses	3
iv. Number of Certificate Programs	1
v. Number of National Visitors to lab	734
vi. Number of International Visitors to lab	12
vii. Number of Live Streaming Hours	48
viii. Number of Average hits per day	1082
ix. Revenue Generated (Rs lakhs)	6.4
x. Other projects being supported	2

The key outcomes of the project are

- TEL protocol standardized with less manpower, high quality and quick production
- Digital course development practices
- Virtual classes to strengthen distance and online education
- MOOCs in an innovative and self-supportive format
- Capacity building of over 2000 personnel and handholding of consortia partners
- Establishment of Revenue Generation Model
- Model Vocational Modules

Some of the **FIRSTs** include

- MOOC (completely in-house)
- Distance online program
- Online certification

Virtual classes for distance programmes

Training Programme on Web-based Educational Content Management (under TELAgE)

The objectives of the programme include sensitize participants on the concept of E-Learning (EL) and its practice with emphasis on open source; to provide an insight into different methods of teaching and learning through Technology Enhancements and impart practical training on effective teaching using e-media and to inculcate the practice of Web-based Learning Management for EL resources using open source methodologies.



NBP/2017/12/22



Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM, addressing participants

More emphasis was given on practical experience on e-learning tools, video production, editing and publishing process, animation techniques, video scripting and open education resources. As part of practical sessions, the participants developed short duration digital content modules on Azolla production, Tomato cultivation, formation of cooperative society, control of mastitis and handling Vetero-legal postmortem examination.

The training programme was organized from October 4-13, 2017 and 13 participants attended the programme, out of which three were females.

2.2.4 Policy Advocacy on Education Systems Management

The Academy is actively engaged in the development of policies focusing on the major theme - Education Systems Management for enriched learning quality & environment regularly. The Academy develops policy briefs by conducting workshops and brainstorming sessions.

Policy Workshop on Digital Learning in Agriculture with reference to Social Media (under TELAgE)

The workshop was organized to update the participants about the knowledge on digital learning tools including social media; and to develop strategies, roadmap for use of digital learning in Agriculture Education. The workshop was organized under ICAR Niche Area of Excellence project on “Technology Enhanced Learning in Agricultural Education”. The initiatives / activities of participant organizations with respect to Social Media and Digital Learning documented. Through brainstorming session, the way forward for use of digital learning in agriculture was documented. The programme was attended by 21 participants, including 2 women, held from 5-7 July, 2017.



WS/2017-18/03
WS/2017-18/11
WS/2017-18/02



Dr. P.K. Joshi and Dr. Ch. Srinivasa Rao with the participants of 'Policy Workshop on Digital Learning'



National Workshop on Technology Enhanced Learning

This workshop was designed to share and sensitize the experiences of TELAgE project and how to take the results of the project for wider use in NARES. The workshop was designed to seek the roadmap in the following areas

1. Technologies for teaching, learning and content development
2. Skill Building Education
3. Distance and Vocational Education

The policy issues, organizational interventions and infrastructural requirements in the aforementioned areas as perceived by various stakeholders have been discussed through panel discussion and thematic brainstorming sessions. About 66 participants deliberated for two days on January 24th and 25th, 2018.



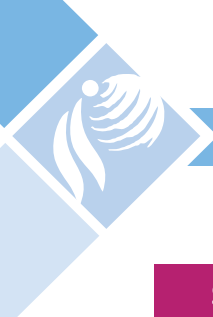
Dr. P.S. Pandey ADG addressing the participants of National workshop on TEL

Review Meeting of Implementation of Recommendation of Fifth Deans Committee on the Community Science (Home Science) & Fisheries Sciences

The ICAR – NAARM organized this programme in collaboration with Education Division of ICAR to review the status of implementation of Recommendations of Fifth Deans' Committee in disciplines of Community Science (Home Science) and Fishery Science. The status of implementation of V Deans Committee recommendations, issues and challenges discussed and resolved to ensure uniformity in implementation of curricula in Community Science and Fishery Science. About 56 Deans of Community Science and Fishery Sciences, official of Education Division, faculty of NAARM participated in the two days' deliberations held on 29th and 30th June, 2017.



Dr. N. S. Rathore, DDG (Edn.), ICAR addressing Deans of Community Science and Fishery Science during review meeting



SIX Thematic Areas

1. Agribusiness Management for Inclusive Growth
2. Education Systems Management for Enriched Learning Quality & Environment
3. **Enhancing Capacities for Leadership and Governance**
4. Extension Systems Management in a Market-driven Environment
5. Information and Communication Management for Promoting Innovation and Governance
6. Mobilizing Science and Technology for Innovation and Sustainable Development

2.3 Enhancing Capacities for Leadership and Governance

Human Resource Management (HRM) deals with issues related to performance management, organization development, safety, wellness, employee motivation, training and others. It plays a strategic role in managing people and the workplace culture and environment. The objective of HRM is the maintenance of better human relations in the organization by the development, application and evaluation of policies, procedures and programs relating to human resources to optimize their contribution towards the realization of organizational objectives.

One of the important domains of HRM is to enhance capacities for leadership and governance in an organization. The strategic framework of this thematic area is:

- Institutionalize a framework for leadership development at all levels of National Agricultural Research and Education Systems (NARES) – early, mid- career, senior professionals and managers across all functions (research, education, extension) and
- Institutionalize good governance in NARES by building capacities for effective management of institutions, performance assessment, accountability, and developing strong organizational value system and work culture that promote innovation.

To realize the above objectives, the following activities related to research and capacity building programs were carried out during 2017-18.

Competency Requirement and Training Need Assessment for the Faculty of Agricultural Universities

The main objective of the study was to find out the relevance and knowledge levels of teaching competencies as perceived by faculty, students and administrators so as to develop a teaching competency framework for the faculty of Agricultural Universities. The study was conducted with 292 respondents consisting of 30 educational administrators/experts, 97 faculty members and 165 students spread over 12 Agricultural and allied sciences (Veterinary, Horticulture, Fishery and Home Sciences) universities. A 28-item Teaching Competency Questionnaire (TCQ) was administered separately for each sample group personally. All participants were asked to rate their perceptions on relevance and knowledge levels of each of the teaching competencies on Likert five-point scale (1 = Low Knowledge/Relevance; 5 = High Knowledge/Relevance). Data were analyzed using descriptive statistics and the ranking procedure described by Edwards & Briers (1999). Borich (1980) needs assessment model was used to identify training needs of the faculty by calculating mean weighted discrepancy scores (MWDS). Results indicated that there are differences in perceptions of educational administrators, faculty and students towards the relevance and teachers' knowledge levels of teaching competencies. However, based on the average perception of all the stakeholders, the competency requirement of faculty by cadre-wise were identified and furnished in table 2.8



SIL/2016/06/116
SIL/2017/12/132
SIL/2017/13/133

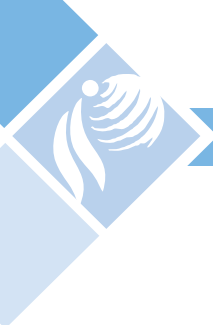
**Table 2.8: Competency requirement of faculty cadre-wise**

Faculty	Competencies required	Mean Relevance
Entry-level (Assistant Professors)	Have a positive attitude towards students	4.64
	Commitment to promote learning of all students	4.52
	Importance of life-long learning	4.45
	Time management skills	4.32
	Classroom management skills	4.31
	Motivating student learning through different teaching methods	4.13
	Command of subject matter knowledge	4.08
	Updating of subject matter knowledge	4.05
	Under graduate advising, guidance and counselling	4.04
Mid-level (Associate Professors)	Knowledge and application of teaching strategies and skills	4.52
	Post Graduate & Ph D guidance	4.44
	Mentoring skills	4.42
	Time management	4.41
	Have a positive attitude towards students	4.40
	Classroom management skills	4.32
	Updating of subject matter knowledge	4.30
	Research and scientific skills	4.23
Senior-level (Professors)	Importance of life-long learning	4.83
	Have a positive attitude towards students	4.73
	Updating of subject matter knowledge	4.66
	Motivating students' learning through different teaching methods	4.63
	Leadership skills	4.60
	Mentoring skills	4.56
	Ability to reflect and evaluate one's own performance	4.53
	Post Graduate & PhD guidance	4.33
	Awareness and knowledge of policies related to education	4.30

Based on the MWDS of each teaching competency, training needs of faculty were prioritized (Table 2.9). The identified and prioritized teaching competencies provide the content and direction for development of capacity building programs for faculty, in service up gradation suggests periodic competency need assessment of faculty at all levels (early, mid-career and senior) and develop appropriate programmes for their development.

Table 2.9: Prioritized training needs for faculty of agricultural universities

Level of priority	Competency	Rank
Very high	Knowledge and application of teaching strategies and skills	1
	Motivation of student learning through different teaching methods	2
	Have a positive attitude towards students	3
	Importance of life-long learning	4
	Updating of subject matter knowledge	5
	Time Management skills	6
	Research /scientific skills	7



High	Under graduate advising, guidance and counselling	8
	Curriculum design, implementation and improvement	9
	Awareness and knowledge of policies related to education	10
	Leadership skills	11
	Ability to reflect and evaluate one's own performance	12
	Sharing and exchange of subject matter knowledge	13
	Command of subject matter knowledge	14
Moderate	Command and application of pedagogical content knowledge	15
	Mentoring skills	16
	Classroom Management skills	17
	Commitment to promote and learning of all students	18
	Evaluation and review of teaching and learning programmes	19
	Updating and sharing of pedagogical content knowledge	20
	Issues of inclusion and diversity	21
Low	Dispositions to change and flexibility	22
	Use of student assessment results (Feedback)	23
	Language proficiency	24
	Cooperating with colleagues, groups and organizations	25
	Use of Multi-media and web based technologies for teaching	26
	Student assessment methods and procedures	27
	Post Graduate & PhD Guidance	28

Team Work and its effect on Organizational Performance in ICAR

This study was undertaken with the objectives of (a) to identify the level of teamwork skills and knowledge on teamwork among the stake holders in NARES. (b) to identify the effects of teamwork on the organizational performance in the NARES and (c) to identify the factors effecting teamwork in the organizations. Out of the 1739 participants studied on their teamwork skills, 77.74 percent have low teamwork skills, 19.15 percent have moderate teamwork skills and 3.11 percent have high teamwork skills. Out of 141 participants in the team, orientation behaviour inventory for task values and task skills, 11 are found to be in lone ranger profile, 24 are in educably retarded profile, 7 are in trainer consultant profile and 99 are in fully functioning team member profile. Similarly, with respect of maintenance of values and maintenance skills profile, 15 are found to be in lone ranger profile, 12 are in educably retarded profile, 14 are in trainer consultant profile and 100 are in fully functioning team member profile. Two ICAR teamwork award winners were interviewed about their teamwork achievement. The study will be continued further and successful cases of teamwork in research will be documented.

Competency Management System for Agricultural Research and Extension

An institutional project was undertaken with the twin objectives to identify the functional and behavioral competencies and to develop a competency framework for scientific cadre for agricultural research and extension.



The available review of literature on Assessment & Development of Competency Framework thoroughly scanned and based on that six functional competencies viz. Discipline orientation, Research attributes (Cognitive), Research Methodologies, Project Management & Governance skills, Research Communication skills and Research Application & Extension attributes and two behavioural competencies viz. Leading & Managing relationship attributes and Networking criteria identified for the study purpose. Developed a questionnaire to study “Competency Framework for Agricultural Research and Extension Scientists” for its relevancy, importance & level of competency.

Executive Development Program on Leadership Development

This program is designed for the newly-recruited Directors, Zonal Project Directors, Assistant Director General, Joint Directors of National Institutions and others who comprise the Research Management Position cadre of ICAR. The focus of the program is on leadership development for transition of National Agricultural Research System to National Agricultural Innovation System.

The objectives of the program include:

- To understand key challenges of leadership and reflect on the ability to tackle such challenges.
- To develop core competencies required for effective decision making
- To develop professionalism and team culture in research system.
- To become leader and develop next-generation leaders.

During the reporting period, one Executive Development Program was organized with 10 participants including one women participant. Personal effectiveness, leadership essentials and system focused areas were the core themes in 5 days long EDP with integration of yoga. The session on topics like inspirational leadership, talent management in research organizations, leadership excellence, good governance, budget planning and implementation, priority setting, monitoring and excellence, institutional management and accountability, institutional building were dealt through case studies, experiential sharing, field visits etc. The flexible framework enabled leaders to reflect on their experiences during the program, which provided an opportunity for the participants to make presentations on their experiences on leadership and to make a self-audit to look at what they would or would not do during their tenure as leaders.

Management Development Program on Leadership Development

These program are planned to catalyze succession planning in National Agricultural Research and Education Systems (NARES). Client group for this program includes Heads of Divisions and other senior-functionaries who are in the pre-RMP cadre of the organization and who are responsible for leading various activities at their level. The objectives of this program are:

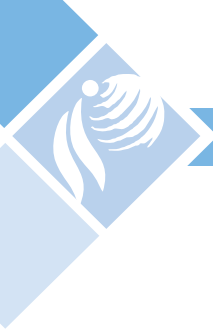
- To help participants understand their own personality dimensions;
- To explore the leadership challenges and core competencies needed in present day context;



EDP/2017/01/09
MDP/2017/01/04
MDP/2017/02/35
NBP/2017/03/08
NBP/2017/01/05
NBP/2018/30/48
NBP/2017/02/06
NBP/2017/24/42
OFC/2017/03/12
NBP/2017/15/28
NBP/2017/22/39
NBP/2017/08/16
NBP/2017/06/13
NBP/2017/27/45



DG, ICAR with EDP participants



- To develop understanding and skills in research and technology management, information and knowledge management, and administration and finance management; and
- To adapt ideas of diverse skills for sharpening capacities to effectively execute various roles in the organization.



Participants involved in management games

During the reporting period, two such programs were conducted. The total number of participants in these two programs include 41 male and 10 female participants. Heads of the Divisions and other senior functionaries, who are in the pre-Research Management positions (pre-RMPs) in the National Agricultural Research and Education System (NARES) play a very important bridge role in the organization and they take up leadership responsibilities in many activities that they perform. The program provided an opportunity for the pre-RMPs, who are in the “middle zone”, to develop smooth and gradual transformation to step into future leadership position. The program covered the areas of Leadership and personality development; Research and technology management; Information and knowledge management; and Administration, finance, and vigilance management.

Need Based Programs

These Programs are organized for mid-career managerial, scientific, technical and administrative personnel of National Agricultural Research and Education System. They are designated for imparting advanced knowledge and skills in human resource management based on the expressed demand from different stakeholders in NARs like Agricultural Universities, Government Departments and ICAR Institutes in enhancing the efficiency of their employees.



Participants of training programme on stress management

A training program on “**Stress Management**” was conducted with a total of 12 participants including three female participants from State Agricultural Universities (SAU's) and ICAR institutes. The participants are either Professors cadre or above cadres from these institutes. This training program facilitated the participants to develop in-depth understanding of personal and psychological causes of stress/anxiety and its effects – physical, social and organizational behaviour. The participants made to understand the mechanism of occurrence of various stress related disorders – cardiac, hypertension, diabetes and other psycho-somatic diseases. Imparted various techniques to develop constructive means for coping up with distress and achieve productive and enriched life-style.

A Competence Enhancement Program on Soft Skills and Personality Development for Technical Staff of ICAR was conducted with total participants of 27 including one female technical participant. The objective of this program is to enhance the efficiency of participants in discharging their functions and providing constructive technical support in meeting the organizational goals and targets through this training on soft skills and personality development. This training program was designed to cover both theoretical and practical aspects on various topics such as Communication skills, Presentation skills, Interpersonal Relationships at work, team work, personality development, time management, inter-team interaction, Official language policies & Hindi word processing, MS Word, PowerPoint, Excel,



Office communication & drafting etc. WhatsApp group on ICAR - Soft Skills trainees has been created to enhance their writing skills, communication skills. Most of the trainees have made communication through this App and used the technology in proper way.

A training program on **“Competency Enhancement Program for Effective Implementation of Training Functions by HRD Nodal Officers of ICAR”** was conducted with a total number of 24 participants including 2 female participants. The objective of the program includes enhancing competencies of the HRD Nodal Officers to provide support and guidance in the implementation of capacity building programs and training functions. The program covered the broad areas in Systematic and Competency Approach to Training, Training Need Analysis (TNA) of employees of all categories, Design and development of training program, Development and Implementation of Cadre Training Plan and Annual Training Plan for all categories of employees and Evaluation and Monitoring of trainings at the Institute level.

Two **Refresher Courses for Section Officers, AAOs, AFAOs and Assistants of ICAR** were conducted at the Academy. The participants were exposed to topics like Establishment Rules, Pension and other Retirement Benefits, Pay Fixation, Reservation in Services for SC/ST/OBC, Ethics and Values in Public Governance, E-Procurement / Purchase Management in Government, E-office, Good Governance and sustainability through HR Practices, Leadership and Performance Management, Communication skills for Manager and Technical Service Rules.

Three training programs were conducted for **Enhancing Efficiency and Behavioral Skills of Stenographers Grade-III, PA, PS, PPS and Sr. PPS of ICAR** at CIFE, Mumbai, NAARM, Hyderabad and at NBSSLUP regional station, Kolkata. The topics covered during these training programs include Noting and drafting, Stenographic Skills, Office Procedures, conduct rules, Leave rules, Official language policy, Communication skills, interpersonal skills, Computer skills, stress management, motivation, positive thinking and personality development.

A **Competence Enhancement Program on Motivation and Positive Thinking for Technical Officers of ICAR (T5 & above)** was conducted at the Academy. The topics covered during this program includes personality development, communication skills, perception, motivation, trust building, positive attitude, assertiveness, teamwork, interpersonal relationships and understanding people's behavior.

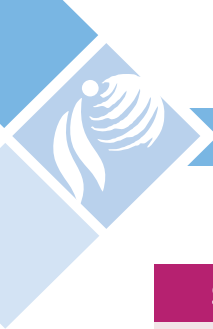
Two training programs were conducted on **Establishment and Financial Matters for Under Secretaries, SAOs, SFAOs, AOs & FAOs of ICAR** at the Academy. This program covers the topics on Establishment Rules, Pension and other Retirement Benefits, Pay Fixation, Reservation in Services for SC/ST/OBC, Ethics and Values in Public Governance, E-Procurement/ Purchase Management in Government, E-office, Good Governance and sustainability through HR Practices, Leadership and Performance Management, Communication skills for Manager and Technical Service Rules.



Technical staff of ICAR participating in training



Group photo of refresher course for administrative finance officer of ICAR



SIX Thematic Areas

1. Agribusiness Management for Inclusive Growth
2. Education Systems Management for Enriched Learning Quality & Environment
3. Enhancing Capacities for Leadership and Governance
4. Extension Systems Management in a Market-driven Environment
5. Information and Communication Management for Promoting Innovation and Governance
6. Mobilizing Science and Technology for Innovation and Sustainable Development



SIL/2017/07/127

2.4 Extension Systems Management in a Market-driven Environment

This theme is one of the core themes of the Academy, which essentially aims to:

- Enhance operational, adaptive and generative capacity of front-line agricultural extension systems to address emerging challenges
- Enhance capacities for use of ICTs to provide customized knowledge, skills and solutions to farmers, farmer groups, rural communities, and for social networking

In order to advance the objectives of this broad theme, the Academy focused on the following broad areas for its research, academic and capacity building activities.

2.4.1 Extension policy, planning and management;

2.4.2 Extension information systems;

2.4.3 Gender mainstreaming in extension.

2.4.1 Extension Policy, Planning and Management

The extension system, which carried the bandwagon of green revolution on its shoulders during 70s and 80s, is in crossroads. There has been a demand for re-examining the architecture of Agricultural Extension research, so that scientists, administrators and private entrepreneurs can together help farmers to bridge the gap between yields that are possible and those that are actually achieved. A strong, vibrant and responsive extension research system which can play a pivotal role in enabling the country to achieve the plan objective of faster, sustainable and more inclusive agricultural growth is the need of the hour, where the academy concentrate.

Research Role Analysis of Extension Scientists in ICAR

Item analysis was done and judges ratings on a 10 point continuum of importance and relevance was elicited from 21 out of the 192 extension scientists of ICAR. The 25 statements scored as above were subject to factor analysis with varimax rotation and we got 3 distinct factors of extension scientists roles for ICAR Extension scientists. The coefficient of concordance revealed a very good agreement among the judges, the reliability was computed by split half method with r value of 0.94, and the varimax rotation has yielded 3 factors to be named as extension research roles.



Management and Impact Assessment of Farmer FIRST Programme

The Farmer FIRST as a concept of ICAR is developed as farmer in a centric role for research problem identification, prioritization and conduct of experiments and its management in farmers' conditions. The focus is on farmer's Farm, Innovations, Resources, Science and Technology (FIRST). Two terms 'enriching knowledge' and 'integrating technology' qualify the meaning of Farmer FIRST in Indian context. 'Farmer FIRST' programme aims at enhancing farmer-scientist interface for technology development and application. ICAR-NAARM as a Project Management Team is involved in delineating and developing research process, methodologies of impact assessment, content development and database creation. It is mainly involved in assessing the impact of technological intervention of Farmer FIRST Programme (FFP), designing & developing the database and decision support system for FFP with the focus on enhancing the capacity of researchers/faculty in methodologies FFP implementation. Five training workshops were conducted on "Methodological Framework for Implementation of Farmer FIRST Project" with 286 participants. Feedback Analysis of workshop was done after completion of each training programme. After three months, extended feedback was administered with the trainee participants for further refinement and upgradation of the training for next coming years.



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NBP/2017/19/34

Good Practices in Extension Research

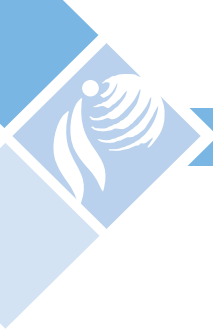
Extension research has long played a significant role in maximizing the utility of socio-technological interventions. Extension research has evolved over the years by integrating scientific research methods and tools from other social sciences. However, the methodological evolution has stagnated in the last few decades thereby producing insignificant outcomes. Quality and relevance of extension research, therefore, has been the subject of serious concern. This training programme aimed at developing the critical abilities of extension professionals to conduct quality research and evaluation. The training was designed with the objectives of equipping the participants with knowledge, skills needed for conducting impact-oriented, quality extension research, and evaluation, providing hands-on practice in writing convincing research proposals in extension and facilitating networking among the participants for inter-institutional research in extension.



Management and Impact Assessment of
Farmer FIRST Programme

Methodological Framework for implementation of Farmers FIRST Program

The Farmer FIRST initiative has been launched by ICAR to move beyond the production and The Project Management Team ICAR-NAARM along with ICAR- NIAP, ICAR-IASRI and ICAR-DKMA are involved in management, capacity building, impact assessment, and content development and showcasing of the technological interventions implemented by 51 FFP institutes all over the country in 23 States. The training workshops on "Methodological Framework for Implementation of Farmer FIRST Programme" for the benefit of 256 participants from 51 FFP institutes of



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WS/2017-18/09
WS/2017-18/01
WS/2017-18/10



Group Photo of Methodological Framework for implementation of Farmers FIRST Program

23 states with the support of five host institutes. The topics on overview of FFP, Participatory Technology Development (PTD), select PRA tools, impact assessment, content development, showcasing of intervention and process documentation were covered. The pedagogy included lectures, interactions, video modules, field visits, case study and discussions.

Brainstorming Workshop on “Collaboration in Research, Education, Extension and Management”

Collaboration and partnerships are the proven strategies to achieve synergy, efficiency and effectiveness by harnessing the mutual strengths of the involved institutions. Hyderabad happens to be an important epicenter of agricultural research and development, as eight (8) ICAR institutes and two (2) regional stations of ICAR institutes, MANAGE, NIRD&PR, NFDB, NIPHM, three (3) State Agricultural Universities and ICRISAT (CGIAR) are located here. Though each of these institutions have specific mandates, they share common framework for research and extension management. It is here that a lot could be learnt by each other to document good research and extension management practices and to identify possible areas within research and extension management for strategic collaborations and partnerships to achieve synergy in performance and magnify the outputs and outcomes. With this backdrop, the workshop was organized with the objectives, i) to identify areas for strategic collaborations in the areas of research and extension management, and ii) to develop action plan for strategic collaborations in research and extension management. Twenty-three Heads of the Hyderabad-based agricultural R&D institutions or their representatives attended.

Let's listen to farmers: A workshop on farmers' feedback on doubling farm income by 2022

In a transition economy like India, consistent performance of farm sector of paramount importance. Ensuring sustainable livelihoods of all sections of the country including farmers is very much a need in times of changing global equations due to global trade and climate change. Thus, Doubling Farmers Income (DFI) by 2022 has become an explicit policy statement and



intent of the government to adapt to these challenges. As a step towards assimilating experiences of farmers and placing the same in front of policy makers, a farmers' workshop was organized by ICAR-NAARM during 22-23 December 2017. The overarching goal of the workshop is to acknowledge the fact that farmers are 'partners' in agricultural development agenda and they do and can contribute immensely in planning and implementation of suitable action plans. Partner farmers of the Farmers FIRST programme were the delegates for the workshop. Two Hundred delegates including about 143 farmers from 25 states participated in this workshop.



Address by Hon. MoS, Shri Gajendra Singh Shekhawat during workshop on 'Let's Listen to Farmers'

2.4.2 Extension Information Systems

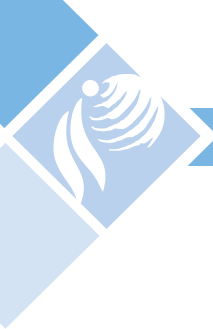
Agricultural extension is the primary delivery system for information to farmers. Improving agricultural production may not be achieved without relevant and reliable agricultural information. Extension organizations have the important role of linking farmers to information on improved agricultural technologies. One of the prime ingredients for rural development in developing countries is information access. Although the Information and Communication Technology (ICT) revolution in these countries has gained momentum, most of the farming communities still have no access to value added information. Academy focus on the agricultural researcher on extension information system to enhance their knowledge by increased stakeholder's participation in research.

Documentation of the Success Stories of Agri-graduates turned Entrepreneurs

The research inculcated with the objectives of documenting the success stories of Agricultural graduates (30) who have turned into entrepreneurs, identifying the processes and Critical Success Factors (CSF) that are responsible for entrepreneurship development among agricultural graduates. Five videos have been made of the Success Stories of promising entrepreneurs having their roots in Andhra Pradesh, Rajasthan and Karnataka from the



SOL/2017/18/138



Participants of training on 'Communication and Management Skills for Extension Professionals'

disciplines of Agriculture, Horticulture & Home Science. Further showcasing and upscaling were streamlined through a book encompassing all the success stories. This research also suggested a roadmap for developing entrepreneurial skills among agricultural graduates in State Agricultural Universities of the country.

Communication & Management Skills for Extension Professionals



NBP/2018/32/50

The participants had cross learning with better diffusion of ideas, and also have enhanced adoption of new knowledge, skills and changed attitudes for better and continuous learning. The training methods include andragogy and pedagogy methods with sufficient training aids, and the mode of instruction was both hands on, and on an interactive experiential mode, to enlarge the cone of experience from abstracts to concreteness. The novelty of social media was that a Whatsapp group was created to keep the trainers and the trainees and supporting resource persons connected 24x7x10 days.

2.4.3 Gender Mainstreaming in Extension

Women are the main farmers or producers, but their roles remain unrecognized. Despite the necessity of utilization of untapped potential of men and women, failure to address gender based differences bring about disparities in development outcomes in general an agricultural extension particular. Thus, the Academy concentrated towards gender mainstreaming.

Gender based ICT-led information management in agriculture



SOL/2016/09/119

This project aimed to investigate the existing use of ICTs in sharing various kinds of data, information and knowledge by farmers and farm women in agriculture and its impact. Knowledge test was conducted for the major crops such as Cotton and Paddy to identify the knowledge gaps wherein the maximum gap is in the plant protection. The preference of media for agriculture information and accessibility were found. Soil Health cards (141) were linked to the cadastral maps and Web based Geo-Spatial Village Information System of S. Lingotam and Tallasingaram was developed in the project. The prototype "Gender disaggregated database on ICT use in agriculture" was developed including 403 number entries of farmers and farm women of various agro climatic zones of the country.



2.5 Information and Communication Management for Promoting Innovation and Governance

Enhancing the research, capacity building and consultancy in the areas of Information Technology based decision support systems; Geographical Information Systems in agricultural management; ICT applications for village knowledge centres, Development of digital multimedia and e-learning resources for agriculture; Geospatial knowledge discovery and management; ICTs in agri-supply chains; ICT policy for NARS; Emerging information technologies and systems; ICTs in participatory technology development and transfer; Designing e-learning systems and Distance training. To realize the above objectives, the following activities related to research and capacity building programmes were carried out during 2017-18.

In order to advance the objectives of this broad theme, the Academy focused on the following broad areas for its research, academic and capacity building activities.

2.5.1 Application of GIS, GPS and remote sensing technologies

2.5.2 Development of Information systems and Knowledge Management

2.5.3 Statistical computing and data analytics

2.5.4 Integration of emerging ICT Technologies

2.5.1 Application of GIS, GPS and remote sensing technologies

Geo-Spatial Knowledge System for Monitoring Climate Change in Agro Climatic Zone-X: Southern Plateau and Hills Region

The main objective of the study was to find out the relevance to Climate variability significantly impact the agricultural growth, stress, cropping pattern, phenophase and its vulnerability. Satellite derived data sets have been used to study the time series trend of NDVI (Normalized Difference Vegetation Index) and LST (Land Surface Temperature) for Agro Climatic Zone-X (States: TS, AP, Karnataka & TN). The study uses Moderate Resolution Imaging Spectroradiometer (MODIS) images of MOD11A2-Terra image

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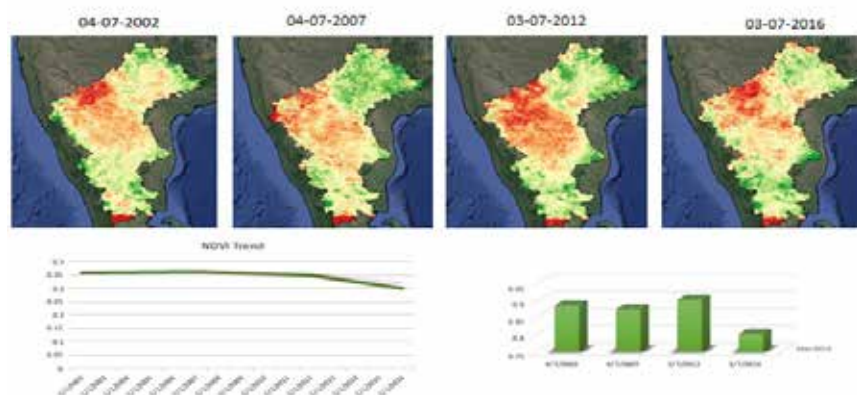
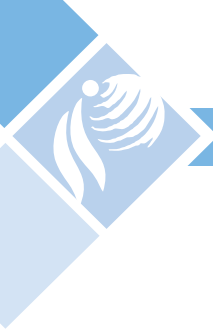


Figure 2.16a: Result of time series NDVI analysis



with 1000m resolution and 8 days' interval (2000-2017), MYD13A1-Aqua image with 500m resolution and 16 days' time interval (2002-2017) and India Meteorological Department (IMD) data sets. The trend analysis of climate was carried out to understand their impact on the study area. Design and development of comprehensive geospatial database for planning and implementation of climate smart agriculture projects in agro climatic Zone-X have been developed under this project.

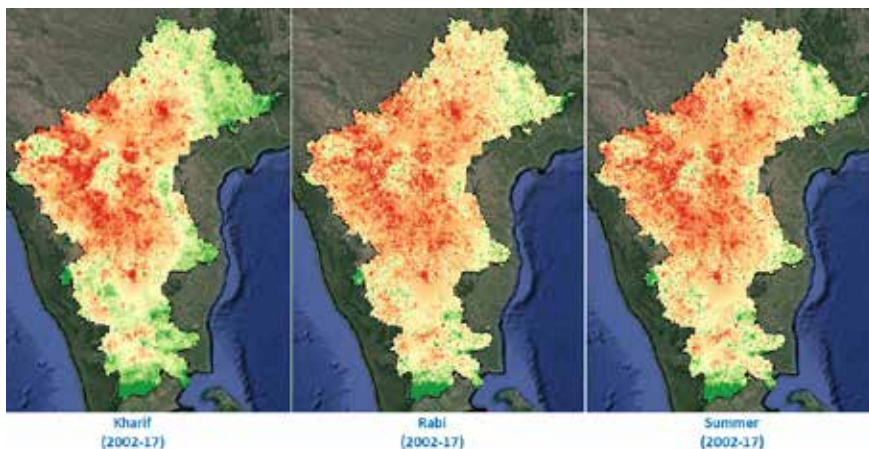


Figure 2.16b: Result of time series NDVI analysis

Geospatial Analysis for Natural Resources Management using Free and Open Source Software's (FOSS) – QGIS, ILWIS & R



NBP/2017/18/33

A training program on “Geospatial Analysis for Natural Resources Management using Free and Open Source Software's (FOSS) – QGIS, ILWIS & R” was conducted during November 16-25, 2017 with a total of 13 participants including 3 female participants from State Agricultural Universities (SAU's) and ICAR institutes. The main objective of this training programme is to enhancing competency levels of NARES professionals in understanding GIS, GPS and RS along with R software applying these principles in Agricultural Sciences.



Participants of training program on “Geospatial Analysis for Natural Resources Management using Free and Open Source Software”



2.5.2 Development of Information Systems and Knowledge Management

Development of portal for Indigenous Technical knowledge (ITK)

In Indian Agriculture, the Indigenous Technical Knowledge (ITK) has immense potential for innovation, especially at the grassroots level. Many of these knowledge and technologies are on par with the modern knowledge and have been provided by the indigenous communities with comfort and self-sufficiency. Indigenous crop cultivation practices are generally organic in nature, do not cause damage to air, water and soil and are safe to human beings. These practices differ considerably from region to region depending on soil type, rainfall, topography etc., and show often modifications by local farmers. In recent years, there has been a growing scientific interest in these indigenous practices as a source of sound ideas that could lead to sustainable use and management of land resources. All the published information is in scattered form and accessing this information is difficult due to its present form in hard copy. In view of the above it has been observed that there is an immediate need to documentation and preservation of the ITK of different agro climatic zones in a database format. This format should contain the description and basic set of information related to ITK resources, such as, information of the communities, beliefs on religious faiths, tools of using in agriculture, materials in house construction work, experimentation in farming and healthcare, natural resources in flora and fauna, human resources and expertise in skilled artisans, education and learning and communication of information. A portal was developed by ICAR-NAARM, using PHP and MYSQL, which provides information resources on ITK resources of pan India for crops viz., cereals, millets, pulses, fruits & vegetables, spices, medicinal & aromatic, plants etc. The portal developed by the academy is user friendly and it provides all the information about ITK resources of India. The Portal named as ITKPA has been developed and it will be updated regularly. This will be a resource base for the multi-stakeholders viz. researcher, academicians, policymakers, farmers, NGOs etc. This portal will be useful for the validation of ITK's with scientific rationale. The Developed ITK Portal is shown in the figure 2.17 and figure 2.18.



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ICAR Research Data Repository for Knowledge Management (KRISHI)

The data management policy of ICAR calls for creation of centralized repository for storing, analyzing and managing experimental, observational and survey data; publications, technologies generated. The centralized repositories are also expected to be base for managing wealth of knowledge available within ICAR system. This project, which is operation from November, 2015, is being executed by ICAR-Indian Agricultural Statistics Research Institute (ICAR-IASRI), ICAR-NAARM, ICAR-Indian Agricultural Research Institute (ICAR-IARI) and ICAR-National Bureau of Soil Survey and Land Use Planning (ICAR-NBSSLUP) along with other ICAR Institutes. During the year, the portal, KRISHI (<https://krishi.icar.gov.in/>) has been

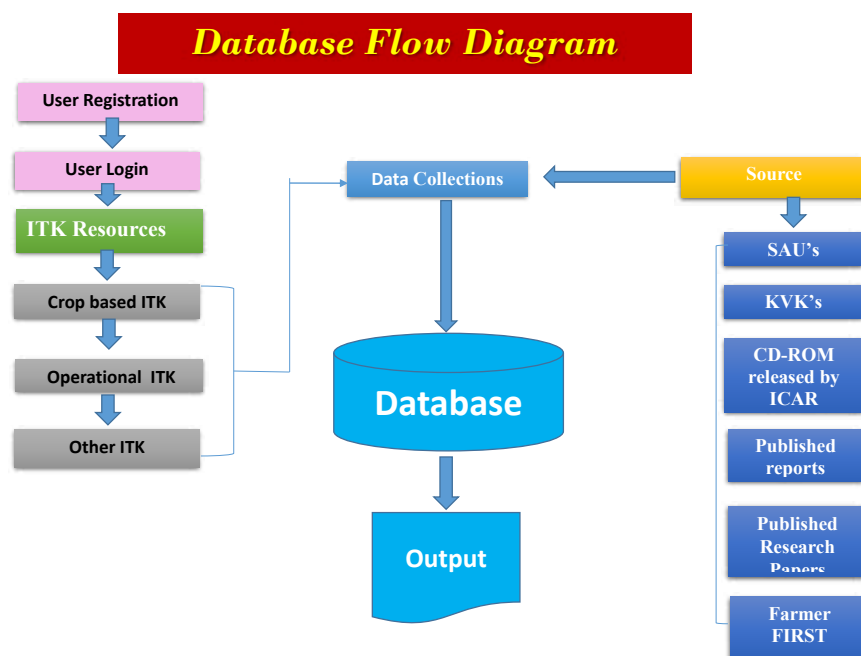


Figure 2.17: Database flow diagram



Figure 2.18: Home page of ITK portal

enriched with Technology Repository. The technology repository of ICAR is created with the purpose of bringing all proven technologies of various ICAR institutes at one place with the aims to popularize it; making it available to all stakeholders; entice prospective buyers and documenting the research outcomes. A common framework for different types of technologies was developed in a collaborative mode with different subject matter specialists. The framework was then implemented. The framework can be used for diverse technologies such as varieties to vaccines to farm implements to software.



The submission to the repository is allowed for all scientific staff of ICAR. However, all submissions undergo a review process within the concerned institute before making it public. The technologies where Intellectual Property issues are involved, provisions are made to provide basic information so as to attract potential customers for the technology. Information system to automate planning, trial creation, data upload and analysis of Coordinated Research Projects, AICRP Automation system was implemented for All India Coordinated Maize Improvement Programme (<http://aicrip.naarm.org.in>). The Rabi trials of 2017-18 season was created using the software and the system is expected to function fully from 2018 season.

A common data usage policy for KRISHI portal was also finalized during the period and now available in KRISHI Portal.

Data Visualization using Tableau

Using the rice production data from Andhra Pradesh and Telangana, a case study was developed (Figure 2.19) and introduced in 107th Foundation Course (FOCARS).

Following are the major advantages;

- Linking data from server database and live data updating.
- Automatic suggestion for data visualization and forecasting.
- Generating story boards.
- The geographical coordinates are generated automatically for the given locations which can be chosen selectively also.
- Clarity and value addition in live presentations.

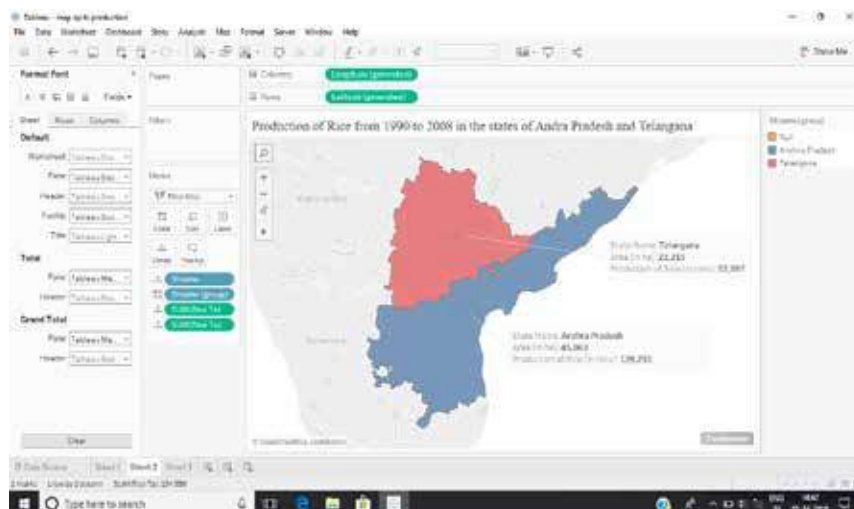
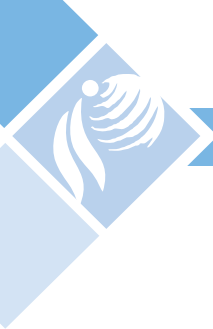


Figure 2.19: Data visualization using Tableau



Decision Support System of Maize Inbred Germplasm

Maize improvement in India has charted a unique success story with respect to expansion in maize area (3.3 to 9 m ha) and increase in production (from 1.7 to 24.4 million ton) and productivity (from 547 to 2583 kg/ha). The larger part of the above success has been attributed to recent surge in the adoption of hybrid technology. The basic elements of the hybrid technology are the maize inbred lines, which are developed by cyclic selection and advancement through selfing in the broad base germplasm like landraces and exotic lines over six (S6) to seven (S7) generations. Presently around 2500 inbred lines with diverse genetic background are being maintained at Winter Nursery Centre, Indian Institute of Maize Research. However, the success of the hybrids developed is determined by getting the choice of diverse inbred lines. In order to facilitate plant breeders, engaged in the development of hybrids, in selection of inbred lines at morphological level, a database has been developed. The database consists of information on 14 traits viz. time of anthesis, time of silk emergence, density of spikelets, number of kernel rows, number of kernels per row, grain type, plant height, ear placement height, anthocyanin colouration at base of glume, anthocyanin colouration of anthers, anthocyanin colouration of silks, grain colour, kernel row arrangement and 1000-kernel weight against each inbred lines. Based on the technical aspects and various symptoms of Maize, a decision support system (DSS) on Maize germplasm has been developed by IIMR in collaboration with NAARM. The main module of DSS is search database, that gives the different options such as Tassel, Ear, Kernel, Plant of maize inbred lines based on multiple traits, where one can select the required options, which searches the entire database and gives the matching records in the next window. From this list, user can select a required record by clicking the plot no. / pedigree and access the detailed information such as pedigree, seed image, year, season, tassel, time of an thesis, time of silk emergence, length of inbred lines, number of rows of grains, type of grains, row arrangement etc., for decision making.

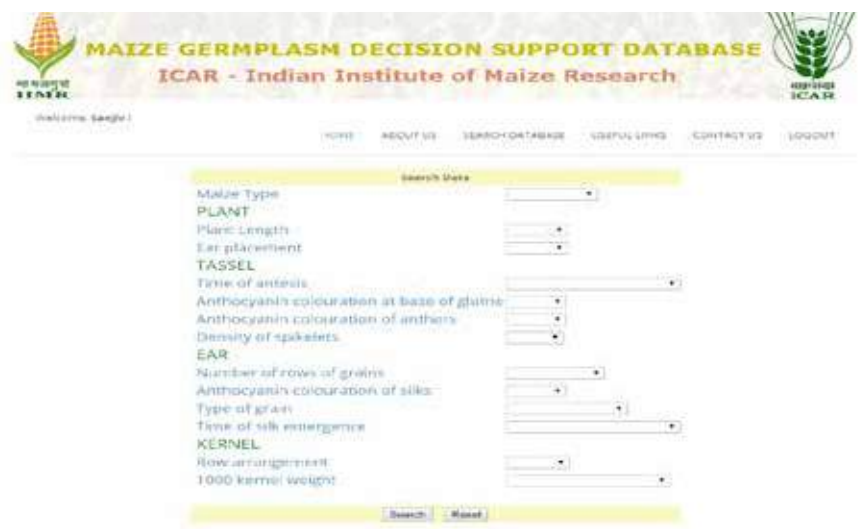


Figure 2.20: Maize germplasm decision support database portal



Web-based Applications for Project Management

Three web-based applications were developed so as to enhance research management efficiency viz., AHP Analyser, Research Concept Writer and Project Logframe Writer. These applications have been integrated in various capacity building programs of the Academy so as to facilitate collaboration, interaction and create online environment of group learning. The applications were developed by using PHP, MySQL, Bootstrap, CSS.

AHP Analyser: Analytic Hierarchy Process (AHP) Analyser is an online tool that facilitates the group decision making by pairwise comparison based on expert judgment values. Its an expert system designed to aid in project prioritization. It's being used by more than 1500 scientists/ academicians in 50 countries including USA, UK, Australia and Netherlands, etc. The application is available at <https://naarm.org.in/ahp/>. (Figure 2.21)

Research Concept Writer: It is a web-based application for preparing research project concepts. The software application provides for a step-by-step approach in preparing the concept note, which can be generated as a report in Microsoft word format, while the same will be stored in server. It's being used by more than 500 scientists/ academicians in six countries. The application is available at <https://naarm.org.in/dwrp/>. (Figure 2.22)

Project Logframe Writer

The Logical Framework Approach (LFA) is a methodology mainly used for designing, monitoring, and evaluating research projects. This Web-based Software application aids in developing a log frame for the research projects, on completion of which, an editable report in MS-Word format could be generated for tailor-made utilities. It's being used by about 400 scientists/ academicians in five countries. The application is available on following link: <https://naarm.org.in/logframe/> (Figure 2.23)

भाकअनुप - राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी
ICAR-National Academy of Agricultural Research Management
(ISO 9001:2008 Certified)

Sign in Downloads Gallery

AHP Analyser - Project Prioritization
(A Group Decision Making Tool for Ranking of Options)

User Name [Forgot](#)

Email Address [Forgot](#)

Sign in Sign up Reset

Analytic Hierarchy Process (AHP) analyser is online tool that facilitates the group decision making by pairwise comparison based on expert judgment values. The Decision Support System (DSS) by web based AHP methodology is converted to Expert System. In developing this online application, the basic of AHP as developed by Dr. T.L. Saaty and other professors has been taken into consideration. The Detailed Documents are available in Downloads.

The AHP decisions are based on Consistency Ratio (CR), the threshold limits of CR values have been modified for agricultural research projects as per the research conducted by Dr. S.K. Soam at NAARM, Hyderabad. The AHP analyser can be used for Project Prioritization, Priority Setting, Research Project Monitoring and Evaluation (PME) and also for Technology Valuation and Technology Management.

Note : In AHP Analyser, there is no provision of saving data on server, it works like a calculator. The data and analysed information from the application removed immediately after signing out from the application.

Online application of AHP methodology.Developed by Dr. S.K. Soam, Ch. Sridharababu, T.RajKumar and B.Raghupathi

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Figure 2.21: Screenshot of AHP Analyser



Figure 2.22: Research Concept Writer

Figure 2.23: Project Logframe Writer



ICAR Sponsored Short Course on “Emerging Tools and Techniques for Agricultural Knowledge communications

A training programme with a total of 13 participants including 2 female participants from State Agricultural Universities (SAU's) and ICAR institutes was conducted to upgrade the knowledge and skills of agricultural professionals for sharing of knowledge management process in research, education. The training programme has been designed by the Academy in the present era of Digital technology led communication, the dissemination of agriculture knowledge faces certain challenges due to involvement of variety of stakeholders such as scientists, farmers, extension functionaries, inputs suppliers etc. The topics covered such as Open access tools for agriculture knowledge communications; Blogs, database creation and databases and internet resources, ICT Role in Agriculture Knowledge Management, Geospatial Knowledge Management, CMS Content Management and E-learning, Expert System language, Biostatistics and Algorithms, VDSA, ODK (Open Data Kit) and data verse, Patent search databases, Bioinformatics and big data and cloud computing in agriculture, etc. Library information system, OERs and on line collaboration tools. Open source tools, social media and their applications. A comprehensive training programme has been designed. It encompasses a range of ICT mediated applications, data management tools, knowledge management initiative in ICAR and web based applications in a networked environment for the data sharing through open source knowledge management tools which will be very much useful to the ICAR/SAU Scientist.



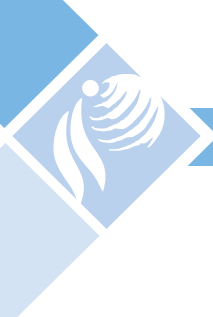
NBP/2017/16/30
NBP/2018/28/46

Koha for ICAR Library Staff'

A training program with a total of 36 participants including 9 female participants from State Agricultural Universities (SAU's) and ICAR institutes was organized. The main objectives of this to make participants understand the application of KOHA in library automation and to provide hands on training on various modules of KOHA related to acquisition, cataloguing, OPAC etc., And also provide the knowledge of broad aspects of KOHA installation and customization as per institute requirements.



Participants of training on “Koha for ICAR Library Staff”



2.5.3 Statistical Computing and Data Analytics

Multivariate Data Analysis



NBP/2017/20/36
NBP/2017/05/11

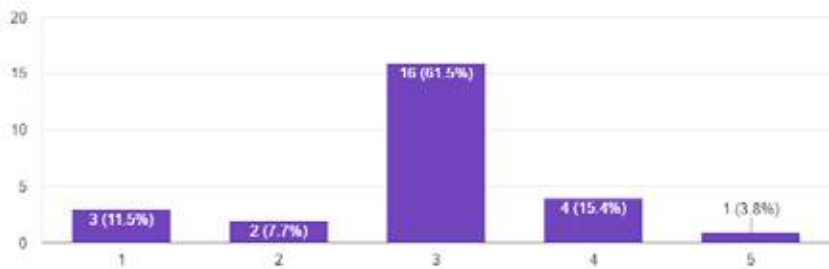
A training program with a total of 24 participants including 3 female participants from State Agricultural Universities (SAU's) and ICAR institutes was organized to expose the participants to different multivariate techniques. In most of the Agriculture experiments observations on multiple variables are being collected. Study of interrelationship among the variables is important as it provide much more valuable information than studying individual variables separately. In this training programme, various multivariate techniques were introduced through R programming language. R programming offers powerful tools for data visualization particularly for the multivariate data. More than one and half a day was spent on R programming language. The multivariate techniques covered include Multiple Linear Regression, Logistic Regression, Principal Component Analysis. Cluster Analysis, Factor Analysis, Multi-Dimensional scaling, Canonical Correlation Analysis besides visualization techniques. Before the starting of the training, the participants were asked about their expectations. The pre-test report is attached as Annexure 'B'. Many participants expressed their desire to learn R and also their understanding of the techniques is mostly moderately clear. At the end of the training programme, a post-test analysis was carried out. During the posttest evaluation, the participants have indicated that their level of understanding has increased in each of the techniques. This shows the training has achieved at least in short term in meeting the expectations and improving the skills of the participants in the area of multivariate data analysis. As suggested during the valedictory programme, another feedback after 6 months would be taken up to understand the effectiveness of the training on (i) application of tools learned in their work environment; (ii) other persons trained or helped by the participant. Two sets of data were used to explain each tool. One data was analyzed while explaining and the other dataset was used by the participants to apply the technique and interpret the results themselves.

(Selected Questions only)

Pre-test

5. Your level of understanding of the technique used(1-not at all clear; 2-somewhat clear; 3 moderately clear; 4 – Very clear; 5 – I can teach others!)

26 responses





Post-test

8. After training, my understanding of Cluster Analysis has improved (1- Strongly Disagree; 2- Disagree; 3- Neither Agree nor Disagree; 4- Agree; 5- Strongly Agree)

25 responses

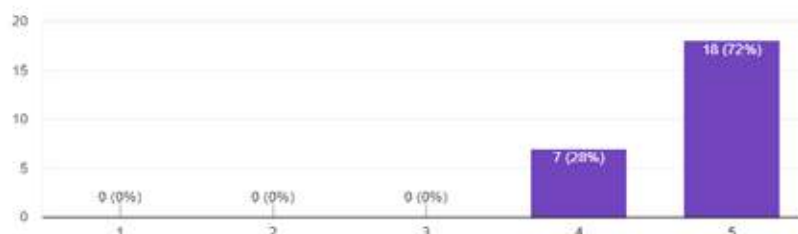


Figure 2.24: Pre-Test and Post-test evaluation of understanding level of trainees of “Multivariate Data Analysis”

Analysis of Experimental Data

A training program with a total of 16 participants including 6 female participants from State Agricultural Universities (SAU's) and ICAR institutes was organized. The main objective of the training programme was to introduce various statistical analysis of experimental data and to give hands-on experience in analyzing different types of experiments. The training programme focuses on how to conduct experiments in a scientific manner and how to interpret using modern statistical software. The participants were exposed with SAS® software which is available with all ICAR/SAU institutions. The course was started with a brief introduction to different experimental layout followed by introduction to SAS®. Analysis of different experimental setup were explained starting with (i) when it should be used (ii) assumptions involved (iii) layout (iv) data file preparation (v) how to do the analysis in SAS® and (vi) interpretation of results. In each case, two examples were used with one to introduce and another one for the trainees to do on their own. The trainees were also taught on getting the desired output by using some of the options which are not available through menu. Participants were also encouraged to sum up the results for presentation point of view.

Policy Workshop on “Big Data Analytics in Agriculture

A Workshop with a total of 48 participants including 8 female participants from State Agricultural Universities (SAU's) and ICAR institutes was organized. The programme was intended to brainstorm about innovative applications of Big Data Analytics in Agriculture. Big data analytics is futuristic and offers tremendous potential in agriculture. Huge quantum of agricultural data in the form of soil health records, geospatial maps, weather information, production and yield records, market data, experimental data, survey data, etc. are being generated continuously. The data thus generated are gradually increasing to unmanageable proportion challenging the capacity



WS/2017-18/12



Policy workshop on “Big Data Analytics in Agriculture”

and computing abilities of the present system. Big data framework can be effectively utilized to capture, store, process and analyze huge voluminous and streaming data. In big data analytics, the data are processed in cluster mode and advanced big data computing techniques can be employed for providing customized services to different stakeholders in agriculture. Case discussions, lectures and demonstrations of portals in various domains were organized during the 2 days programme. A panel discussion on various applications of big data in agriculture was also organized as part of the programme. Experts in the field of agriculture, veterinary, fisheries, social scientists, statisticians and computer science have participated in the panel discussion. Panelists of the view that though there is plenty of data generated, there is hardly any big data platform developed in agriculture. Developing such a platform would enable the policy makers making better decision-making which will benefit the farming community. Further, there is a greater need for capacity development programme to implement big data analytics. Senior scientists, Professors from SAUs, Senior managers and professionals from different parts of the country have participated. The participants also requested organization of more number of training programmes in this line with longer duration. The Academy has assured the participants of providing support for development of big data platform for agricultural solutions.

2.5.4 Integration of emerging ICT Technologies

Application of Bioinformatics in Agricultural Research and Education



NBP/2017/10/18



Participant receiving certificate from Director, NAARM

A training program with a total of 12 participants including 4 female participants from State Agricultural Universities (SAU's) and ICAR institutes was organized. The main objectives of the training programme was to create an awareness with respect to the basic tools and techniques used in bioinformatics in agriculture. The training programme was conducted to upgrade the knowledge and skills of agricultural professionals for sharing of knowledge management process in research, education. The training programme has been designed by the Academy. It encompasses a range Bioinformatics is an innovative developing field that helps in collecting, linking, and manipulating different types of biological information to discover new biological insights. This specific genetic knowledge could then be used to produce stronger, more drought resistant, disease resistant, insect resistant and more productive crops. Keeping in view the changed scientific scenario in the post-genomic era and to expose the agricultural scientists to the latest computational biology tools for biological research. The training includes theory and hands-on sessions on different aspects. This training programme is conceived with an idea to provide an insight to the participants about the enormous potential of Bioinformatics and its tools management to explore it in the area of agriculture.



2.6 Mobilizing Science and Technology for Innovation and Sustainable Development

This theme lies in the centre for the Academy's multifaceted activities, which essentially aims to:

- o Enhance capacities for research policy, strategy, priority setting, planning, management, monitoring and evaluation for a more pluralistic NARES, in the emerging contexts of food security, climate change and globalization
- o Enhance capacities for technology foresight and strategic management of intellectual property and commercialization of technologies

In order to advance the objectives of this broad theme, the Academy focused on the following broad areas for its research, academic and capacity building activities.

2.6.1 Research Management

2.6.2 Technology and Innovation Management

2.6.3 Research in the context of food security, climate change and globalization

2.6.4 Research, Technology & Innovation: Preparing for future

2.6.1 Research Management

Framework for Measuring Research Efficiency

The Academy has undertaken systematic effort to develop a framework for enhancing efficiency of Agricultural Research Management in NARES. It is envisioned to aid in institutionalization of new guidelines for research prioritization, monitoring and evaluation (PME) in all ICAR institutions and to enhance efficiency and infuse transparency in research management in ICAR. A draft framework is being developed for profiling the research productivity and identifying the drivers of research efficiency, based on review of indicators, to aid in subject-neutral assessment of research output against resources used at various levels viz., scientists/ divisions/ institutions. A prototype for developing a Research Management Application has been prepared and steps have been initiated to empanel professional engineering firms for the Academy in order to aid in co-development of research solutions, scalable for the entire country. In order to strengthen the analytical framework for measuring research efficiency, two web applications viz., SciVal of Elsevier and Web of Science of Clarivate Analytics have been subscribed.

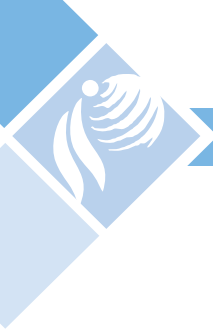
Series of consultations were held with Department of Science and Technology (DST), New Delhi and International Food Policy and Research Institute (IFPRI), Washington who undertake periodic surveys among national and international agencies, respectively on various agricultural research and technology indicators (ASTI). Consensus has been built for jointly building a National Dashboard for evidence-based S&T Governance, in India for real-time data collection, analysis and visualization.

SIX Thematic Areas

1. Agribusiness Management for Inclusive Growth
2. Education Systems Management for Enriched Learning Quality & Environment
3. Enhancing Capacities for Leadership and Governance
4. Extension Systems Management in a Market-driven Environment
5. Information and Communication Management for Promoting Innovation and Governance
6. Mobilizing Science and Technology for Innovation and Sustainable Development



SIL/2017/09/129



Agricultural research efficiency hugely depends on the research infrastructure viz., laboratories, research farms, resources, manpower etc. In order to understand the significance of physical infrastructure on the research outcome a pilot study has been undertaken in Andhra Pradesh and Telangana to assess the status and gaps of infrastructure and resources available for agriculture research and also to assess its impact on agricultural research and farmers. Interim results indicate that nine per cent of ICAR institutes are located in these two states. The agricultural research stations of state agriculture universities (52 Nos) were driving the cutting edge level agricultural research. The data showed that 36 % of scientific and technical manpower positions are vacant in ICAR-CTCRI, mandated for research on tobacco. In Telangana where eight ICAR institutes are located the gap in scientific manpower ranged between 10 (CRIDA) and 47 percent (NAARM) (Table 2.10). It was also observed that the budget for R & D has been more or less stable and in some institutions, it is coming down in the recent past.

Table 2.10. Scientific Manpower Status in Andhra Pradesh and Telangana Agricultural R & D Institutions

Andhra Pradesh			
Institute	Sanctioned	In position	Gap (%)
IOPR	82	66	20
CTRI	190	122	36
ANGRAU	843	534	37
Telangana			
NAARM	60	32	47
IIOR	42	38	10
DPR	16	19	(19)
IIMR	48	35	27
CRIDA	67	60	10
NRCM	14	10	29
IIRR	71	62	13

Note: Figures in parentheses indicate surplus

Foundation Course on Agricultural Research Management (FOCARS)



FOC/2017/01/07
FOC/2018/02/38
NBP/2017/13/24

FOCARS is the flagship program of the Academy and is offering the same twice a year. The Foundation course aims at building the competencies of the scientist probationers of Agricultural Research Service so as to nurture and develop the skills required to meet the professional challenges. The competency frame work of FOCARS comprises the development of core discipline research skills, personal effectiveness, information management and business orientation.

The 106th Foundation Course for Agricultural Research Service (FOCARS) started on 5th July 2017 and completed on 4th Oct 2017. Total **60** young scientists- **17** women and **43** men, joined the FOCARS representing 22 states and 28 disciplines.



The first phase of three months of training is organized in three stages 1) Orientation and Capacity Building 2) Field Experience Training and 3) Multidisciplinary perspectives, out of which first and third stages were organized at the Academy and the second stage at identified centers pan India for 21 days. During the on-campus capacity building programs, major emphasis was laid on modular approaches interwoven around themes viz. 'Agricultural Policy Perspectives', 'Intellectual Property Rights and Technology Management', 'Human Resource Management', 'Documentation, Presentation and Communication Management', 'Participatory Technology Development and Dissemination', 'Developing Winning Research Proposals', 'Futuristic and multidisciplinary perspectives', including 'Financial and Administrative Management etc.

First phase of 30 working days had total 42 Sessions. Besides, there were practical sessions in the fields of- Video Production, Digital Presentation and Team Building. We also had special 3-day module on 'Developing Winning Research Proposals'. In this phase, they also had opportunity to interact with the top management of two CGIAR institutes. Dr. PK Joshi, Director, IFPRI South Asia and Dr. David Bergvinson, DG, ICRISAT gave comprehensive perspective of global trend in agricultural research.

Second phase of 21 working days were of Field Experience Training (FET), during which 10 groups of these scientists were deputed to 10 different KVKs across 9 states. They laid strong foundation during this time for their future growth.

Third phase of about 25 working days had 33 Sessions and 3 practical sessions each in the field of - Data Analytics, IP Informatics and Personality Development. This phase offered more of experiential learning through several presentations like FET report, Group seminar and RPP presentation. They also had 2-day module on 'Scientific Writing'. The young probationers were also exposed to working environment of different types



Inaugural session of 106th foundation course FOCARS



FOCARS participants undergoing Mock FET



106th foundation course participants with Dr. T. Mohapatra, DG, ICAR and NAARM faculty members



Certificate distribution of 106th foundation course

of organizations like large corporate sector (Advanta India), CGIAR institute (ICRISAT), Fisheries Board and other ICAR institutes such as CRIDA, PDP, IIMR and NRC Meat.

As a part of Social Responsibility, the FOCARS probationers participated overwhelmingly in **Blood Donation camp** organised with the support of Red Cross Society. They also conducted QUIZ for more than 300-farmers during *Kisan Mela* organized by the IIOR. Their active participation in flagship programmes of the government, like *Haritha Haram*, *Hindi Pakehwada*, *Swachh Bharat Mission* and *Swachhata Pakehwada* are also noted.

Several new initiatives were introduced during this FOCARS, which would help in improving the credibility of the programme:

- **360° evaluation of training programme:** Training-Learning process can really be progressive when we take feedback from the learner. Keeping this in view, we introduced Online Feedback on all the 100+ sessions of the training programme from all the 60 probationers.
- **Writing Blogs:** The FOCARS probationers wrote and published their own Blogs. All the 10 FET groups narrated their experiences of rural immersion in that Blog. The Blogs will become an important mode of communicating science for the new generation researchers. <http://focars106.blogspot.in/>
- **Success stories from the FET villages:** The probationers were asked to scout for one success story from the respective FET village. Some of those are really interesting and worth noting. These Success Stories were screened and one has been adjudged as the Best Success Story.
- **Baseline Database of FET villages:** ARS probationers inadvertently spend 2-3 weeks in one particular village for FET. There can't be any better opportunity to capture quality information of those villages and



107th foundation course participants with Dr. T. Mohapatra, DG, ICAR and NAARM faculty



convert it into a global public good. With this perspective, the geo-referenced database of Baseline information of all the 10 FET villages has been created and will be uploaded on NAARM website. It has lots of scope for scalability & value addition.

The 107th Foundation Course for Agricultural Research Service (FOCARS) started on 4th January 2018 and completed on 3rd April 2018. A total 24 scientists-probationers, comprising 15 men and 9 women, joined the FOCARS representing 11 states and 14 disciplines. Majority of them were post-graduated from IARI, followed by NDRI and UAS-B.

The complete course of 3 months-duration was organized in three phases for 42, 22 and 16 working days on orientation and capacity building, field experience training and multi-disciplinary perspectives, respectively.

The probationers of multidiscipline were deputed to 4 KVKs falling under all the four major regions of the country, viz. Udaipur, Gulbarga, Bhopal and Coockbehar to undergo field experience training in the second phase, where they also collected basic village information for uploading on the NAARM website.

Many eminent personalities in agricultural research and administration in the country, including our Hon'ble Secretary-DARE and Director General-ICAR Dr. T. Mohapatra, Dr. A.K. Sikka, Indian Representative-IWMI, New Delhi, Mr. Chhabilendra Roul, Special Secretary-DARE and Secretary-ICAR, Dr. B. Rajender, Joint Secretary, MoA, GoI and Dr. Mangala Rai and Dr. R.S. Paroda, Former Secretaries-DARE and DG-ICAR interacted with the scientis-probationers and motivated them during the course.

Management Development Programme on PME

A training programme on, "Priority setting, Monitoring and Evaluation (PME) of Agricultural Research Projects" was held at NAARM during 06-11 Oct 2017 in order to sensitize the participants on the concepts of PME in ICAR and equip them with various techniques of PME in different areas of agricultural research. A total of 15 participants from various ICAR institutions and State Agricultural Universities participated in the program. The participants were introduced to "Participatory Action Plan Approach (PAPA)" for identifying actionable points after return their work place.

Need-based Programs on Research Management

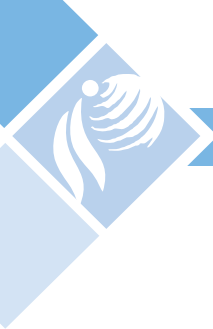
Eight training programs were organized with special focus on various aspects of research projects management viz., developing winning research proposals (6 nos, including 3 off-campus programs), and one each on impact assessment and research communication.



107 FOCARS trainee receiving certificate from Dr. R.S. Paroda



Participant receiving certificate from Director, NAARM



SIL/2017/08/128

2.6.2 Technology & Innovation Management

The study was undertaken at the Academy for understanding the Intellectual Property landscaping of emerging agri technologies for sustainable value chain development. Presently, sensors play an important role in data collection, automation and in remote sensing applications for agriculture. The recent developments in the different types of sensor enable quick, robust and accurate methods for sensing various soil parameters. The innovations in the field of sensors for agriculture has been on a rise. This present patent landscape study was aimed at capturing patent and non-patent references disclosing the different types of sensors used in agriculture mainly to measure three parameters: Moisture, Nutrients, Biomass. (Figure 2.25).

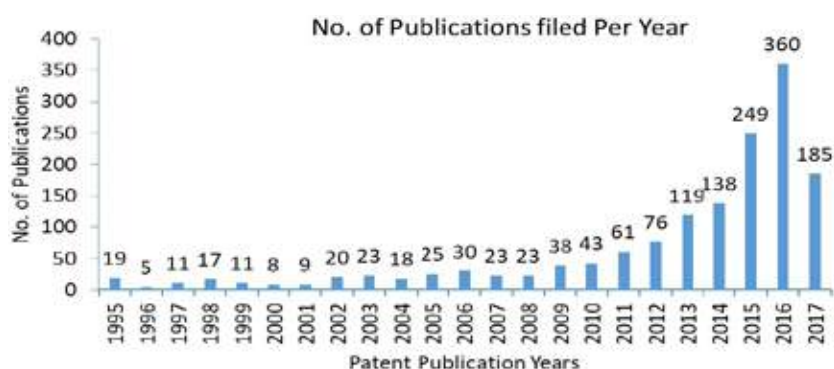


Figure 2.25: Patent publication trend from 1995-2017 for Moisture/ Nutrients/ Biomass Sensors

Out of all the publications from 1995 till 2017, about 95% of the publications mention the use of sensors for moisture detection of soil and 12% for nutrients detection. (The patent text search was performed using the Thompson Innovation database on 25th October 2017). Few of these sensors are common for detecting moisture and nutrients. Most of the patent documents were related to sensors detecting Nitrogen (35%) followed by Phosphorus (18%), Potassium (15%), and Carbon (6%) as depicted in figure 2.26. Many of these sensors are used to detect multiple nutrients. It is observed that popular & widely used sensors are the electric sensors followed by the optical sensors and mechanical sensors (Figure 2.27). The mechanical sensors include the Tensiometer that measures the soil water tension.

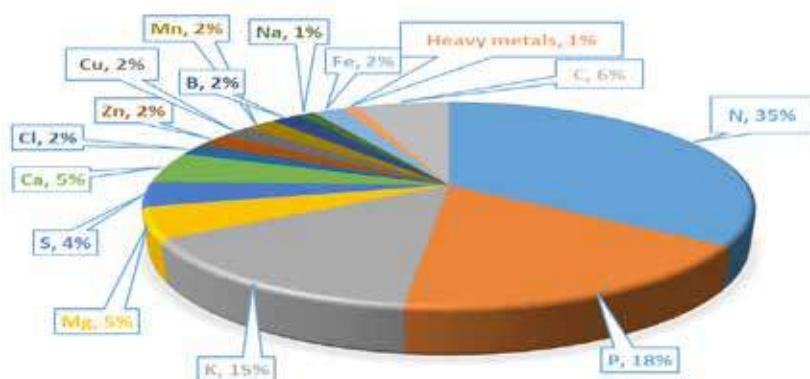


Figure 2.26: Different types of nutrients that were monitored by sensors

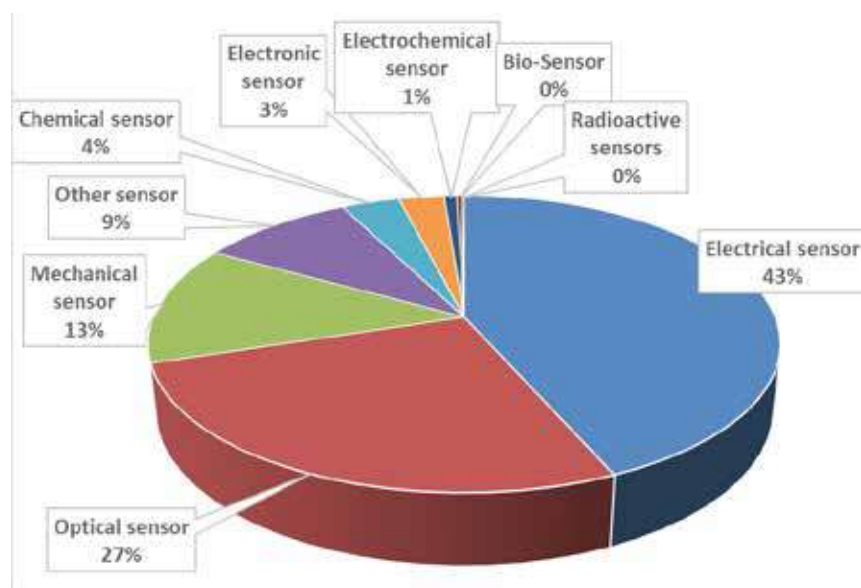


Figure 2.27: Broadest categorization of types of sensors for different types

Post Graduate Diploma in Technology Management in Agriculture

The One Year Post Graduate Diploma in Technology Management in Agriculture (PGD-TMA) is a two- semester course jointly offered by NAARM and University of Hyderabad (UoH), on a distance learning mode. The aim of the course is to develop the skills of students, researchers, policy makers, IP and technology transfer practitioners as professionals to enable them to handle successfully Intellectual Property and Technology Management assignments in agriculture and related enterprises. The no. of students admitted in 2017 were 68 in which 19 are girls. Many students are in-service candidates representing various Governmental and Non-Governmental Organizations such as ICRISAT, Agricultural Universities, ICAR, Food Corporation of India (FCI), Steel Authority of India, GVK Group, Syngenta, Bharat Petroleum, NABARD and State departments.

The impact of the effectiveness of various capacity building programmes on IP & TM and the PGDTMA programme initiated by NAARM was analyzed through a study. The response of the participants suggesting the utility of the course in handling the IP & TM portfolio, career progression are illustrated below (Figures 2.28 & 2.29). It is evident that the training had led to development of large human resources in the NARES, which is in alignment with the ICAR IP policy 2006 as well as National IPR policy 2016. The activity is important for developing higher-level professionals in the evolving domain of IP & TM in alignment with the New IPR Policy 2016. The full time programme of the Academy has developed 16 hardcore IP professionals.



ACA/2017/02/56

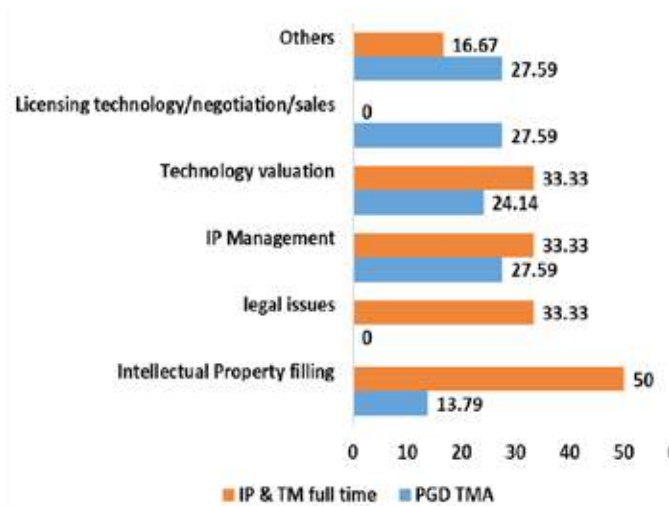
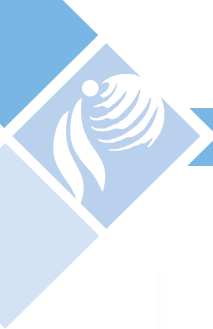


Figure 2.28: Utilization of knowledge gained through the course in profession by PGD TMA and full time respondents

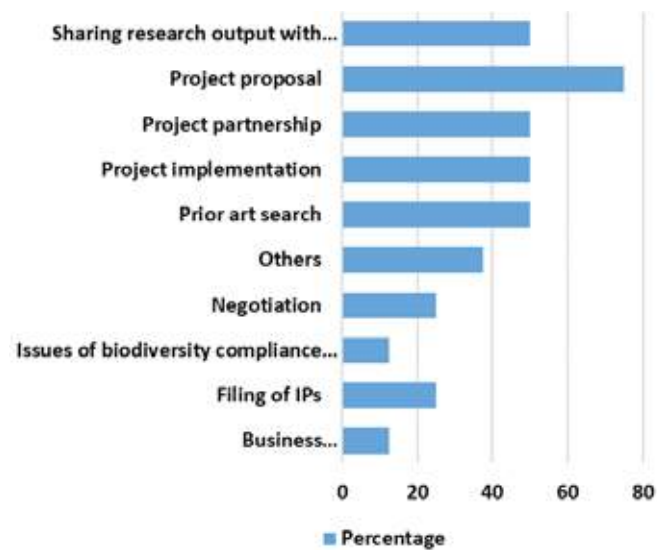


Figure 2.29: Utilization of knowledge gained through the course in profession by IP & TM training respondents

2.6.3 Research in the context of food security, climate change and globalization

Market signals for Indian Trade during WTO regime

During WTO regime markets are liberalized. The market economy depends on price signals to correctly allocate its scarce resources. Sufficient and reliable information is prerequisite for proper decision making be it domestic market or international marketing. Market analysis content helps us discover profitable opportunities. Market export signals at a general level, all commodity prices are affected by the market fundamentals of demand and supply. In the past decade, India has emerged as a major agricultural exporter \$5 billion in 2003 to a record of more than \$39 billion in 2013. India trails only the United States in cotton exports. India became the world's seventh-largest exporter of agricultural products in 2013, surpassing Australia. India emerged as a top bovine meat exporter. India has become a very important player on the global market, especially for rice, cotton, sugar, and beef (buffalo). India has also become a sizeable exporter of soybean meal, guar gum, corn, and wheat, as well as a diverse range of other products.

This study was undertaken to analyse market signals of exports and, prices of major India's agricultural commodities and identification of their destinations; track information on demand, policy changes, tariffs, safety and standards of exports for Indian major crops and estimates the effects of exports coefficients signals on production, trade and poverty of India's agriculture commodities and suggest R&D strategies and policies for boosting Indian agriculture exports and foreign earnings.

Decade wise growth rates and elasticity's were estimated for cereals, pulses, oilseeds and fiber crops (1990-2016) international policies effecting on market



SOL/2017/04/124
COP/2017/06/126
SCP/2017/02/122
CCP/2016/10/120



& price signals of India's Agricultural Trade were studied. The effects of global political economy of agriculture in general and continents & country-wise in particular from 37 to 170 major countries of 6 continents policies and insights were analyzed.

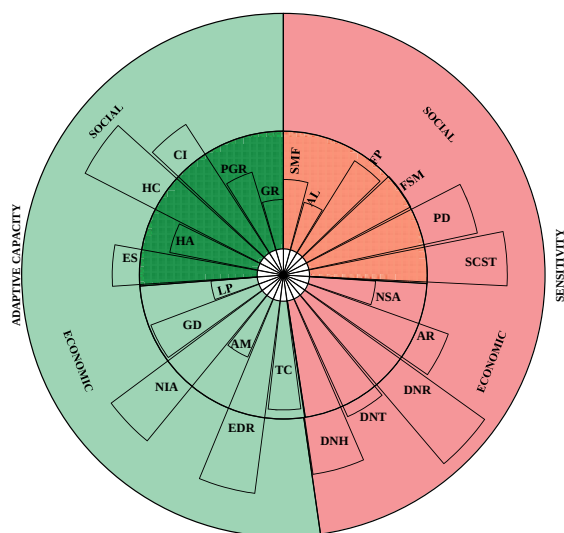
The key findings include the following:

- Import from inelastic countries should be exempted from any ban (without any quantity restriction) which would help in good relationships. Multilateral trade relationship with high CAGR countries would help in smooth trade of agricultural crops.
- During the period 2000 to 2016, Russia, India, China and New Zealand are given more importance to agricultural sector than compared to other WTO member countries. Technical Barriers to Trade, Sanitary and Phytosanitary and Anti-dumping were the most prominent non-tariff measures in world.
- World trading prices of cereals, pulses, oilseeds wheat & cotton export prices are more than import prices indicating India has comparative advantage in cereals, pulses, oilseeds wheat & cotton export these crops. India witnessed the comparative advantage in exports of cereals, pulses and cotton to Bangladesh, Saudi Arab and Vietnam respectively.
- India's major exports destinations for cereals, pulses, cotton, jute, groundnut, soybean & Niger seeds, sesamum, and sunflower are Bangladesh, Sri Lanka, China, Nepal, Indonesia, USA, South Korea and Philippines respectively. The major India's imports destinations are Australia, Canada, USA, Bangladesh, Germany, USA, Nigeria and Ukraine for cereals, pulses, cotton, jute, groundnut, soybean, sesamum, and sunflower respectively.

The study findings have important implications in designing WTO Agreements or policies or programs to boost agricultural development and promotion of foreign trade.

Framework for Coastal Vulnerability Assessment

The study was undertaken in the Academy in order to demonstrate science-driven policy making. A holistic and robust framework to assess the vulnerability of coastal India at different levels was developed under the study, wherein Cumulative Vulnerability Index (CVI) can be estimated as a function of exposure, sensitivity and adaptive capacity, at the village-level, using nationally comparable and credible datasets (Figure 2.30). The exposure index (EI) was determined at village level by decomposing the spatial multi-hazard maps while sensitivity and adaptive capacity indices were estimated using 23 indicators, covering social and economic aspects. The Socio-Economic Vulnerability Index (SEVI) was constructed as a function of sensitivity and adaptive capacity. The framework was piloted in Sindhudurg, a coastal district in Maharashtra, India. It comprises of 317 villages, spread across three taluks *viz.*, Devgad, Malvan and Vengurla. The villages in Sindhudurg were ranked based on this multi-criteria approach. Based on CVI values, 92 villages (30%) in Sindhudurg were identified as highly vulnerable.



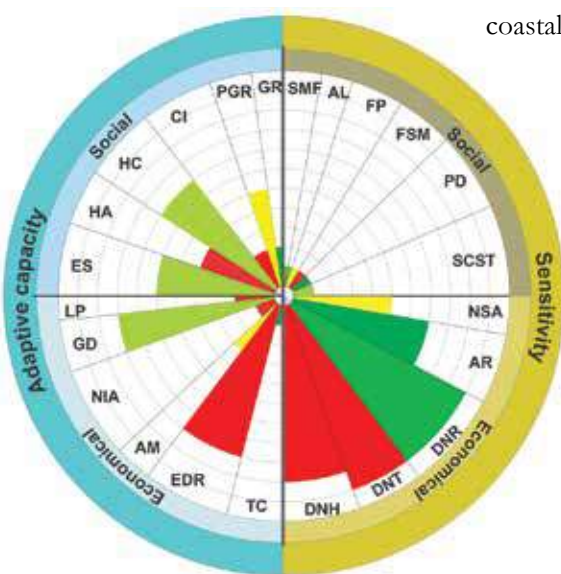
SMF-Small and Marginal Farmers, AL-Agricultural Labourers, FP-Fisher Population, FSM-Food Sufficiency/ Malnutrition, PD-Population Density, SCST-Schedule Castes/Schedule Tribes Population, NSA- Net Sown Area, AR-Annual Rainfall, DNR-Dependence on Natural Resources, DNT-Distance to Nearest Town, DNH-Distance to Nearest Hospital, TC-Transport and Communication, EDR-Economic Dependency Ratio, AM-Access to Market, NIA-Net Irrigated Area, GD-Groundwater Development, LP-Livestock Population, ES-Education Status, HA-Household Amenities, HC-Housing Condition, CI-Community Infrastructure, PGR-Population Growth Rate, GR-Gender Ratio

Figure 2.30: Framework for assessment of Socio-economic vulnerability depicting the prioritized indicators based on their weightage for SEVI (%)

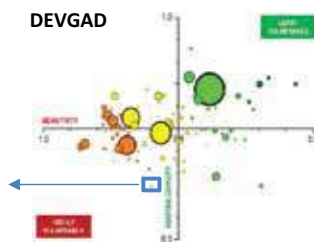
A decision tool for identifying villages vulnerable to changing climate has been developed, based on their level of sensitivity and adaptive capacity in a two-dimensional matrix, thus aiding in planning location-specific interventions. Here, vulnerability indicators are classified and designated as 'drivers' (indicators with significantly high values and intervention priority) and 'buffers' (indicators with low to moderate values) at the village level.

The current framework would provide the policy makers to prioritize target areas for intervention, plan appropriate interventions based on need and thus aid in strengthening the implementation of National Coastal Mission launched by the Government of India as a sub-mission of National Action Plan for Climate Change (NAPCC), which has made scientific assessment of coastal vulnerability, a prerequisite for adaptation and mitigation planning.

VILLAGE: WADEJAR



DEVGAD



SMF-Small and Marginal Farmers, AL-Agricultural Labourers, FP-Fisher Population, FSM-Food Sufficiency/ Malnutrition, PD-Population Density, SCST-Schedule Castes/Schedule Tribes Population, NSA- Net Sown Area, AR-Annual Rainfall, DNR-Dependence on Natural Resources, DNT-Distance to Nearest Town, DNH-Distance to Nearest Hospital, TC-Transport and Communication, EDR-Economic Dependency Ratio, AM-Access to Market, NIA-Net Irrigated Area, GD-Groundwater Development, LP-Livestock Population, ES-Education Status, HA-Household Amenities, HC-Housing Condition, CI-Community Infrastructure, PGR-Population Growth Rate, GR-Gender Ratio

Figure 2.31: Illustration of status of contributing factors for a representative village in Devgad for intervention planning (red most important, green least important)



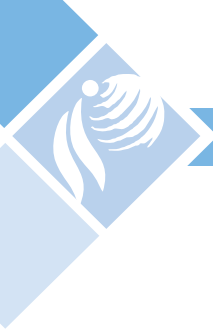
Environmental Impact Assessment on Tobacco Curing

India is the world's 2nd largest producer of tobacco with export earnings of around ₹6050 crores. Nearly 80 per cent of the total tobacco production is from Andhra Pradesh, Karnataka, and Gujarat. Flue curing is the predominant method being adopted for processing tobacco and wood requirement of tobacco curing is presumed to be at large scale. As per the estimates of PTC (Pakistan Tobacco Company), firewood burning at farmer's barns accounted for 2617 tCO₂e by consuming 796 44 tons of firewood in 2011. This may lead to environmental criticality if left unchecked. It is further to be noted that a high proportion of the tobacco growing area in developing countries lie within parts of the world identified by the FAO as being in wood deficit or prospective wood deficit situations (ITGA, 1997). The present study has been conducted to make an assessment of impact of tobacco curing in Andhra Pradesh and Karnataka with the following specific objectives of (i) surveying the sources and types of wood used for tobacco curing and determining wood consumption for flue curing; (ii) assessing the impact of tobacco curing on loss of forest and environment; (iii) studying the supply chains of wood fuel for curing, demand and supply patterns, price sensitivity and purchase patterns of wood fuel and (iv) exploring the feasibility of identifying energy efficient tobacco curing to reduce the dependence on fuel wood.

The survey was conducted in Prakasam district of Andhra Pradesh state wherein tobacco is largely grown. Four blocks (Kandukur, Tangutur, Voletivaripalem, Zarugumilli) were selected considering the predominance of tobacco cultivation. Two hundred and seventy-two barn owners were interviewed from sixteen villages encompassing four blocks. Well-structured questionnaire was prepared and pretested with the pilot survey and key informants/tobacco barn owners. Information on sources and types of fuel for curing, crop/ variety of forest wood, plantation, cost incurred for curing, quantity of wood used for tobacco curing were collected through the survey. Based on comprehensive survey, a few preliminary observations were made. The dominant tobacco variety was Siri occupying 98% of total area. The average yield of the variety was 759.70 kg/ac. Farmers owned barns at an average of 1.5 barns and average age of tobacco barn owners was 57.4 years with an experience in curing was 28.7 years. Wood species available in waste lands and farm forestry like Prosopis, Casuarina, Mango, Cashew besides stocks of Eucalyptus spp, were used for flue curing. Quantity of wood fuel used per season was found to be at 22.63 ton and the requirement of wood for curing 1 kg of tobacco leaves was estimated at 4.07 kg of fuel wood.

Socio-economic assessment of Community based Seed Producers (CBSPs)

ICAR-National Academy of Agricultural Research Management (ICAR-NAARM) undertook a socio-economic study of women SHG based seed-production groups of Rajiv Gandhi Mahila Vikas Paryojana (RGMVP). The key objectives of the study were (i) to assess the impact of integrating informal seed systems across different groups of SHG members on intra-household bargaining power with respect to agriculture, (ii) to assess the impact of SHGs on intra-household bargaining power with respect to agriculture, (iii)



to explore whether seed production is an economically profitable activity. In addition, the study has attempted to capture the impact of the model in short run. A modest attempt has been made towards quantifying the willingness to pay of the non-seed producer SHG members for participating in the programme. The focus of the study was to understand the working of the model and to suggest strategies for mid-course correction if necessary.

The summary of socio-economic and farm characteristics shows that the member households are mostly from socially backward classes, but economically well-off among their peers.

Overall, the role of women in agricultural decision making had improved in seed and non-SHG households. In non-SHG households, conflicts were due to male dominance, while in SHG-households, it was due to female dominance in decision making. The study shows that the conflicts were lesser in seed-producer households compared to non-seed producer households in seed-SHG.

Members who are socio-economically better (elite) among the peers (SHG members in a group) and those in occupying positions (president or secretary) have a higher chance of being selected as a seed producer. This might not be an issue as the interest of the group is to produce good quality seed. But our study shows that those people may not end up with producing good quality seeds. The study showed that the non-seed SHG women are willing to pay higher quantity of seeds than the current arrangement (3X quantity of seed in case of Paddy and 2X in case of Wheat, where X stands for seed provided for seed production) for participation in the seed production.

2.6.4 Research, Technology & Innovation: Preparing for future

The Academy organized series of Workshops and Consultations in order to strengthen its focus on research management, technology and innovation management.

Revisiting FOCARS: Reflections and Feedback of Trained Scientists



WS/2017-18/04
WS/2017-18/13

The workshop was conducted to review the present model of three-staged FOCARS which is implemented since December, 2011 (94th FOCARS batch), assess perceptions of trainees on the effectiveness of FOCARS, and to suggest improvements in the design and implementation of FOCARS to enhance its effectiveness. The participants for this workshop were selected in such a way to have representation from different Discipline/Subject Matter Division (SMD), Agro-ecosystem and gender and belonged to FOCARS 94th to 103rd batches. The participants included 74 scientists and 6 special invitees including FET coordinators, PAT coordinators and orientation module coordinators.

Brainstorming was conducted using card method. The participants were randomly distributed into three groups and break-out sessions were conducted to elicit maximum points from the participants on various issues and suggestions. The points written by participants were discussed and



Inaugural function and participants of National workshop on Revisiting FOCARS

validated at each break-out session and were put forward for discussion in the open house through rapporteurs and validated.

Workshop on “Agri Innovations”

Two days consultation workshop was organized “Agri Innovations” with the main aim to identify the factors that influence the innovations in agricultural R&D systems and to construct a framework for indexing of agri-innovations for R&D system and agricultural technology. Twenty eight experts participated from various ICAR institutes, State Agricultural Universities, industries, start ups, etc. The Agriculture Innovation Index will highlight agricultural technology evolution in the country, research and innovation capacities across the agricultural ecosystem to identify strengths and existing gaps within the clusters. It is envisioned to bring out policy level interventions needed to intensify performance and relating quantum investments and funding in the startups for their potential commercialization and impact creation.

2.7 List of Ongoing Research Projects

2.7.1 Institute Projects

No	Project code	Title of the project	Team	Duration	
				From	To
1	SIL/2016/04/114	Development of portal for Indigenous Technology Knowledge (ITK)	M Balakrishnan SK Soam P Venkatesan	Apr-16	Mar-18
2	SIL/2016/05/115	Asynchronous learning in agricultural higher education through MOOCs: A case Study	D Thammi Raju GRK Murthy	Apr-16	Mar-18
3	SIL/2016/06/116	Development of teaching competency framework for quality agricultural education	P Ramesh D Thammi Raju	Apr-16	Mar-18
4	SIL/2016/07/117	International market signals for Indian agricultural trade during WTO Regime: R & D strategies and policies	MB Dastagiri K Kareemulla	Apr-16	Mar-17



5	SIL/2016/08/118	Strategic and financial analysis of livestock feed industry in India	Sanjiv Kumar B Ganesh Kumar PC Meena	Apr-16	Mar-18
6	SIL/2016/09/119	Gender based ICT led knowledge management in agriculture	N Sandhya Shenoy VKJ Rao PD Sreekanth	Apr-16	Jun-18
7	SIL/2017/07/127	Research role analysis of extension scientists in ICAR	VKJ Rao Bharat S Sontakki Surya Rathore	Apr-17	Mar-18
8	SIL/2017/08/128	IP landscaping of emerging agricultural technologies for sustainable value chain development	Manju Gerard K Srinivas SP Subash	Apr-17	Mar-20
9	SIL/2017/09/129	Developing a framework for enhancing agricultural research management efficiency	P Krishnan I Sekar B Ganesh Kumar SK Soam	Apr-17	Mar-18
10	SIL/2017/10/130	Financial inclusion of smallholders and landless households in the underdeveloped state.	PC Meena Ranjit Kumar Sanjiv Kumar	Apr-17	Mar-19
11	SIL/2017/11/131	Structural Changes in Agricultural Marketing in India: Effect on Inclusiveness, Efficiency & Profitability	Ranjit Kumar PC Meena B Ganesh Kumar N Sivaramane Sanjiv Kumar	Apr-17	Mar-19
12	SIL/2017/12/132	Team work and its effect on organizational performance in NARES.	RVS Rao Alok Kumar KH Rao	Apr-17	Mar-19
13	SIL/2017/13/133	Competency management system for agricultural research and extension.	K H Rao Alok Kumar RVS Rao P Ramesh	Apr-17	Mar-20
14	SIL/2017/14/134	Geo-Spatial knowledge system for monitoring climate changes in Agro-climatic Zone X: Southern Plateau and Hill Region.	PD Sreekanth BS Yashavanth	Apr-17	Mar-19
15	Pilot Study	Assessment of technological interventions in teaching, learning process	S. Senthil Vinayagam GRK. Murthy D. Thammi Raju	Apr-17	Sep-17
16	SIL/2017/15/135	Decision Support database of maize inbred germplasm	N Sunil (IIMR) JC Shekar(IIMR) N Srinivasa Rao	Jul – 16	Jul - 20

(Characters in Project Codes: S- Intra Institutional, C- Inter Institutional, I- Institute Funded, O- Externally Funded, C- Consultancy, L- Institute is Leader, P- Institute is Partner)



2.7.2 Externally Funded Contract Research (Sponsored/Collaborative/Grant-in Aid Projects)

No	Project Code	Title of the project	Funding Agency	Team	Budget (₹ in Lakhs)	Duration From To
1	Grant-in Aid SOL/2008/03/067	IP Management and Transfer/ Commercialization of Agricultural Technologies Scheme	ICAR Scheme	K Srinivas Manju Gerard SP Subash	25	Apr-08 Mar-18
2	Grant-in Aid SOL/2014/09/100	Technology Enhanced Learning in Agricultural Education (TELAge)	ICAR, Education Division under Niche Area of Excellence (NAE) Scheme	GRK Murthy D Thammi Raju S Senthil Vinayagam P Ramesh M Krishnan	425	Jan-14 Dec-17
3	Grant-in Aid SOL/2015/05/108	Socio Economic Analysis of Finger Millets in India	DBT, Govt. of India	B Ganesh Kumar N Sivaramane	43.88	Apr-15 Mar-18
4	Grant-in Aid SOL/2015/07/110	KRISHI: ICAR Research Data Repository for Knowledge Management	ICAR Hqrs	A Dhandapani SK Soam	49.76	Nov-15 Mar-18
5	Grant-in Aid SOL/2016/02/112	Establishment of Agri- Business Incubation (ABI) Centers under XII Plan Scheme for National Agriculture Innovation Fund (NAIF)	ICAR Scheme	K Srinivas Manju Gerard SP Subash	79.15	Jan-16 Mar-18
6	Sponsored CCP/2016/10/120	Socio-economic assessment of Community Based Seed Producers (CBSP) Groups by Women	Rajiv Gandhi Mahila Vikas Pariyojana, Rai Barcilly	SP Subash K Srinivas	2.50	Sep-16 Aug-17
7	Grant-in Aid COP/2017/01/121	Management and Impact Assessment of Farmer FIRST Project	Agri. Extn. Division, ICAR	P Venkatesan Bharat S Sontakki N Sivaramane	144.05	Feb-17 Mar-18
8	Sponsored SCP/2017/02/122	Environmental Impact Assessment of Tobacco Curing	Tobacco Board	I Sekar, P Krishnan K Kareemulla	20.87	Mar-17 Jun-18
9	Grant-in Aid SOL/2017/03/123	Agricultural Research and Development Infrastructure in select Indian States	NSMIS Scheme of DST, GoI	K Kareemulla S Ravichandran	44.7	Apr-17 Mar-19



10	Grant-in Aid SOL/2017/04/124	International market signals for Indian agricultural trade during WTO Regime: R & D strategies and policies	DST, GoI	MB Dastagiri	12.42	Apr-17	Mar-18
11	Collaborative COP/2017/06/126	Assessment of socio-economic Sensitivity and adaptive Capacity of Sindhudurg District	NCSCM, Chennai	P Krishnan I Sekar K Kareemulla	8	Apr-17	Aug-17
12	Sponsored CCP/2017/16/136	Estimating consumption of different food commodities in Telangana state	Govt of TS	Sanjiv Kumar Ranjit Kumar A. Dhandapani PC Meena N.Sivaramane	42.00	Jul-17	Mar-18
13	Grant-in Aid SOL/2017/18/138	Documentation of Success Stories of Agri-graduates turned Entrepreneurs (ICAR Extra-mural funding	ICAR	Surya Rathore V.K. Jayaraghavendra Rao Bharat S Sontakki	2.90	Aug-17	Mar-18
14	Grant-in Aid SOL/ 2017/19/139	Rising Start Ups of Agri-entrepreneurs: A Success story	ICAR	Manju Gerard K Srinivas	3.96	Aug-17	Mar-18
15	Collaborative SCP/2018/01/140	Strengthening and Integrating Agricultural Science and Technology Indicators (ASTI) in Indian NARES.	IFPRI	Ch. Srinivasa Rao B Ganesh Kumar P Krishnan	\$45,000	Jan-18	Dec-18
16	Consultancy CCP/2017/005/00125	Developing guidelines and methodologies for IP policies of PJTSAU	PJTSAU	Manju Gerard SK Soam SP Subash	5	Apr-17	Mar-18
17	Consultancy CCP/2017/ 015/00135	Exploring Potential for Medicinal Plants Cultivation in Telangana	TS Medicinal Plant Board	N Sivaramane Ranjit Kumar	1.00	Jul-17	Aug-17

(Characters in Project Codes: S- Intra Institutional, C- Inter Institutional, I- Institute Funded, O- Externally Funded, C- Consultancy, L- Institute is Leader, P- Institute is Partner)



2.8 Capacity Building Programmes Organized during 2017-18

Programme Code	Title of the Programme	Starting Date	Ending Date	Programme Director/s	No. of Participants	Type of the programme	Type of Participants
OFC/2017/01/01	Administrative Training for SC/ST Teachers of Veterinary College, Shimoga and Hassan, Karnataka	24 April 2017	26-Apr-2017	Ashish Roy D Thammi Raju	17	Off-Campus	Scientific Personnel
FOF/2017/01/02	4th Foundation Course for Faculty of Agricultural Universities (FOFAU)	2-May-2017	31-May-2017	S Senthil Vinayagam M Krishnan	36	FOFAU	Assistant Professors
OFC/2017/02/03	Competency Enhancement for Efficiency and Effectiveness at Workplace” for faculty and scientists of NDRI, Karnal	25-May-2017	30-May-2017	K.H.Rao D. Thammi Raju	25	Off-Campus	Scientific Personnel
MDP/2017/01/04	MDP on Leadership Development	13-Jun-2017	24-Jun-2017	KH Rao I Sekar	13	MDP (Pre RMP)	Scientific Personnel
NBP/2017/01/05	Competency Enhancement Programme on Soft Skills and Personality Development for Technical Staff of ICAR T-1 to T-5	15-Jun-2017	24-Jun-2017	N Srinivasa Rao MA Basith	27	Need Based Programmes	Technical Staff
NBP/2017/02/06	Refresher Course for Section Officers, AAOs, AFAOs and Assistants of ICAR	23-Jun-2017	29-Jun-2017	Ashish Roy JNL Das	23	Need Based Programmes	Administrative Staff
FOC/2017/01/07	106 FOCARS	5-Jul-2017	4-Oct-2017	Ranjit Kumar Surya Rathore	60	FOCARS	Scientific Personnel
NBP/2017/03/08	Training Programme on Stress Management	12-Jul-2017	15-Jul-2017	A Debnath P Ramesh	12	Need Based Programmes	Scientific Personnel
EDP/2017/01/09	EDP on Leadership Development	28-Jul-2017	1-Aug-2017	SK Soam R Kalpana Sastry	10	EDP	Scientific Personnel
NBP/2017/04/10	Training Programme on Developing Winning Research Proposals in Agricultural Research	1-Aug-2017	5-Aug-2017	SK Soam P Krishnan	22	Need Based Programmes	Scientific Personnel
NBP/2017/05/11	Training Programme on Analysis of Experimental Data	3-Aug-2017	9-Aug-2017	S Ravichandran A Dhandapani	22	Need Based Programmes	Scientific Personnel
OFC/2017/03/12	Enhancing Efficiency and Behavioral Skills of Stenographers Grade-III, PA, PS, PPS and Sr. PPS of ICAR at CIFE Mumbai	3-Aug-2017	9-Aug-2017	WS Bhat PG Kohad	26	Off-Campus	Administrative Staff
NBP/2017/06/13	Administration and Financial Management for Under Secretaries, SAOs, SFAOs, AOs & FAOs of ICAR	17-Aug-2017	23-Aug-2017	Ashish Roy JLN Das	19	Need Based Programmes	Administrative Staff
OFC/2017/04/14	Training Programme on Competency Enhancement for Efficiency and Effectiveness at Work-place at Parbhani	4-Sep-2017	8-Sep-2017	D Thammi Raju KH Rao	17	Off-Campus	Scientific Personnel
NBP/2017/07/15	Training Programme on Impact Assessment of Agricultural Research and Technologies	12-Sep-2017	16-Sep-2017	I Sekar B Ganesh Kumar	10	Need Based Programmes	Scientific Personnel



NBP/2017/08/16	Competence Enhancement Programme on Motivation and Positive Thinking for Technical Officers of ICAR (T5 & above)	13-Sep-2017	22-Sep-2017	Laxman Maharu Ahire M Shekhar Reddy	37	Need Based Programmes	Technical Staff
NBP/2017/09/17	Training programme on Public Financial Management system	13-Sep-2017	13-Sep-2017	Jagan Mohan Rao N Vijayalakshmi	40	Need Based Programmes	Administrative Staff
NBP/2017/10/18	Training Programme on Application of Bioinformatics in Agricultural Research and Education	14-Sep-2017	23-Sep-2017	M Balakrishnan N Srinivasa Rao	12	Need Based Programmes	Scientific Personnel
NBP/2017/11/20	Developing Business Proposal for Producer Companies and Startups in Agribusiness	19-Sep-2017	23-Sep-2017	Ranjit Kumar K Srinivas	12	Need Based Programmes	Scientific Personnel
NBP/2017/12/22	Training Programme on Web based Educational Content Management (under TELAgE)	4-Oct-2017	13-Oct-2017	S Senthil Vinayagam D Thammi Raju	13	Need Based Programmes	Scientific Personnel
OFC/2017/07/23	Training Programme on Developing Winning Research Proposals' at College of Veterinary Sciences, Tirupati	4-Oct-2017	6-Oct-2017	SK Soam Ch Srinivasa Rao	40	Off-Campus	Scientific Personnel
NBP/2017/13/24	Training Programme on Priority Setting, Monitoring and Evaluation (PME) of Agricultural Research Projects	6-Oct-2017	11-Oct-2017	K Srinivas P Krishnan	15	Need Based Programmes	Scientific Personnel
NBP/2017/14/26	Training Programme on 'Developing Winning Research Proposals'	13-Oct-2017	16-Oct-2017	S.K. Soam Ch. Srinivasa Rao	25	Need Based Programmes	Scientific Personnel
NBP/2017/15/28	Enhancing Efficiency and Behavioral Skills of Stenographers Grade-III, PA, PS, PPS and Sr. PPS of ICAR (at NAARM)	25-Oct-2017	31-Oct-2017	G Aneeja KR Ghanshyam	34	Need Based Programmes	Administrative Staff
MOO/2017/01/29	Massive Open online Course on Teaching Management (MOOC)	1-Nov-2017	30-Nov-2017	D Thammi Raju G R K Murthy	988	MOOC	Others
NBP/2017/16/30	ICAR Short Course on "Emerging Tools and Techniques for Agricultural Knowledge communication"	1-Nov-2017	10-Nov-2017	M Balakrishnan SK Soam PD Sreekanth	13	Need Based Programmes	Scientific Personnel
NBP/2017/17/32	Winter School on Advanced Analytics in Developing Agricultural Market Intelligence	14-Nov-2017	4-Dec-2017	N Sivaramane A Dhandapani	19	Need Based Programmes	Scientific Personnel
NBP/2017/18/33	Training Programme on Geospatial Analysis for Natural Resources Management using Free and Open Source Softwares (FOSS) - QGIS, ILWIS & R	16-Nov-2017	25-Nov-2017	PD Sreekanth SK Soam	13	Need Based Programmes	Scientific Personnel
NBP/2017/19/34	Training Programme on Good Practices in Extension Research and Evaluation	11-Dec-2017	16-Dec-2017	Bharat S Sontakki Surya Rathore	19	Need Based Programmes	Scientific Personnel
MDP/2017/02/35	MDP on Leadership Development	12-Dec-2017	23-Dec-2017	RVS Rao GRK Murthy	28	MDP (pre-RMP)	Scientific Personnel



NBP/2017/20/36	Training Programme on Multivariate Data Analysis	14-Dec-2017	20-Dec-2017	A Dhandapani N Sivaramane	27	Need Based Programmes	Scientific Personnel
NBP/2017/21/37	Communication and Scientific Writing for Technical Officers of ICAR	3-Jan-2018	9-Jan-2018	Bharat S Sontakki P Krishnan	13	Need Based Programmes	Technical Staff
FOC/2018/02/38	107 FOCARS	4-Jan-2018	3-Apr-2018	I Sekar B Ganesh Kumar	24	FOCARS	Scientific Personnel
NBP/2018/22/39	Enhancing Efficiency and Behavioral Skills of Stenographers Grade-III, PA, PS, PPS and Sr. PPS of ICAR at NBSSLUP Regional Station, Kolkata	5-Jan-2018	11-Jan-2018	WS Bhat N Raghunath	34	Need Based Programmes	Administrative Staff
FOF/2018/02/41	5th Foundation Course for Faculty of Agricultural Universities (FOCFAU)	17-Jan-2018	15-Feb-2018	K Kareemulla VV Sumanth Kumar	48	FOCFAU	Assistant Professors
NBP/2018/24/42	Refresher Course for Section Officers, AAOs, AFAOs and Assistants of ICAR	18-Jan-2018	23-Jan-2018	Ashish Roy S George	50	Need Based Programmes	Administrative Staff
NBP/2018/25/43	Technology Enhanced Learning in Agricultural Education: The Roadmap (under TELAgE)	24-Jan-2018	25-Jan-2018	GRK Murthy M Krishnan	66	Need Based Programmes	Scientific Personnel
NBP/2018/26/44	Capacity Building Programme on Enhancing Efficiency and Effectiveness at Workplace for Faculty & Scientists of UAHS, Shivamogga (Off-Campus)	29-Jan-2018	2-Feb-2018	KH Rao Alok Kumar	30	Need Based Programmes	Scientific Personnel
NBP/2018/27/45	Establishment and Financial Matters for Under Secretaries, SAOs, SFAOs, AOs & FAOs of ICAR	1-Feb-2018	7-Feb-2018	Ashish Roy S George	14	Need Based Programmes	Administrative Staff
NBP/2018/28/46	Training Programme on KOHA for Library Staff of ICAR	5-Feb-2018	9-Feb-2018	SK Soam N Srinivasa Rao	36	Need Based Programmes	Technical Staff
NBP/2018/29/47	Faculty Development Programme on Enhancing Efficiency and Effectiveness at Workplace at Dhanwate National College, Nagpur (Off-Campus)	6-Feb-2018	10-Feb-2018	KH Rao D Thammi Raju	35	Need Based Programmes	Scientific Personnel
NBP/2018/30/48	Competency Enhancement Programme for Effective Implementation of Training Functions by HRD Nodal Officers of ICAR	15-Feb-2018	17-Feb-2018	RVS Rao KH Rao	24	Need Based Programmes	Scientific Personnel
NBP/2018/31/49	Training Programme on Analysis of Experimental Data	19-Feb-2018	24-Feb-2018	A Dhandapani S Ravichandran	17	Need Based Programmes	Scientific Personnel
NBP/2018/32/50	Training Programme on Communication and Management Skills for Extension Professionals (ICAR Short Course)	19-Feb-2018	28-Feb-2018	VKJ Rao N Sandhya Shenoy Surya Rathore	17	Need Based Programmes	Scientific Personnel
NBP/2018/33/51	Foundation Course for Newly Recruited Assistant Professor, Scientists of ANGRAU	12-Mar-2018	10-Apr-2018	P Krishnan PD Sreekanth VV Sumanth Kumar	49	Need Based Programmes	Scientific Personnel



NBP/2018/34/52	Motivation for Excellence in Work Hands on Experience on MIS&FMS& AEBAS, for Skilled Staff (SSS)	12-Mar-2018	17-Mar-2018	WS Bhat S Rajukumar	27	Need Based Programmes	Skilled Support Staff
NBP/2018/35/53	Training Programme on Agricultural Research Management for the faculty of JNKVV	14-Mar-2018	20-Mar-2018	P Krishnan SK Soam	17	Need Based Programmes	Scientific Personnel
NBP/2018/36/54	Knowledge Workshop on Research Project Management, JNKVV Jabalpur, MP	24-Mar-2018	26-Mar-2018	P Krishnan SK Soam	34	Need Based Programmes	Assistant Professors

Total No. of Participants

2209

Educational Programmes (ongoing programmes)

ACA/2016/01/55	Post Graduate Diploma in Management in (Agriculture) 2016-18- Second Year	1-Jun-2016	30-Apr-2018	M Balakrishnan	30	PGDMA 2016-18	Students
ACA/2017/02/56	Post Graduate Diploma in Technology Management in Agriculture-PGTMA	1-Apr-2017	31-Mar-18	Manju Gerard	68	PGDT-MA2017	Students
ACA/2017/03/57	Post Graduate Diploma in Management in (Agriculture) 2017-19- First Year	20-Jun-2017	30-Apr-2019	S Ravichandran P Krishnan	34	PGDMA 2017-19	Students

Total No. of Students

132

(Characters in Programme Codes: OFC- Off-Campus, FOF- FOCFAU, MDP- Management Development Programme, NBP- Need Based Programme, FOC- FOCARS, EDP- Executive Development Programme, MOO- Massive Open Online, ACA- Academic)



2.9 Workshops/Meetings Organized during 2017-18

Programme Code	Title of the Programme	Programme Directors	No. of Participants	Starting Date	Ending Date
WS/2017-18/01	Brainstorming Workshop on Collaboration in Research, Education, Extension and Management	Bharat S Sontakki	20	23-Jun-17	23-Jun-17
WS/2017-18/02	Review Meeting of Implementation of Recommendation of Fifth Deans Committee on the Community Science (Home Science) & Fisheries Sciences	D Thammi Raju	56	29-Jun-2017	30-Jun-2017
WS/2017-18/03	Policy Workshop on Digital Learning in Agriculture with reference to Social Media (under TELAgE)	D Thammi Raju S Senthil Vinayagam	21	5-Jul-2017	7-Jul-2017
WS/2017-18/04	Workshop on Agri-innovations	K Srinivas Manju Gerard	21	17-Aug-2017	19-Aug-2017
WS/2017-18/05	Training Workshop on “Methodological Framework for Implementation of Farmer FIRST Project” at IISS, Bhopal	P Venkatesan Bharat S Sontakki	49	18-Sep-2017	21-Sep-2017
WS/2017-18/06	Training Workshop on “Methodological Framework for Implementation of Farmer FIRST Project” at CISH, Lucknow	P Venkatesan Bharat S Sontakki N Sivaramane	45	3-Oct-2017	6-Oct-2017
WS/2017-18/07	Training Workshop on “Methodological Framework for Implementation of Farmer FIRST Project” at TANUVAS, Chennai	P Venkatesan Bharat S Sontakki N Sivaramane	43	10-Oct-2017	13-Oct-2017
WS/2017-18/08	Training Workshop on “Methodological Framework for Implementation of Farmer FIRST Project” at IARI, New Delhi	P Venkatesan Bharat S Sontakki N Sivaramane	42	23-Oct-2017	26-Oct-2017
WS/2017-18/09	Training Workshop on “Methodological Framework for Implementation of Farmer FIRST Project” at IISWC, Dehradun	P Venkatesan Bharat S Sontakki	57	6-Nov-2017	9-Nov-2017
WS/2017-18/10	Let's listen to farmers: A workshop on farmers' feedback on doubling farm income by 2022	P Venkatesan Bharat S Sontakki Ch Srinivasa Rao	180	22-Dec-2017	23-Dec-2017
WS/2017-18/11	National Workshop on Technology Enhanced Learning	G R K Murthy M Krishnan	66	24-Jan-18	25-Jan-18
WS/2017-18/12	Policy Workshop on Big Data Analytics in Agriculture	S Ravichandran A Dhandapani	48	8-Feb-2018	9-Feb-2018
WS/2017-18/13	Workshop on “Revisiting Foundation Course for Agricultural Research Service (FOCARS) Reflections and Feedback of Trained Scientists”	N Sivaramane B. S. Yashavanth M. B. Dastagiri Bharat S. Sontakki	60	15-Mar-18	16-Mar-18
Total No. of Participants			648		

(Characters in Programme Code: WS- Workshop)



2.10 PGDMA Students' Guidance by NAARM Faculty

2.10.1 PGDMA Dissertation submitted during the year 2017-2018

Name of the Student	Title of the Project	Major Guide
Abhishek Jain	Value Chain Analysis of Finger Millet in Karnataka	Ganesh Kumar
Aditya Pandey	Identifying Key performance Indicators (KPIs) for Farmer Producer Organization FPO in India	Ranjit Kumar
C H Srikar	Study on Implementation of NAFED e-SAMRIDHI	Ranjit Kumar
Chittimalli Akhilesh	Economics of Selected crops in Prakasam and Nellore Districts, AP	K Kareemulaa
Eerla Ramesh	Diagnosing the market performance of APC nutrients products in Surat (Gujarat)	P Venkatesan
Harshit Joshi	Impact of Digital Initiative on farmers Income through samrudh krishi	K H Rao
Jaju Gautam	Customers' perception towards P3401-corn hybrid	S Ravichandran
Kalambe Parag	Mapping of Existing distribution network of Tirumala and Enhancing performance of Tirumala milk	Alok Kumar
S Kavinraaj	Design framework for Machine readable crop calendar in Association with vasudhaika and Kalgudi	M.Balakrishnan
S S Kavya	Study on competitors channel network and pricing strategies in Karnataka for Cotton Seed	V.V.Sumanth Kumar
Konka Prabhu Kumar	Mapping of Existing distribution network of Tirumala and Enhancing performance of Tirumala milk	Sanjiv Kumar
Kotaiahgari Mallika	Study of competitor channel network & market development strategies for APC nutrients in AP	Manju Gerard
Kritika Sharma	Technical Feasibility and Financial Viability of live carp handling FARM to Fork	K Srinivas
Linsu Cherian	Analyzing the customer behavior and formulating strategies for doubling the sale of cut fruits and vegetables	P D Sreekanth
Manisha	Problems and prospects of Kangra Tea in Himachal Pradesh	RVS Rao
Meenakshi Chauhan	Design framework for machine readable crop calendar	I. Sekar
Naushaba Khatoon	Innovation to Enterprise- Assessment of Farmers Innovation	Bharat S Sontakki
M Pavithra	Study on Farmers Knowledge on Input usage and its effect on environment under major crops grown in Nellore and Prakasam Districts of AP	N.Srinivas Rao
Praveen N Shivalli	Innovation to Enterprise- Assessment of Farmers Innovation	Bharat S Sontakki
Raipure Shubham	Mapping of Existing distribution network of Tirumala and Enhancing performance of Tirumala milk	Ch.Srinivasa Rao
Madhukar	Business Feasibility of Urban waste management through NCSL innovative strains	S.K. Soam
Ramagoni Mukesh	Understanding market dynamics of Agro chemical Industries in India	K Srinivas
P Sagar Athithiyan	Assessing the Performance of Mango Pulp Processing units in Karnataka	N.Sivaramane
V Santhosh Reddy	Perception of stakeholder for evaluating market opportunities for VAMSO products in Aqua segment in coastal Andhra Pradesh	M.Krishnan
Shashank Reddy Belanagari	Socio-Economic Assessment of Farmers of Mysuru District	A.Dhandapani
Sibin C Baby	Scope of Specialty nutrients markets in India and understanding marketing strategies of Greenovate Agrotech	M.Balakrishnan
Siddharth Bhattacharya		



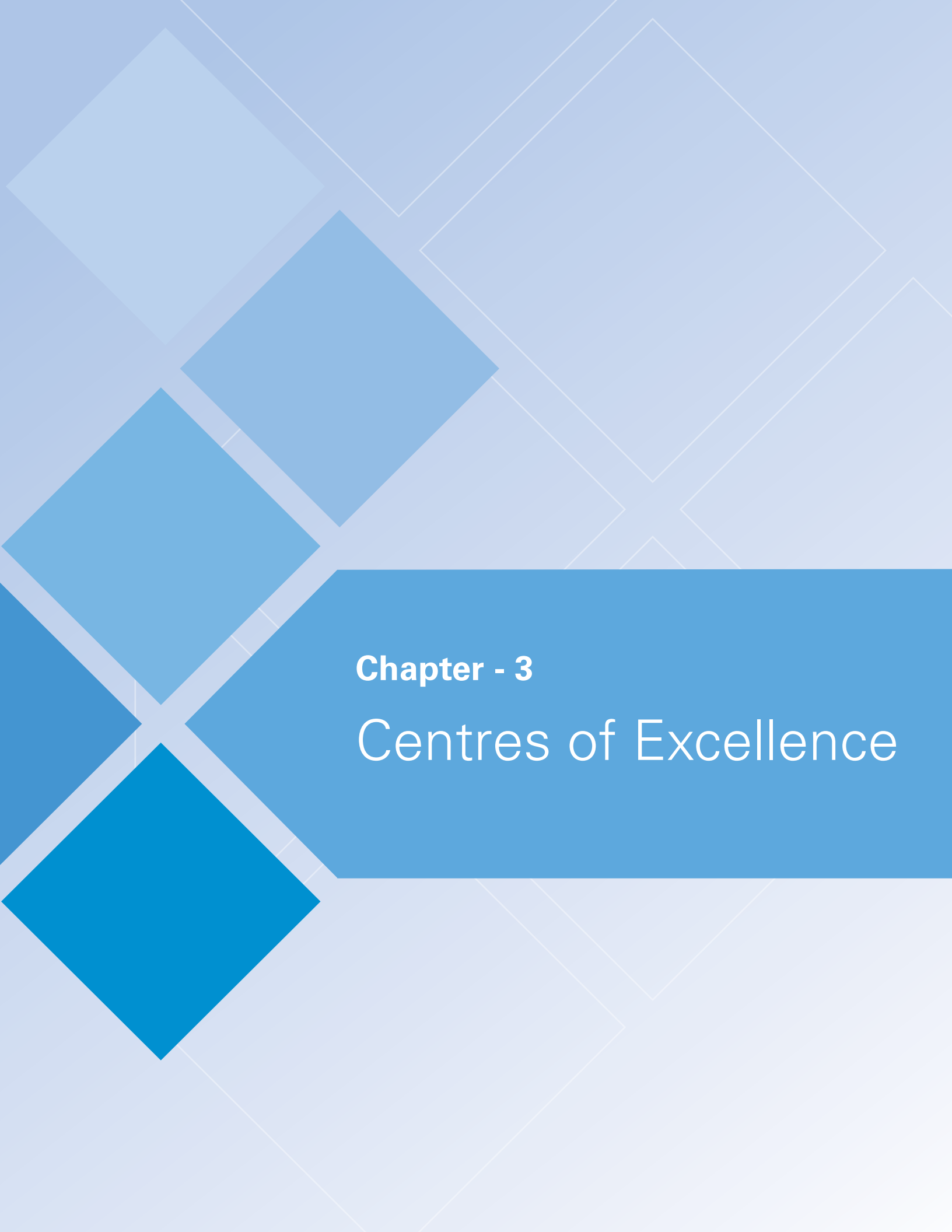
Sneha Mishra	Study on Competitor channel network and pricing analysis of cotton in Maharashtra	P.C.Meena
V Subha Shree	Study on Farmers Knowledge on input usage and its effect on environment under major crops grown in Mysuru Districts of Karnataka	S Senthil Vinayakam
B Vaishali	Study on Bio product market in Tamil Nadu	A.Dhandapani
Godbole Mohanish Arunrao	Value Chain Analysis of Nagpur Orange	VKJ Rao

2.10.2 Other Research Students Guided as Major Guide/Supervisor

Name of the faculty	Student	Title of the Thesis	Programme	University/ Institute
B. Ganesh Kumar	Anees Fathima	Evaluation of Performance of Important Horticulture Lending Schemes of Agricultural Development Bank of State Bank of Hyderabad, Mahbubnagar	MBA (ABM)	PJTSAU, Hyderabad
Bharat S. Sontakki (Co-Guide)	Bhuvana Rao	Assessment of farmer innovation system	PhD in Agricultural Extension	PJTSAU, Hyderabad
Ch. Srinivasa Rao	Sumanta Kundu	Improving nutrient use efficiency and profitability through conservation tillage and improved nutrient management in maize (zea mays)-horsegram (macrotyloma uniflorum) cropping sequence in rainfed alfisol.	PhD	Department of Agronomy, University of Kolkata
	Sharanbhoopal Reddy	Nitrogen management and greenhouse gas (ghg) emissions in rainfed maize (zea mays l.) - horsegram (macrotyloma uniflorum) system in semi-arid alfisols".	PhD	Department of Soil Science and Agricultural Chemistry, PJTSAU, Hyderabad
K. Kareemulla	Sk Mahmood	Market Performance of Vijaya Telangana Milk Products Brand vis a vis Competitors in Hyderabad	MBA (ABM)	PJTSAU, Hyderabad
M. Krishnan	P. Gomathi	Livelihood Assessment of Bereaved Fisher Families Ramanathapuram district, Tamil Nadu	MFSc (Fisheries Economics)	ICAR-CIFE, Mumbai
Manju Gerard (Co-Advisor)	Shyla Haqq	Synthesis and characterization of silver nanoparticles biofabricated from medicinal plants: <i>Jasminum auriculatum</i> and <i>Chrysanthemum coronarium</i>	PhD	SHUATS, Allahabad
	Rayees Ahmad Malik	Synthesis of novel organic pyridine tetrazole based transition metal complexes for advanced photonic applications	PhD	SHUATS, Allahabad
M. Balakrishnan	Mrs. Anidath	Development of an Expert System for Coconut Management (Thesis submitted)	Ph.D	PRIST University, Tanjavur



M. Balakrishnan	Md. Atheeq Sultan Ghor	Pre harvest forecasting of Turmeric crop Yield in Telangana State using Data Mining and Neural Network Techniques	Ph.D	Bharathiar University , Coimbatore
	M. Naveen Kumar	Improvising production and cultivation techniques of sugarcane yield in Telangana State by developing an Android Application using mobile networking	Ph.D	Bharathiar University , Coimbatore
	D. Karunakaran	Development of Algorithm for prediction of fish yield in reservoirs using Artificial Neural Network	Ph.D	Bharathiar University , Coimbatore
N. Sivaramane	Kamuni Shravani	Evaluation of Digital Platform for Agriculture of eFresh Agribusiness Solutions Pvt. Ltd.	MBA (ABM)	PJTSAU, Hyderabad
P. Krishnan	Dasari Bhoomaiah	Mapping the fisheries research base and productivity of fisheries scientists in India	PhD (Social Science)	PRIST University, Tanjavur
	Siddhika V Meher	Designing a Mobile App Prototype for Online Seafood Sale and Purchase	MFSc.	ICAR-CIFE, Mumbai
S. Senthil Vinayagam	Ms. Kirti Jamwal	Precision Farming for Horticulture Crop Production	PGDTMA	ICAR-NAARM
Sanjiv Kumar	Prashant Kumar Soni	Current Trends of Food and Beverage Service Industry in India	PGDTMA	ICAR-NAARM



Chapter - 3

Centres of Excellence



Centres of Excellence

In accordance with the EFC plan, the Academy has successfully set up two Centres of Excellence, viz. Centre for Agri-Innovation (CAI) and Centre for Open and Lifelong Learning in Agriculture (COLLAge).

3.1 Centre for Agri-Innovation (CAI)

The current challenges facing agriculture include not only enhancing productivity and production of agricultural produce but also building opportunities for agricultural goods in the production-consumption continuum. These call for the skills in management of research outputs towards technologies and up-scaling these to processes and products of commercial value. Therefore, the creation of a one-stop Center for Agri-Innovation was set up where in tech management, entrepreneurship and incubation, acceleration and investment with strong foresight study are promoted. The initial handholding attempts made by this centre including ICAR-funded Agri-Business Unit has led towards creating institutional culture for creating value chains through technology transfer from ICAR institutes towards entrepreneurship channels. The Centre provides hand-holding support to the Agri-business incubators by inter-alia developing modules, guidelines and other forms of learning material for capacity building for the start-ups; and also provides need based support for analysis/valuation of the commercial potential of technologies; negotiation skills; monitoring of IP-rights and license agreements, marketing instruments. The main objectives of the Centre are: to organize training & capacity building programs on technology transfer, IP management, business incubation, entrepreneurship development and investment; to develop foresight and policy framework for governance and coordination of technology development and commercialization in NARES and incubation development for agribusiness entrepreneurs, and to provide a platform for up-streaming and down-streaming of technologies through advisory services for IP and technology management leading to entrepreneurship development.

The Centre for Agri-Innovation of the Academy functions through the technology business incubator centre called a-IDEA (Association for Innovation Development of Entrepreneurship in Agriculture). a-IDEA is a registered society, supported by Department of Science & Technology (DST), Government of India (GoI). It also supports activities of BioNest-Agri- Bio-incubator for Agri-Biotechnology Startups. BioNest has received the financial support from BIRAC, DBT, GoI, which runs in partnership with Agri-Biotech Foundation (ABF), Rajendranagar, Hyderabad.



The aIDEA is presently focusing on 13 sub-sectors of agriculture innovations across the value chain in agriculture:

- | | |
|---|------------------------------|
| (i) Sustainable inputs | (viii) Animal husbandry |
| (ii) Precision/smart agriculture | (ix) Urban/ vertical farming |
| (iii) Innovative food technology | (x) Agri-biotech |
| (iv) Supply chain technology | (xi) Post-harvest technology |
| (v) Fisheries | (xii) Farm fresh retail |
| (vi) Soil, water & weather technologies | (xiii) Farm mechanization |
| (vii) ICT & IoT in agriculture | |

Till date 36 incubatees have been inducted and out of that four were graduated in 2017-18.

S. No.	Name of Incubatee	Year of induction	Company Profile
A. Incubatee Startups			
1.	Millets n Cereals Food Products	Dec 2015	Millet based value added products for social impact www.milletbowl.in
2.	Saro Agri and Dairy	Dec 2015	Supply chain of A2 grade cow milk http://www.sarofarmfresh.com/
3.	BharatRohan Innovations	Apr 2016	Hyper spectral aerial (UAV) imagery https://bharatrohan.in/
4.	Gen Agritech	Aug 2016	Naturally derived bio-repellants, nutrient & other plant growth supplements. www.genagritech.com
5.	Fresco Greens	Sept 2016	Modular rooftop kits plug & play model for organic vegetable & fruit cultivation. www.homecrop.in
6.	Greenovate Agrotech	Feb 2017	Manufacturing and marketing of crop specific micronutrient technology a technology licensed from ICAR www.greenovateagro.com
7.	Stamp IT Business Solutions	Feb 2017	IT business process automation for Agri companies (B2B) www.stampit.biz
8.	Vertical Farming Technologies	Nov 2017	Enabling city dwellers to grow safe, better and fresh produce by using Innovative technology like Vertical farming & Hydroponics https://citygreens.in/
9.	Zoofresh Foods	Nov 2017	Transport & Retailing live fish using high –tech logistics to bring down wastages in fish supply chain https://zoofreshfoods.com
10.	Cocogreen Foods	Dec 2017	Linking farmers and primary producers directly across India with retailers and consumers https://www.cocogreen.co.in/
11.	Transity Digital Solutions	Jan 2018	Offer a SaaS and technology platform that automates the workflows, vehicle allocation, scheduling and routing http://transity.co/
12.	Skycraft Aerospace	Jan 2018	Addresses commercial and enterprise drone needs http://www.skykrafts.com/



B. Accelerator Startups			
1.	Gen Agritech	Aug 2016	Naturally derived bio-repellants, nutrient & other plant growth supplements www.genagritech.com
2.	Muddy Puddle Foods	Aug 2017	Provides the products that innovate around good nutrition and taste without any additives & preservatives https://www.smoodies.in/
3.	Delmos Research	Aug 2017	Reagent strips that provides a fast, convenient way of testing milk for presence of adulterants http://www.delmosresearch.com/
4.	Grow Yield Technologies	Aug 2017	Offer variety of products and solutions to enhance the crop yield and reduce the cost with innovative technology https://www.growyi.com/
5.	Intello Labs	Aug 2017	Uses Artificial intelligence based commodity testing app and equipment for quality assessment http://www.intellolabs.com/
6.	Yukti Harvest	Aug 2017	Developed a unique value chain model for agriculture produce marketing by an IT enabled platform to link farmers to end customers http://yuktiharvest.com/
7.	Jivabhumi Agritech	Aug 2017	A technology platform for connecting farmers directly with institutional buyers and consumers https:// bhumi.com/
8.	RF Wave Technologies	Aug 2017	Regulate cost and profit in crop productivity with respect to irrigation and get paid for the losses due to uncertainties http://www.rfwavetek.com/aboutus.htm
9.	Agricxlab	Aug 2017	Assessment & reporting the quality of Agri-produce by a user-friendly mobile app through Computer Vision and Artificial Intelligence http://www.agricx.com/aboutus.php
10.	Odaku Fishery Platform	Aug 2017	Developed a platform for fishermen communities for seafood traceability using GPS, Big Data, Geofencing, Sharing Data, E-Commerce and Block chain technology http://www.odaku.in/
C. Graduated Startups (in 2017-18)			
1.	Inner Being Wellness	Oct 2014	Quinoa & healthy foods supply chain & branding www.innerbeing.in
2.	Input Agri Tech	Oct 2015	ICT platform for farm input supply chain www.agrowbook.com
3.	Mother Biotech	Mar 2017	Agri biotech startup working on teak Plant tissue culture to revolutionize timber industry www.motherbiotech.in
4.	Navaratnaa Cropscience	Feb 2017	Agri Biotech company focusing on Biopesticide, Biostimulants, Plant growth regulators etc www.navaratnacropscience.com



Incubatee Startups



Accelerator Startups



Graduated Startups (2017-18)



Events conducted by a-IDEA, NAARM during 2017-2018

a-IDEA has conducted various programs in different parts of the country so as to reach as many Agri Entrepreneurs as possible. Listed here are the events:

1. AGRI UDAAN – Food & Agribusiness Accelerator
2. Road Shows – 6
3. Pitch Perfect – 1
4. Capacity Building workshop - 1
5. Startup 360 – 1
6. Demo Day – 2
7. Startup Samvaad – 1
8. Sensitization Workshop - 3



Glimpses of events conducted by a-IDEA, NAARM during 2017-2018



Launch of AGRI UDAAN in New Delhi



Road Show in Bengaluru



Pitch Perfect in ICAR-NAARM



Capacity Building workshop in ICAR-NAARM



Start up 360 workshop in ICAR-NAARM



Demo Day in Hyderabad



Startup Samvad in Bhubaneswar



Sensitization workshop in BHU, Varanasi



"Agri Udaan", Food & Agribusiness Accelerator 2.0 is a program for scaling up of Agritech startups. This is the first ever accelerator program funded by DST. The objective of Agri Udaan 2.0 is to help Agri startups to scale their operations on a bigger canvas on fast track mode.

Agri Udaan was planned in such a way that it covered 7 Cities in different parts of India including its launch at New Delhi. The presence of different organizations like NITI Aayog, ICAR, DST, Investors and other stakeholders on one platform made it unique. Multiple Roadshows in Chandigarh, Ahmedabad, Bangalore, Pune, Kolkata, Hyderabad to bring awareness and have a bigger pool of startups for effective cohort selection was done. Nearly 1000 startups and Agri stakeholders attended the program launch and roadshows, 579 startups had applied for scaling up. Rigorous three tier selection procedure was adopted to select top 10 startups as a cohort.

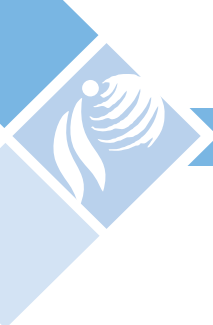
These top 10 startups had to go rigorous capacity building workshops, mentor matching and milestone setting. Three months continuous mentoring and monitoring of the startups and final pitching training gave opportunity to startups to face investors during Demo Day/Investor Meet in Hyderabad and Mumbai. Out of the 10 startups in the accelerator cohort 3 have raised funding to the tune of approx.. 6.25 Cr INR and remaining are in different stages of discussions with investors

Mr. Bill Gates felicitates the a-IDEA Incubatee

BharatRohan, an Agriculture technology start-up supported by a-IDEA (Technology Business Incubator of NAARM) has been selected amongst the Top 10 Startups in AP AgTech Summit 2017 out of 259 entries from 61 countries. BharatRohan was felicitated by Sri Nara Chandrababu Naidu, Hon'ble Chief Minister of Andhra Pradesh and Mr. Bill Gates, Chairman Bill & Melinda Gates Foundation. BharatRohan is presently working for the farmers by reducing their losses through the prediction of pest attacks and disease outbreaks using UAV-based Hyperspectral Imaging. BharatRohan collects valuable data by flying our UAV (or drone) platforms equipped with Hyper Spectral Cameras and transforms it into actionable information to generate early pest alerts, conduct crop nutrition diagnostics, detect weed anomalies and perform yield predictions. This ensures the reduction in losses for the farming community and increases in profit margins. Young Amandeep Panwar, Rishabh Choudary and KD Singh are the founders of the start-up.



Bill Gates felicitating TBI incubatee

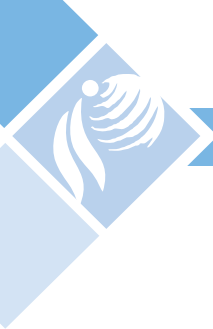


Activities of a-IDEA for promotion of Agripreneurship and Startups

S. No.	Name of Events	Date	Venue	Number of Participants
1	Sensitization workshop for Assistant Professors from State Agricultural Universities of ICAR regarding Agri startup ecosystem and Incubation	31 st May 2017	a-IDEA, NAARM-TBI, Hyderabad	30
2	Sensitization workshop at MANAGE towards Incubation & Entrepreneurship in agriculture	8 th June 2017	a-IDEA, NAARM-TBI, Hyderabad	45
3	1 st Startup Samvaad under Agri BioNest- Bio Incubator of a-IDEA NAARM & Agri Biotech Foundation	30 th June 2017	a-IDEA, NAARM-TBI, Hyderabad	80
4	Launch of Agri Udaan – Food & Agribusiness Accelerator 2.0	4 th August 2017	NASC Complex, New Delhi	200
5	Agri Udaan – Food & Agribusiness Accelerator 2.0 Roadshow	8 th August 2017	ISB, Mohali, Chandigarh	45
6	Agri Udaan – Food & Agribusiness Accelerator 2.0 Roadshow	19 th August 2017	CIIE, IIM – Ahmedabad	40
7	Agri Udaan – Food & Agribusiness Accelerator 2.0 Roadshow	23 rd August 2017	MCCIA, ICC Tower, Pune	135
8	Agri Udaan – Food & Agribusiness Accelerator 2.0 Roadshow	26 th August 2017	Banquet Hall, The Sahar Pavilion, Bangalore	100
9	Agri Udaan – Food & Agribusiness Accelerator 2.0 Roadshow	29 th August 2017	British Club, British High Commission, Kolkata	70
10	Agri Udaan – Food & Agribusiness Accelerator 2.0 Roadshow	9 th September 2017	Katriya Hotel & Towers, Somajiguda, Hyderabad	92
11	Ideation event for the students of biotechnology by Agri-BioNest, Bio Incubator of a-IDEA, NAARM & Agri Biotech Foundation	12 th September 2017	ABF, Hyderabad	40
12	Sensitization workshop for foodpreneurs	27 th September 2017	Vellore	60



13	AGRI UDAAN cohort finalization event where top 40 startups pitched in front of experts comprising of partners, mentors from business, agriculture and technology	9 th -10 th Oct 2017	a-IDEA, NAARM-TBI, Hyderabad	80
14	Agripace 2017 - An International conference & Expo on Agriculture and veterinary Science: Research & Technology Session on "Agri startups as change makers"	23 rd -25 th Oct 2017	PJTSAU, Hyderabad	200
15	Pitch Perfect - Startups pitching event for incubated at a-IDEA, NAARM - TBI.	1 st Nov 2017	a-IDEA, NAARM-TBI, Hyderabad	25
16	As part of National Entrepreneurship Day an ideation event was conducted under Agri BioNest, Bio Incubator of a-IDEA, NAARM - TBI & Agri Biotech Foundation for the students of PJTSAU, University - College of Science & Osmania university.	9 th Nov	ABF, Hyderabad	40
17	Agri Udaan- Capacity Building workshop	13 th -16 th Nov 2017	a-IDEA, NAARM, Hyderabad	17
18	One-day workshop on "Entrepreneurship Development Programme for Prospective Incubatees" has been organized by a-IDEA, NAARM - TBI and Agri Biotech Foundation.	15 th December 2017	ABF, Hyderabad	50
19	a-IDEA, NAARM – TBI has hosted the fraternity of incubators, Telangana at NAARM campus	29 th Jan 2018	a-IDEA, NAARM, Hyderabad	25
20	"Startup 360" a mentoring & networking session with the mentors and incubatees	6 th February 2018	a-IDEA, NAARM, Hyderabad	10
21	AGRI UDAAN - Food & Agribusiness Accelerator 2.0, Demo Day-Hyderabad	12 th Feb 2018	Lemon Tree Premier, Hyderabad	35
22	AGRI UDAAN - Food & Agribusiness Accelerator 2.0, Demo Day-Mumbai	17 th Feb 2018	Hilton, Mumbai	25
23	Startup Samvaad	3 rd March 2018	XIM, Bhubaneswar	82
24	Sensitization Workshop	14 th March 2018	Banaras Hindu University, Varanasi	200
25	Unconvention Speaker Series	20 th April 2018	a-IDEA, NAARM, Hyderabad	30



3.2 Center for Open and Lifelong Learning in Agriculture (COLLAge)



Parliamentary Standing Committee visiting COLLAge

Academy has spearheaded the e-learning movement during last decade to enhance quality and access of educational content in agricultural universities. Its next level follow-up of the initiative has resulted in establishing a state of art technology enhanced learning facility which has been instrumental in spreading high end digital content production and management. Learning in Agriculture is very diverse with varying levels of stakeholders- students to learn in academic framework, policy makers to update their knowledge, entrepreneurial personnel who look out for short and simple course content, farmers who learn in simple nontechnical content in local language. Considering the rapid developments in the area of technology enhanced learning, the **Centre for Open and Lifelong Learning in Agriculture (COLLAge)** at the Academy has been set up. The COLLAge will aim to function as a Centre to provide guidance, consultancy and support in initiating advanced digital learning activities across various institutions in NARES. The centre will provide support in latest advances in technology pertaining to learning processes through its continual updating and upgrading in technological advancements.

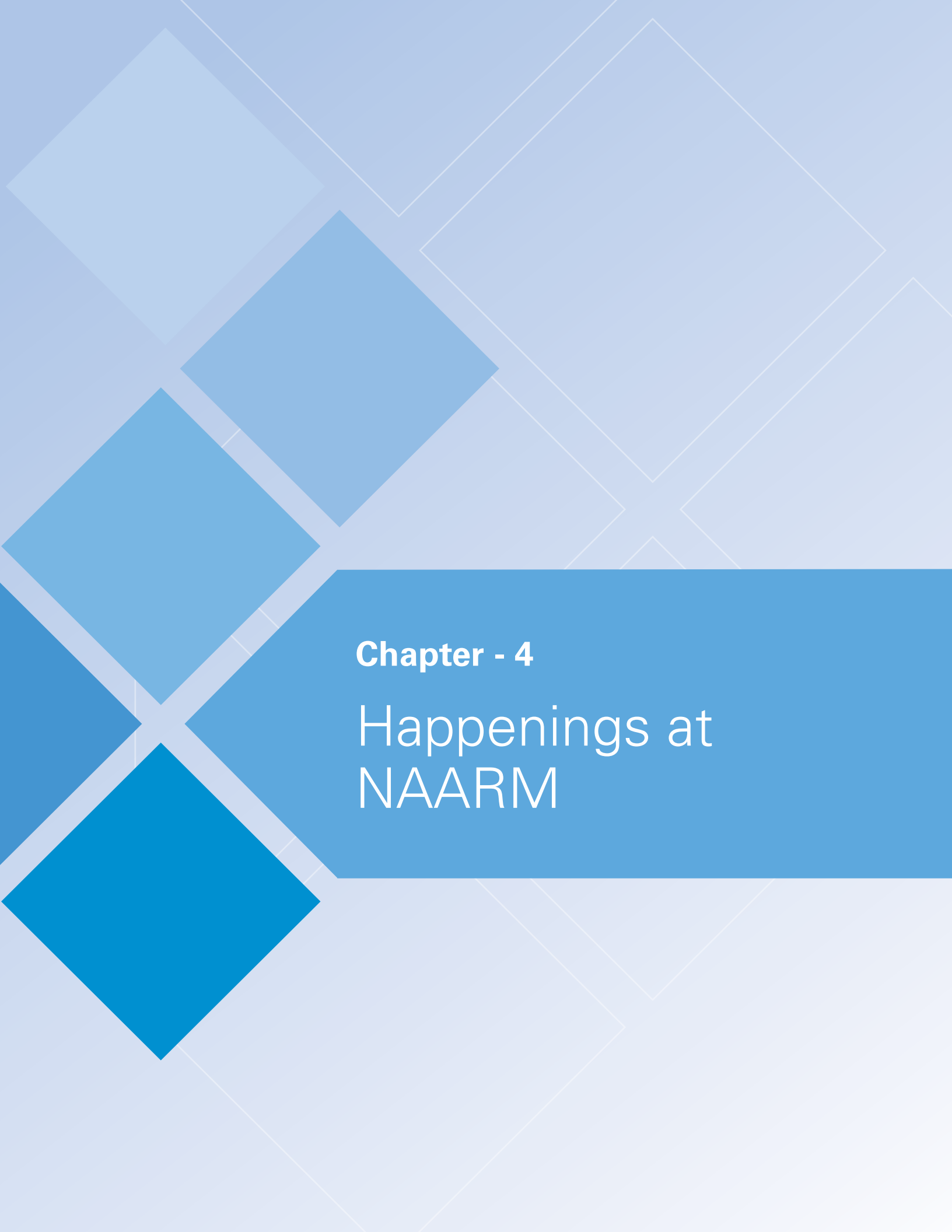
A state of art facilities has been established which is scalable, easily implementable, cost-effective and with less manpower-intensive. The protocols have been constantly refined and upgraded to enhance better quality in content. The protocols include:

- practice of technology enabled pedagogy that contours along the various natural styles of teaching
- methods of content capture
- processes of refinement and storage in appropriate digital formats
- mode of delivery to the end-user either offline/online
- integration of interactivity to the digital content
- repurposing methods of the granular content for purposes like distance, vocational, open access, massive open online courses.



Ethiopian team visiting the centre

The protocols were used to develop model digital courses, offer distance and online programmes, conduct virtual classes, and support some of the inhouse programmes. A total of 6 digital courses were developed. A distance education programme was offered completely in online mode with 18 participants. A total of about 100 streaming hours were provided for virtual classes to support contact sessions of another distance education programme.



Chapter - 4

Happenings at NAARM



Happenings at NAARM

4.1 Accolades

4.1.1 ICAR-NAARM Awarded Rajarshi Tandon Rajbhasha Purashkar



NAARM awarded Rajarshi Tandon Rajbhasha Purashkar

The ICAR-National Academy of Agricultural Research Management (ICARNAARM), Hyderabad, was awarded the ICAR “Rajarshi Tandon Rajbhasha Puraskar Award” for doing the maximum work in the Official Language ‘Hindi’ among the Institutes situated in ‘C’ Region for the year 2015-2016. Dr Ch. Srinivasa Rao, Director, ICAR-NAARM received the Award from Shri Radha Mohan Singh, Hon’ble Union Minister of Agriculture and Farmers Welfare, during the ICAR Foundation Day Programme at New Delhi on 16 Jul 2017. Shri Sudarshan Bhagat, Hon’ble Union Minister of State, Ministry of Agriculture and Farmers’ Welfare, and Dr T. Mohapatra, Secretary, Department of Agricultural Research and Education (DARE) and Director General, ICAR were also present.

4.1.2 NAARM Awarded for Digital Media Initiatives



NAARM awarded digital media initiative

The Public Relations Society of India (PRSI) recognised the digital initiatives of ICAR- NAARM. The academy won the second prize of PRSI National Award – 2017 in the category of “Digital Media – Best Viral Marketing Campaign – 2017”. The award was presented during its 39th All India Public Relations Conference – 2017. Sri Ganta Srinivasa Rao garu, Hon’ble Minister for Human Resources Development, Govt. of Andhra Pradesh presented the award in the presence of Shri M. Srinivasa Rao, Hon’ble Member of Parliament, Anapapalli of A.P. and other officials of PRSI. The award function was held on 15th December 2017 at Hotel Novotel, Visakhapatnam.

4.1.3 NBA Accreditation of PGDMA @ NAARM

The two-year full time Post Graduate Diploma in Management- Agriculture (PGDMA) programme is successfully running at the Academy since 2009. This is AICTE-approved programme. 211 students (including ongoing two batches) got admission in the course over eight years. In its constant endeavour, the Academy is working on all fronts to create a niche and a brand in the area of Agribusiness Management education in the country. In this process, the Academy submitted the Self-Assessment Report (SAR) for its accreditation from National Board of Accreditation (NBA) to validate its quality assurance and continuous improvement. Three-member committee of NBA visited the Academy on Sep 16-17, 2017 to reaffirm the process, structure and governance of the PGDMA programme. The



committee interacted with all the stakeholders i.e., current students, alumni, their parents, recruiters (industry representatives), all the faculty members of the Academy as well as Guest Faculty, technical and administrative staff members. Finally, the PGDMA programme of the Academy got the much-awaited NBA accreditation.

4.2 Special Events

4.2.1 Meeting with Chairman of 7th Quinquennial Review Team (QRT)

The Secretary, DARE and DG, ICAR nominated Dr. Mangala Rai, Former Secretary, DARE and Director General, ICAR as Chairman of 7th Quinquennial Review Team (QRT) of NAARM. A preliminary meeting was conducted on 3rd January 2018. Director, NAARM, Joint Director, HoDs and Head, PME attended the meeting. During the meeting, the QRT Chairman and faculty discussed several issues related to research, education, extension and training.

The Chairman emphasized that research in management is essential to strengthen the Executive Development Programmes (EDP), Management Development Programmes (MDP) and also general management programmes. He stressed upon that NAARM has a very specific role in conducting policy research. Research being greatly linked to higher education, therefore, the process to obtain 'Deemed-to-be University' status for NAARM should be speeded up. Another aspect linked to education is entrepreneurship and development of start-up culture in the country. Therefore, PGDM(Agriculture) was initiated and its yielding very good results. It was also discussed that in all of actions of the Academy, farmers must be the focus area.

4.2.2 Research Advisory Committee (RAC) Meeting

The first meeting of 7th RAC was held on August 24, 2017 to review the research and capacity building activities of the Academy. The chairman of the committee, Dr. S. K. Sopory gave the opening remarks and suggested to include 'Farmers Welfare' in research and training agenda because ultimate goal is food and nutritional security. The achievements on the ongoing research and capacity building activities were reviewed. The chairman in his concluding remarks suggested to take up research in tune of 'New India Initiative' for innovative technologies such as impact of GM technology, agri-waste management, gap in consumer preference and farmers' practices, farmers' constraints, last mile connectivity in extension, poultry feed industry



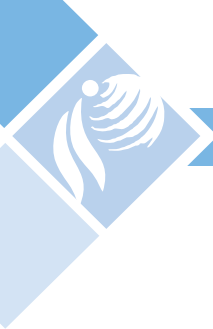
NBA team visit to NAARM



Dr. Mangala Rai interacting with faculty



Dr. S.K. Sopory chairing RAC meeting



mapping, application of drone etc. The following key recommendations emanated from the meeting:

1. The Academy can take up impact studies of capacity building programmes on priority basis, and can publish documents thereof.
2. The Academy can strengthen its research activities through initiating PhD in collaboration/affiliation with University of Hyderabad.
3. The RAC may co-opt a member from the industry.
4. Farmers welfare can be included in vision/mission and subsequently addressed through various capacity building initiatives.
5. NAARM can go from regional to global level and may have collaboration with IFPRI for training programmes on prioritized areas for African and Latin American countries.
6. An interactive workshop can be conducted with farmers.

4.2.3 TIFAC Stakeholders' Consultation Meeting at ICAR-NAARM



Delegates of TIFAC stakeholders' consultation meeting

NAARM holds 'TIFAC Stakeholders' Consultation Meeting under Climate Resilient Agriculture on July 13-14, 2017. Dr Himanshu Pathak, Chairman of the Global Technology Watch Group (WTWG) under the Climate Change Programme of the Technology Information, Processing and Assessment Council (TIFAC), Department of Science and Technology (DST), Government of India delivered the introductory remarks. Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM gave a background of the meeting and said that the meeting of stakeholders is being held across India to formulate policy and action plans for implementation across India. The objectives of this meeting were: Technology scouting and its prioritization in Indian context to support the Government and industry in accelerating actions that can help in the early adoption of climate-friendly technologies in agriculture and allied sectors; and Assessment of technology needs in the adaptation and mitigation areas dealt by other National Missions on climate change under NAPCC particularly related to agriculture mission. The 2nd Stakeholders Consultation Meeting had participants from ICAR Institutes, State Agricultural Universities, MANAGE, International Crops Research Institute for the Semi-Arid Tropics (ICRISAT), NIPHM, NABARD, the Seed Industry, Jain Irrigation, Agri-entrepreneurs and progressive farmers.

4.2.4 Silver Jubilee Conference of Agricultural Economics Research Association (AERA)

The Agricultural Economics Research Association (AERA) based at New Delhi that has a membership of over 1000 agricultural economists and social scientists organized its Silver Jubilee Annual Conference at ICAR- National Academy of Agricultural Research Management (NAARM), Hyderabad during 7-9 November 2017. The theme for the conference was "Doubling Farmers' Income: Options and Strategies".

Shri S. Niranjana Reddy, Vice Chairman, State Planning Board of Telangana



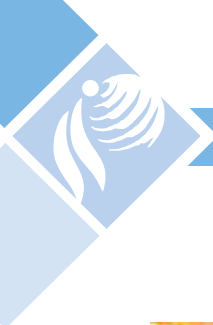
State, inaugurated the conference. In the inaugural speech, he highlighted the various initiatives being taken up by Telangana State Government such as Mission Kakatiya, Vegetable markets, focus on e-NAM etc. which are all aimed at increasing the income of the farmers and also complement efforts of Central sector schemes.

Dr. P. K. Joshi, Director, International Food Policy Research Institute (South Asia) and also the president of AERA highlighted the scope of the conference such as special lectures/ sessions by eminent economists of the country. The conference president Dr. P. G. Chengappa spoke on the experiences of agricultural value chains as an important intervention for effectively increasing farmers' income. He quoted several case studies of basmati rice, poultry, sugarcane, rubber, coffee, tea, maize etc. which have succeeded at various scales in value chain management. He also emphasized on capacity building of value chain players and focus on agribusiness extension.

Dr. Ch. Srinivasa Rao, Director, NAARM spoke on various technology and policy alternatives for achieving the Prime Minister's vision of doubling the farmers' income by the year 2022. Earlier, Dr. R. Kalpana Sastry, Joint Director, NAARM welcomed the dignitaries and the delegates of the conference. Eminent agricultural scientists and economists such as Dr. S.S. Acharya, Dr. K. Mahendra Dev, Dr. Mrutyunjaya, Dr. D.K. Marothia participated in the conference deliberations besides 120 other agricultural economists of ICAR, agriculture universities and other institutions of the country. Dr. Madhur Gautam from World Bank gave special lecture on "Doubling Farmers' Income: Transforming India's Food System". The Conference was overall coordinated by Dr. K. Srinivas and Dr. Ranjit Kumar with the help of various organising committees.



Conference delegates outside NAARM auditorium



4.2.5 ASRB Foundation Day – 2017



Dignitaries lightening the lamp

The 44th Foundation Day of Agricultural Scientists Recruitment Board (ASRB) of Indian Council of Agricultural Research (ICAR) was celebrated on November 1, 2017 at the Academy. The Chairman of the Board Dr. AK Srivastava welcomed the dignitaries such as Dr. R. S. Paroda former Secretary, Department of Agricultural Research and Education, Government of India & DG ICAR, former chairmen and members of ASRB, former member of UPSC, Vice Chancellors of agriculture universities from Andhra Pradesh, Telangana and Maharashtra, Directors of ICAR institutes of AP and Telangana, scientists from various Institutes who graced the foundation day celebrations. Dr. Srivastava also made an emphatic presentation on the historical achievements of ASRB in selecting scientific staff for the ICAR institutes. Later Dr. Gurbachan Singh and Dr. M. Mahadevappa, former chairmen of ASRB and Dr. K. V. Raman former member ASRB spoke on the contributions of ASRB besides making suggestions for reforming the ASRB to adapt to the changing demands of Indian agriculture. The Foundation Day lecture was delivered by Dr. R.S. Paroda, who is also the chairman of the Committee on revamping ASRB. The foundation day celebration was attended by around 200 scientists from various ICAR institutes of the region.

4.2.6 Policy Dialogue on Mainstreaming Biodiversity into Inland Fisheries



Policy dialogue on mainstreaming biodiversity into inland fisheries

A policy dialogue on mainstreaming biodiversity into inland fisheries was held at NAARM during 28-29 Nov 2017. The meeting was jointly organized by the National Biodiversity Authority, Chennai and NAARM, wherein eminent researchers from ICAR, State Agricultural Universities, other central and state research institutions; senior functionaries from state fisheries departments, state biodiversity boards; NGOs and policy makers deliberated on the core biodiversity concerns in the inland fisheries sector. Dr. Dilip Kumar, Former Director, CIFE and Chairman of the Technical Committee for drafting the Inland Fisheries Policy for Government of India presented an overview of the draft policy and indicated that the current dialogue would help in identifying and accommodating specific concerns pertaining to biodiversity. The focus area of deliberations on the first day was inland capture and culture fisheries, while on the second day it was cold water fisheries. Dr. SD. Tripathi, Dr. Maniranjana Sinha, Dr. KK. Vass, Dr. VR. Chitranshi, Prof. Dinesh K. Marothia, Dr. VV. Sugunan, Dr. SK. Soam, Dr. KK. Lal and Dr. AK. Singh shared their experience and expertise during the deliberations. Prof. Brij Gopal, Dr. RS Chouhan and Dr. Bhalki provided their insights on the policy gaps based on their studies. Dr. Thompson Jacob, Consultant with Centre for Biodiversity Policy and Law, NBA provided an overview of the enabling provisions of the Biodiversity Act and moderated the deliberations. Dr. P. Krishnan, Principal Scientist coordinated the event at NAARM.



4.2.7 NAAS- Hyderabad Chapter Meeting

An interactive meeting of National Academy of Agricultural Sciences (NAAS), Hyderabad Chapter was conducted at ICAR-National Academy of Agricultural Research Management, Hyderabad on 12th January 2018. The meeting was Chaired by Dr Panjab Singh, President, NAAS. The Hyderabad based NAAS Fellows viz. Dr EA Siddique, Dr NH Rao, Dr Shikhamani, Dr JS Bantur, Dr Sammi Reddy attended the meeting. The faculty members from NAARM also attended the meeting. The event was coordinated by Dr SK Soam, Joint Director (I/c), NAARM. Dr Sammi Reddy, Director, CRIDA presented the activities done by Hyderabad Chapter. Following are major recommendations:

- Bring out more visibility to NAAS through collaborations with local ICAR institutions and other organizations
- The localized policy level research studies can be conducted by NAAS Fellows
- Contribution to national level planning and policy activities, which are of current national requirement
- The expertise of the retired NAAS Fellows to be harnessed to the maximum level
- The regional chapters to be more proactive in conducting several activities related to local farming/ farmers needs



**Dr. Panjab Singh, President of NAAS with
NAAS-Hyderabad chapter members**

4.2.8 Institute Research Council Meeting

The 20th Institute Research Council meeting was held during March 03-05, 2018 under the chairmanship of Dr. Ch Srinivasa Rao, Director, NAARM. The meeting started with briefing about the proceedings of 19th IRC meeting by Dr. P. Krishnan followed by remarks from the director. Each divisional presentation started with Head's presentation and brief about the research outputs of the respective division. Total 32 research projects were carried out in the current year comprising of 12, 7, 4, 4, 3 and 2 projects from RSM, ABM, XSM, ICM, HRM and ESM divisions respectively.



20th IRC of the Academy

4.3 Visit of Dignitaries at NAARM

4.3.1 Visit of Parliamentary Standing Committee on Agriculture

The Parliamentary Standing Committee on Agriculture headed by Hon. Sh. Hukumdev Narayan Yadav, Member of Parliament along with eleven members of Parliament who are part of the Committee visited ICAR- National Academy of Agricultural Research Management on 11th November 2017. The team comprised of the following:

1. Shri Hukumdev Narayan Yadav (Chairperson)
2. Shri Sanjay Dhotre
3. Prof. Ravindra Vishwanath Gaikwad
4. Md. Badaruddoza Khan
5. Dr. Tapas Mandal



NAARM welcoming parliamentary standing committee with Bathukamma, festival of Telangana

6. Shri Janardan Mishra
7. Shri Devji M. Patel
8. Shri Mukesh Rajput
9. Shri Arjun Charan Sethi
10. Shri Jai Prakash Narayan Yadav
11. Shri Megraj Jain
12. Shri Mohd. Ali Khan
13. Shri Shankarbhai Vegad

Dr. Ch. Srinivasa Rao, Director, NAARM made a presentation to the Standing Committee members on National and International collaborations of ICAR. During the presentation Dr. Rao highlighted various pioneering and path breaking research collaborations with various partners of National Agricultural Research and Education System, memoranda of understanding that are in force with Consultative Group of International Agricultural Research Institutions and leading global universities. He also emphasized on the major achievements of the ICAR institutions in solving issues of food security for all, poverty and income sustainability among farmers and other stakeholders. The members and the Chairman of the committee complemented the efforts of ICAR in general and NAARM in enabling the progress of agricultural research for the benefit of Farmers. The Committee members gave suggestions for making the system more responsive and effective such as enabling better coordination of units/ partners of ICAR and agriculture universities, integrated and single window solutions to farmers' problems etc. Other issues such as shortage of scientific manpower,



Parliamentary standing committee with NAARM faculty members



slow rate of technology diffusion were also discussed in the interactions of the scientists with the Parliament members. The Chairman of the Standing Committee released two publications namely “Forty Years of NAARM” and “NAARM brochure (Hindi version) on this occasion. Hon'ble Sh. Hukumdev Narayan Yadav inaugurated exhibition organized in this connection.

4.3.2 Visit of Hon'ble Union Minister of State for Agriculture and Farmers' Welfare, Shri Sudarshan Bhagat

Hon'ble Union Minister of State for Agriculture and Farmers' Welfare, Shri. Sudarshan Bhagat visited ICAR National Academy for Agricultural Research Management (ICAR-NAARM), Hyderabad on 24 June 2017 and participated in the valedictory function of two training programmes viz., Management Development Programme on, “Leadership Development” and Competence Enhancement Programme on, “Soft Skills and Personality Development for Technical Officers”. Speaking during the interaction meeting at the Academy, Shri. Sudarshan Bhagat said that Indian agriculture is facing newer challenges from the vagaries of changing climatic conditions, leading to availability of excess water for cultivation in some regions and drought-like situation in others. He called upon the agricultural researchers to evolve new technologies to tackle these challenges and to transfer the technologies to the farmers so as to benefit them. Shri. Sudarshan Bhagat distributed certificates to the participants of the ICAR-NAARM training programs. He expressed satisfaction on the feedback of the trainees and complimented the NAARM faculty for their efforts. The Hon'ble Minister also released two of the Academy's publications on the occasion.



Hon'ble MoS, Shri Sudarshan Bhagat releasing PGDMA handbook 2017-18

4.3.3 Hon'ble Union Minister of State for Agriculture and Farmers' Welfare, Shri G.S. Shekhawat Gracing the National Workshop- Let's Listen to Farmers

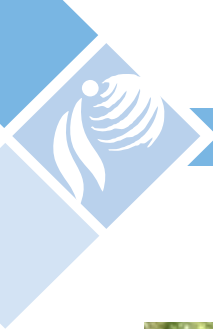
Hon'ble Union Minister of State for Agriculture and Farmers' Welfare, Shri G.S. Shekhawat presided the plenary session of National Workshop-Let's listen to farmers: A workshop on farmers' feedback on doubling farm income by 2022 on Dec 23, 2017. He highlighted the importance of the day, as it coincided with the birth anniversary of former Prime Minister of India Sh. Chaudhary Charan Singh and is celebrated as “Kisan Diwas”. A total of 200 delegates including about 139 farmers from 25 states participated in the workshop.



Hon'ble MoS, Shri G.S. Shekhawat felicitating farmer

4.3.4 Visit of Ethiopian Team of Ministers

H.E. Dr. Eyasu Abraha, Cabinet Minister and H.E. Mr. Tesfaye Mengiste Dori, State Minister, Ministry of Agriculture and Natural Resources, The Federal Democratic Republic of Ethiopia visited the ICAR- National Academy of Agricultural Research Management on 12th March 2018. The main purpose was to discuss the issues of mutual benefit and find out the opportunities of collaboration with ICAR and Ministry of Agriculture and Farmers Welfare, Govt. of India. Dr. Peter Carberry, DDG (ICRISAT) and Dr. Anthony Whitbread, Research Program Director – Innovations Systems in the Drylands, ICRISAT also participated in the interactions.



Ethiopian ministers' walk-through the campus

Dr Ch. Srinivasa Rao, Director, ICAR-NAARM made presentation on success path of DARE, ICAR and Ministry of Agriculture and Farmers Welfare, Govt. of India from green revolution to present day food exporting status of the country. Role of ICAR-NAARM in capacity building and knowledge transfer activities as per the MoUs of ICAR and Indo-African Summit was also highlighted.

Honourable Ministers of Agriculture explained about the agriculture and rural development in Ethiopia and said that 80% of farmers are dependent on agriculture, mostly for subsistence, where illiteracy, climate change are influencing the country. He sought the collaborations with ICAR for higher productivity with technology support, poverty reduction, agri-entrepreneurship. Self-sufficiency in food production is important challenge which can be overcome by capacity building and empowering scientists and farmers of Ethiopia. Some areas like farm mechanization for small farms, eLearning and MOOC courses with joint certification, IPR and technology management, farming demonstration centres were identified as strategic potential collaborations. They visited various infrastructure and laboratory facilities of the Academy.

4.4 Memorandum of Understanding Signed

4.4.1 MoU for Professional Collaboration

The Academy signed MoUs with many professional government as well as private institutions in the year 2017-18. An MoU was signed with TISS, Hyderabad on Sep 25, 2017 to develop co-operation and collaboration in research for capacity building, training and education. On 12th February 2018, a Memorandum of Understanding (MoU) was signed with Indira Gandhi National Forest Academy (IGNFA), Dehradun. The purpose was to enhance technical collaboration in mutual learning in the capacity building areas of agriculture, agroforestry, forestry etc. Under this MoU, the probationers of Indian Forest Service (IFS) will be imparted two-day training on PRA and communication strategies etc, and similarly NAARM trainees will go to IGNFA on consensus basis. The Academy has also signed an MoU with Administrative Staff College of India (ASCI), Hyderabad on 20th March 2018 for the purpose of strengthening training, project monitoring and impact assessment. Under the MoU, the officers of Civil Services will be imparted one-day training at NAARM, similarly, the participants of EDP/MDP will be exposed in one-day module at ASCI on mutually agreed terms.



MoU with Administrative Staff College of India (ASCI), Hyderabad



List of MoUs in 2017-18

S. No.	Other party signing MoU with NAARM	Date	Purpose
1	Tata Institute of Social Sciences (TISS)	25 Sep, 2017	To develop co-operation and collaboration in research
2	Telangana State Medicinal Plant Board (TSMPB)	21 Jun, 2017	For project on “Market Intelligence and Medicinal Plants in Telangana”
3	State Government of Andhra Pradesh	16 Oct, 2017	For development of food processing sector in Andhra Pradesh
4	Indira Gandhi National Forest Academy (IGNFA), Dehradun	02 Feb, 2018	To develop co-operation and collaboration in research
5	Administrative Staff College of India (ASCI), Hyderabad	20 Mar, 2018	Strengthening training, project monitoring and impact assessment of the trainings and projects.

4.5 Celebrations

4.5.1 Foundation Day of ICAR-NAARM

The Academy celebrated 42nd Foundation Day on 1st September 2017. Dr. S.M. Virmani, a renowned natural resources expert, who was the Chief Guest for the function, delivered a ‘Foundation Day Lecture’ on “Reshaping the Indian Agriculture for Doubling Farmers Income”. During his lecture he stressed upon that NAARM should work as instrument of change through a demand driven ‘New NAARM’. The Academy is a unique institution, which deals with human resources development, therefore curriculum needs to address those requirements, which help the scientists to work in such a way that they are able to contribute for doubling the farmers’ income by 2022. Dr Ch. Srinivasa Rao, Director, NAARM in his presidential remarks said that staff has to be committed par excellence with highest work efficiency. The training content has to be innovative and address the changed scenario in the country. He also informed that specialized workshops have been planned and the academy is working on various policy papers. On the occasion, the best worker awards were given by the Chief Guest.



NAARM celebrating its 42nd foundation day

Best Worker Award- 2017

S.No.	Name	Category
1.	Shri MK Samson	Administrative (Male)
2.	Smt Radha Sujatha	Administrative (Female)
3.	Shri K. Verannarsaiah	Technical (Male)
4.	Shri K. Pentiaiah	SSS (Male)
5.	Smt. B. Santhamma	SSS (Female)
6.	Shri L. Narasimha	TSW (Male)
7.	Smt. Andalu	TSW (Female)

Best Employee of the Decade 2006-2016

S.No.	Name	Category
1.	Smt. KK Rukmani Ammal	Administrative
2.	Shri MK Shamshuddin	Technical
3.	Shri R. Sattaiah	SSS
4.	Shri GD Ramulu s/o Danaiah	TSW



Haritha Haram at the Academy

4.5.2 Haritha Haram at ICAR-NAARM

Shri Jogu Ramanna, Hon'ble Minister for Environment, Forests, Science and Technology and BC Welfare, Government of Telangana, planted saplings in the campus of ICAR-NAARM in Hyderabad on Jul 22, 2017, as part of the third phase of Haritha Haram. This is one of the flagship programmes of the Telangana State Government, with the objective of increasing the tree cover in the Telangana State.

He was accompanied by Shri T. Prakash Goud, MLA, Rajendranagar Assembly Constituency and Officials from the Greater Hyderabad Municipal Corporation (GHMC) and the State Department of Environment, Forests, Science and Technology. Dr Ch. Srinivasa Rao, Director, ICAR-NAARM and Dr. R. Kalpana Sastry, Joint Director, ICAR-NAARM received the Hon'ble Minister. The Staff of ICAR-NAARM, Trainees of FOCARS 106, and students of the ICAR-NAARM Post-Graduate Diploma in Management (Agriculture) also participated in the programme.



SPICMACAY concert at NAARM

4.5.3 SPICMACAY Concert of Warsi Brothers (Qawwali)

The ICAR-NAARM hosted the Qawwali concert of WARSI Brothers at NAARM academy on August 28, 2017 via the SPICMACAY platform. The WARSI Brothers Qawwali program enthralled the audience of students of various institutions located at Rajendranagar besides faculty and families. The program was inaugurated by Smt. Usha Rani, IAS, Director General, MANAGE, Rajendranagar. She spoke extensively on the virtues of inculcating the habit of imbibing classical arts. Dr. R. Kalpana Sastry, Joint Director, ICAR-NAARM in her welcome address, pointed out the efforts of the Academy to enable the overall personality culture and development of the students.

4.5.4 Hindi Pakhwada

The Academy celebrated "Hindi Pakhwada" during 1 – 14 September, 2017. On September 14, the "Samapan Samaroh" was organized. Dr. R. Kalpana Sastry, Joint Director and President of occasion spoke about power of the language and motivated all the staff to use Hindi in office work. During the fortnight various competitions were organized and prizes were distributed to all the winners. The essay competition was organized on "Swachha Bharat: How to make country clean.

4.5.5 Swachhta Hi Seva Campaign

Swachhta campaign was organized in the academy during Sep 15- Oct 2, 2017. In connection with the "Swachhta Hi Seva" campaign, essay and elocution competitions were held on September 27, 2017. The competitions were open to all the staff, students and trainees in the academy. A total of 27 contestants participated in the essay competition on the topic "Swachhta Abhiyan: Relevance of Gandhian Philosophy", both in Hindi and English languages. The speech competition was conducted on the topics: 'Cultural mindset change on Swachhta', 'How can I contribute for Swachh Bharat?' and 'Swachhta and ODF' which was contested by 10 participants, in Hindi, English and regional languages.



Winner receiving prize from Director, NAARM



4.5.6 Vigilance Awareness Week in NAARM

The Academy observed the Vigilance Awareness Week during Oct 30-Nov 04, 2017. Various competitions were organized on the occasion. A sensitization workshop on vigilance was organized in which the Chief Guest-Shri PVK Ramana Prasad, Principal Chief Legal Advisor, Anti-Corruption Bureau, Govt. of Telangana delivered a lecture and distributed certificates to winners of various competitions. About 100 persons including staff from other ICAR institutes participated in the workshop. The vigilance awareness campaign was also done at village Vemulanarva by Dr. N. Sandhya Shenoy and Dr. V.K. Jayaraghavendra Rao. The integrity pledge was taken in 'Telugu' language, and farmers were sensitized about digital initiatives of Govt. of India, which helped in curbing corruption in public life.



A glance of participant's drawing by the Director, NAARM

4.5.7 Women Farmers' Day & World Food Day Celebration at Village Balijarala Tanda

Women Farmers' Day & World Food Day was celebrated on October 15, 2017, by Extension Systems Management division of the Academy. An event was organized in Village Baljirala Tanda under Vermulanarva Village, Keshampet Mandal of Ranga Reddy district of Telangana. On this occasion, women farmers were enlightened with knowledge related to the involvement of nutritive food in their diet and various non-farm entrepreneurial ventures that women farmers can go for in order to supplement their farm income. The resource persons were invited from National Institute of Nutrition, Indian Institute of Millet Research and National Institute of Rural Development & Panchayati Raj along with Dr. Surya Rathore & Dr. V.K. Jayaraghvendra Rao participated in the event. Around 65 women and 20 men attended the programme.

4.5.8 Krishi Kala Utsav

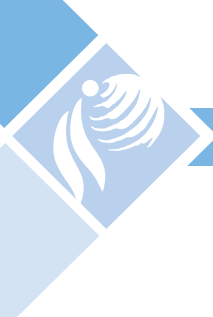
The Academy organized 5 days "Krishi Kala Utsav" during Oct 30 – Nov 3, 2017 with the objective to create works of art to capture agricultural, livestock and fisheries related activities which can adorn the walls of NAARM buildings – Faculty Blocks, Halls of Residences, Guest Houses and Offices etc. The team of student artists led by Shri Deepak Khogre, ACTO (Artist), ICAR-CIFE, Mumbai and Dr. Anant Nikam, Professor, JJ School of Arts, Mumbai were invited to create customized works of art - paintings. Twenty student artists with a background of painting, murals and art design etc. attended the workshop and produced 44 canvas paintings depicting various themes in agriculture and allied sector.



Canvas painting by the participant

4.5.9 Agriculture Education Day

The Academy celebrated Agriculture Education Day on 3rd December 2017 to commemorate the birthday of *Bharat Ratna* Dr. Rajendra Prasad, the First Union Agriculture Minister and first President of Independent India. Inter-school competitions on painting, debate and elocution were done involving seven High Schools in Rajendranagar, Hyderabad in which 200 school students participated actively in these competitions. In the morning 3K Run was organized on PDP to Himayat Sagar Road in Rajendranagar, about 400



3K run organized on Agriculture Education Day



World soil day organized at Vemulanarva village



School children participating in National Science Day competition

college and school students participated in the run.

Dr. David Bergvisnson, Director General, ICRISAT, the Chief Guest reiterated that India has the challenges of malnutrition, lack of micronutrients etc for which top brains are required for innovative agricultural research, therefore talented students must join agriculture education, he said 'Jai Jawan Jai Kisan'. Dr. Ch. Srinivasa Rao, Director, NAARM in his address emphasized the role of teachers and parents for motivating students for taking up agriculture education as a career option. He also stressed that talented students can be an asset in providing quality food, livelihood security and develop secondary agriculture through innovative agri-business ideas.

4.5.10 World Soil Day

World Soil Day was celebrated by ICAR-NAARM, Hyderabad at Vemulanarva Village, Kesmpet Mandal of Ranga Reddy district on 5th December 2017 to stress the importance of soil as a critical component of the natural system and as a vital contributor to the human commonwealth through its contribution to food, water and energy security and as a mitigator of biodiversity loss and climate change. In this programme, about 100 farmers (Vemulanarva and Baljarala Thanda) including NAARM PGDMA students and scientists along with Agricultural department personnel have participated. In this programme awareness about application of soil health card based recommendation of fertilizer was created. Video on soil health was screened before the farmers ensuring its importance. Mr. Anjaiah Yelganamoni MLA of Shad Nagar graced the occasion and Dr. Venku Reddy, President and Executive Director, PRDIS and the guest of honour created awareness about the Government sponsored soil health based programme. Dr. Ch. Srinivasa Rao, Director of ICAR-NAARM presided over this occasion with the remark on the importance of soil testing. He emphasized on integrated management of soil, water, vegetation and biodiversity to enable sustainable agricultural production that is good for the environment and farm profits.

4.5.11 National Science Day

The Academy celebrated 'National Science Day' on 28th February 2018. About 150 students of Sri Chaitanya Techno Schools participated in various scientific learning activities at the Academy on this occasion. The importance of NAARM, its role in the development of the agricultural research management in the country were apprised to the students besides orientation towards the agriculture-related career options. Technology Enhanced teaching learning in agricultural education facility, Technology Incubation Centre along with nursery visit were arranged for the students. On the occasion, essay writing and elocution competitions were organized for the students.



4.5.12 International Women's Day

The Academy celebrated International Women's Day on March 8, 2018. Dr. Nalini Gangadharan, Founder, CAP foundation, Hyderabad, was the chief guest on the occasion. She advocated women not to depend on men and stressed the fact that men like independent women. Through her life experiences, she elicited the learning points. While addressing the gathering, Dr. Usha Verma, Scientist G, ASL Hyd., DRDO, said leadership is about one life influencing the other. Dr Ch. Srinivasa Rao, Director, NAARM said women are so compassionate, adjustable and adoptable to any given situation. Agriculture is no exception wherein women are encouraged to take up the entrepreneurship activities. He advised the working women employees to raise up to the emerging responsibilities on both fronts. Women should get quality time for themselves, which bring them happiness, he added.



International Women's Day at NAARM

4.6 Sports, Yoga and Meditation

4.6.1 Outstanding Performance by NAARM Team in Various Sports Meet

NAARM players Mrs. Rukmini, Mr. MK Samson and Mr. Sham Bahadur participated in ICAR Inter-Zonal Sports Meet held at IARI, New Delhi during April 25-29, 2017. Mrs. Rukmini was adjudged winner in Shuttle Badminton (Singles) and stood second in Shotput and Discus throw. Mr. Samson stood third in Discus Throw and Shotput.

NAARM Contingent participated in ICAR Inter-Institutional Sports Meet at SBI, Coimbatore during October 9-13, 2017. NAARM team participated in Volleyball (Smashing & Shooting), Kabaddi, Table Tennis, Shuttle Badminton, Basketball and Athletic events. Mrs. Rukmini became Winner in Shuttle Badminton (Singles) as well as in Shotput; and stood runner in Table Tennis (Singles), Carom (Singles) and Discuss throw. Mr. MK Samson stood first in Discus Throw and in Shotput; and third in Javelin throw



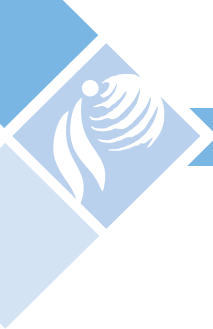
Director, NAARM felicitating the winner

4.6.2 ICAR Inter-Zonal Sports Meet 2017 Organized by ICAR-NAARM

The Inter Zonal Sports Meet of ICAR Institutes was organized by ICAR-NAARM at Hyderabad during 21-25 February, 2018. A total of 430 sportspersons participated in this championship. Events like Foot Ball, Volley Ball (Smashing & Shooting), Basket Ball, Kabaddi were held in the Railway Sports Complex (RSC), South Central Railway, Hyderabad. Indoor games like Badminton, Table Tennis, Chess and Carom were organized in the Sports Complex of the Academy. Athletic events such as running, jumps and throws were held in the RSC ground. The meet was inaugurated on 21st Feb, 2018 by Shri. Alloysius Edwards, Olympian Hockey Player, Indian Hockey Team, Atlanta Olympics. The Indian Agriculture Research Institute won the overall championship award by winning majority events. The overall champion (Men) was Mr. Sumer Katoch of CAZRI, while, in the women category it was Mrs. Rukmani Ammal of ICAR-NAARM. Similarly, the champion athlete (Men) was Mr. E. M. Anees from CPCRI and Mrs. K. Preethi, from CPCRI was the champion (women) athlete. Shri. Maxwell Trevor, Indian International



Oath taking during inaugural function of sports meet



Cyclist was the chief guest for the concluding ceremony. Dr. Ch. Srinivasa Rao, Director ICAR-NAARM presided over the inaugural and valedictory functions.

Performance of NAARM Players

Name	Winner/First	Second	Third
Mrs. Rukmani Ammal	Shuttle Badminton (Singles) Shotput	Table Tennis (Singles) Carrom (Singles) Discus throw	
Mr. MK Samson	Shotput	Discus throw	Javelin

4.6.3 International Yoga Day – 2017

The 3rd International Yoga Day was celebrated at ICAR-NAARM on 21 June 2017. The faculty, students, trainees and staff members of the Academy participated in the yoga from 6.00 to 7.30 am. Dr. Debnath, Chief Medical Officer and Yoga Instructor of the Academy administered the Common Yoga Protocol (CYP) and highlighted the features as well as importance of Yoga in everybody's daily life. The students from Bhartiya Vidhya Bhavan, NIRD also participated in the session. Dr. Debnath highlighted the features as well as the importance of Yoga in everybody's daily life. Dr. Ch. Srinivasa Rao, Director of the Academy, who also participated in the Yoga, addressed the gathering on the occasion and motivated the children and students, in particular, to adopt the basic tenets of yoga and wellbeing in their life.



International Yoga day organized at NAARM



4.7 Campus Beautification of the Academy

Landscape Garden and Nursery Management:

The existing landscape gardens of Hostels, Faculty block, Academic block, Admin. block, Director's office, Auditorium and Lake side garden were taken care. Special focus was given on beautification of landscape gardens of Lake side garden, Auditorium, Director's Office, Admin. Block & Conference hall. The lawns and ornamental plants were maintained properly. The Lake side garden has become an ideal venue for Yoga programmes, Official get-togethers, dinners etc.

Nursery activities include multiplication of ornamental plants like *hibiscus*, *golden durantha*, *bougainvillea*, *almanda* etc. Potted plants consisting of indoor plant spp. Viz. *ferns*, *dieffenbachias*, *dracaena*, *pedilanthus*, *palms*, *aralias*, *eranthemums*, *senciveria*, *kalaonche* (total 1000 Nos.) were developed and arrangement of the same was taken up in the academy during different functions. All the office buildings were decorated with good quality indoor plants.

Orchard Crops:

The orchard crops viz. mango (2.0 ha), sweet lime (1.5 ha), sapota (0.25 ha), aonla (0.25 ha), custard apple (1.0 ha), phalsa (0.5 ha), and other miscellaneous fruits were grown in an area of 6.5 hectares.

Floriculture:

This block consists of *Rose* garden, cut flower species like *tube rose*, *gladioli*, *chrysanthemums* and *golden rod* cultivated in an area of 0.5 ha. The farm has provided its services of floral arrangement during various office functions from this area.

Rainwater Harvesting

Initiative was taken up to harvest the maximum quantity of rain water in the lake situated at the main entrance of the Academy. Expert advice from the CRIDA scientists was taken in this aspect. The recommendations like clearance of obstructions in the catchment area, diverting the water harvested from roof tops, rain water drains etc. were implemented and the lake was full to its maximum capacity by the end of monsoon. Rainwater harvesting

- helped in charging the nearby bore wells during dry season by enhancing the ground water table.
- helped in irrigating the landscape gardens around the lake.
- the lake water and the surrounding lush green lawns have been adding to the beauty of the campus and created an aesthetic ambience in the Academy. It helped in fish culture and revenue generation.

Swatch Bharat Abhiyan (Clean India Mission)

The surroundings of residential quarters, guest houses, surroundings of office buildings, sports complex etc. were cleaned by conducting special drive programmes involving staff, students, trainees, TSWs, contractual workers etc. The campus was declared a plastic free zone.



Roses bushes in bloom in the rose garden



Lake with full of water in monsoon



Mass cleaning programme in the campus

The other cleaning activities included the campus roads and pavements cleaning, cleaning sports ground, tennis court, hostel blocks inside areas. Other activities viz. cleaning of building roof top, painting of avenue tree trunks was also taken up with lime and red oxide, cleaning of all the residential quarters. The contractual garbage collection scheme with the help of *GHMC* is also implemented in the Academy.

Recycling of solid wastes

The leaf litter, grasses, weeds and all the bio-degradable wastes are collected and heaped in the old nursery area. After segregation the farm waste is used for vermicomposting in the old nursery area. The vermin compost so produced is used for nursery plants and landscape garden.

4.8 Glimpses of Celebrations in the Academy



New Year 2018 celebrations at NAARM



Independence Day celebrations at NAARM



Director, NAARM hoisting flag on Republic Day-2018



Ganesh Chaturthi celebration at NAARM

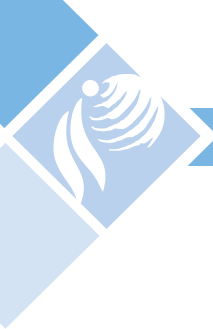


Cultural night, Abhisaar by 106 FOCARS



4.9 Distinguished Guest Faculty from Other Institutes

Guest Faculty Name	Title of the lecture
Dr. B Pandu Reddy Former Comptroller, University of Hyderabad, Hyderabad	Finance Management in AU's
Mr. D. Mallikarjuna Rao Matrusree Studio, Hyderabad	Video editing in Adobe and Camtasia
Dr. D. Rama Rao Former Director, NAARM, Hyd	Developing Winning Research Proposal, Digital Learning in Agricultural Education
Dr. David Bergvisnson Director General, ICRISAT	Achieving Sustainable Development Goals through Partnership in Agricultural Research
Dr. G. Mohan Professor, ASCI, Hyderabad	Blog writing and social media for learning and engagement
Dr. G. Purushotham Rao Professor, College of Vet Sc., Hyderabad	New VCI Regulations: prospects
Dr. Gopal Krishna Director, CIFE, Mumbai	Strategies for enhancement of quality education
Dr. Inigo Arulselvan Assistant General Manager, Corporate Planning Department, NABARD, Hyderabad	Institutional Finance for Agribusiness
Dr. K. Narendar Head, EMRC, Osmania University, Hyderabad	Content Management for Mainstream Media
Dr. K. Pradeep Kumar Dy. Comptroller, PVNRTVU, Hyderabad	General Administration in AUs including Service, Conduct and Leave rules
Dr. K. Veeranjanyulu University Librarian, PJTSAU, Hyderabad	Agricultural Information Resources in the Digital Era
Dr. K. Vijayaragavan Former JD, IARI	Case study Analysis & Preparation
Dr. M. B. Chetti ADG, ICAR	Research and Capacity Development Opportunities for SAUs in ICAR
Dr. M. G. Sivakumar Professor, Tamil Nadu Agricultural University	Activities of market intelligence cell of TNAU -DEMIC and methodology & for forecasting Marketing Intelligence Cases in India
Ms. Piya Bahudur Regional Officer, USIEF	Opportunities & Challenges in Higher Agrl. Education
Ms. Sushumna Faculty, Freelancer, Hyderabad	Useful content creation tools for e-learning
Dr. N. S. Rathore DDG (Education), ICAR	Higher Agricultural Education
Dr. N.H. Rao Former Joint Director, NAARM	Writing and Publishing a Scientific Paper: General Principles

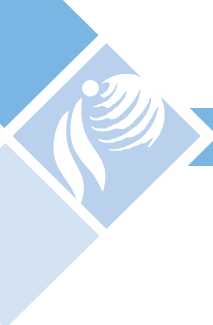


Dr. Neeraj Raj MD, MedSkills Learning Lab	Technology Enhanced Learning in Medical Education
Dr. P. Chandrashekara Director, MANAGE, Hyderabad	Research priorities in extension in agriculture and allied sectors
Dr. P. G. Chengappa Former Vice Chancellor of UAS Bengaluru and Independent Director of M/s Tata Coffee Limited	Researchable Issues in Agricultural Marketing
Dr. P. Manikandan Ex. Head, HRM Division, NAARM, Hyderabad	Human Resource Management in Education
Mr. P. Srinivasulu Naik Comptroller, SVVU, Tirupati	Financial Management and Purchase Procedures in AUs (PSN)
Dr. R. Suresh Senior Consultant, NFDB, Hyderabad	Institutional Arrangement for Agricultural Research and Development
Mr. Ramakrishna Kumar Assistant Manager, A-VIEW Team, Amrita University	Virtual learning, Online classroom through AVIEW
Dr. Rasheed Sulaiman Director, CRISP	Writing for Science Communication: Issues and Challenges
Dr. Ravi B. Gaddagimath Dean, Acharya Institute of Technology, Bangalore	Netiquette
Mr. Ravinder Reddy OPEL Multimedia, Hyderabad	Video module Editing
Dr. S.M. Virmani Retired Eminent Scientist, ICRISAT	What is a Research Project Proposal (Winning Type)
Dr. Sandhya Khode Director, Centre for Education Technology, IIT, Hyderabad	Recent Advances in Digital Education
Dr. Saravanan Raj Director (Agricultural Extension), MANAGE	Social Media for Teaching & Learning
Dr. Suresh Dhamodharan Director, Kisangates Agro Informatics	Mobile Digital Content Development in regional languages : A Case study
Dr. T. Balaguru Former Principal Scientist, NAARM	National Agricultural Research System
Dr. V. Praveen Rao VC, PJTSAU, Hyderabad	Doubling Farmers Income

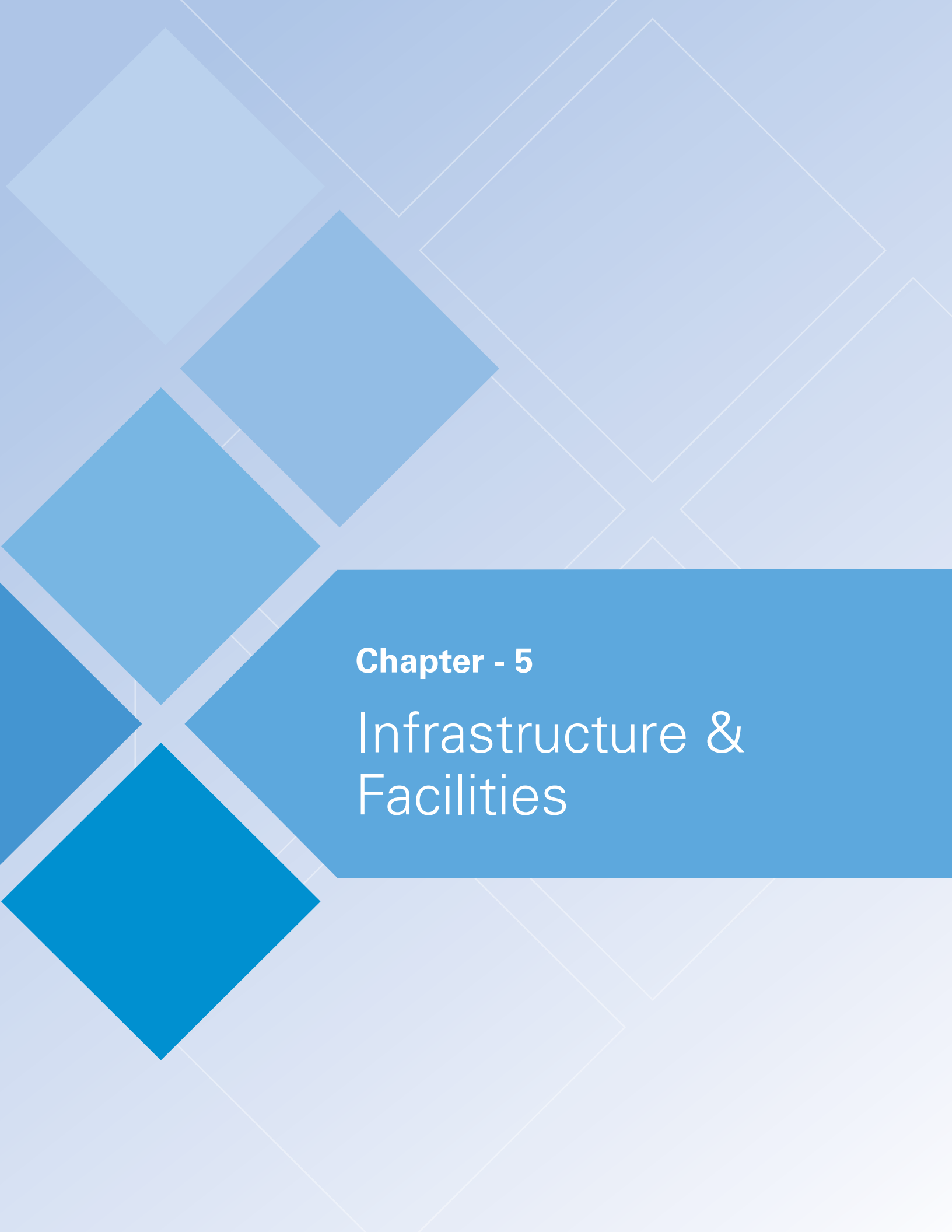


4.10 Visits of Students from Various Agricultural Universities/Colleges

Date of Visit	University/Institute	No. of Students
25-May-2017	3 rd Year B.Sc. (Hons.), Kerala Agricultural University	58
21-Aug-2017	4 th Year B.Sc. (Agri.), PGP College of Agricultural Sciences, Namakkal	83
28-Aug-2017	Final Year B.Sc. (Agri.), RVS Agricultural College, Thanjavur	83
5-Sep-2017	Final Year B.Sc. (Agri.), Imayam Institute of Agriculture and Technology, TNAU	89
7-Sep-2017	Final Year B.Sc. (Hort.), RVS Padmavathy College of Horticulture, Sempatti, TNAU	34
7-Sep-2017	Final Year B.Sc. (Agri.), College of Agricultural Technology, Theni TNAU	124
11-Sep-2017	Final Year B.Sc. (Agri.), JKK Munirajah College of Agriculture Science T.N. Palayam	55
12-Sep-2017	Final Year B.Sc. (Agri.), Agriculture College and Research Institute, Kudumiyamalai Pudukkottai, TNAU	52
13-Sep-2017	Final Year B.Sc. (Agri.), Don Bosco College of Agriculture, (DBCA) Vellore District, (Affiliated to TNAU)	46
13-Sep-2017	Final Year B.Sc. (Agri.), College of Agricultural College and Research Institute TNAU	112
16-Sep-2017	Final Year B.Tech (Bioinformatics), Tamil Nadu Agricultural University	80
25-Sep-2017	Final Year B.Sc. (Agri.), Agriculture College and Research Institute, Madurai TNAU	98
4-Oct-2017	Final Year B.Sc. (Horti), Horticulture College and Research Institute Periyakulam TNAU	86
26-Oct-2017	Final Year B.Sc. (ABM), Tamil Nadu Agricultural University	45
28-Oct-2017	Final Year B.V.Sc. & AH, Veterinary College and Research Institute Namakkal	62
31-Oct-2017	Final Year B.Sc. (Agri.), Agricultural College and Research Institute, Vizhavachanur, Thruvannamalai, TNAU	53
9-Nov-2017	Third Year B.Sc. (Agri.), College of Horticulture, KAU, Vellanikkara	68
14-Nov-2017	Final Year B.Sc. (Home Science), College & Research Institute, Madurai, TNAU	26
15-Nov-2017	Final Year B.Sc. (Agri.), Thanthai Roever Institute of Agriculture & Rural Development TNAU	47
17-Nov-2017	Final Year B.Sc. (Agri.), Adhiparasakthi Agricultural College, Kalavai TNAU	88
20-Nov-2017	Final Year B.Sc., (Kumaraguru Institute of Agriculture Agri.)	58
27-Nov-2017	Final Year B.Sc. (Agri.), Agricultural College and Research Institute, TNAU	52
1-Dec-2017	Final Year B.Sc. (Hort.), University of Agricultural & Horticultural Science Shivamogga	58
3-Dec-2017	Final Year B.Sc. (Hort.), UAHS & College of Horticulture, Hiriyyur	54
6-Dec-2017	Exposure Visit to ATMA by SAMETI UAS, Dharwad	25



15-Dec-2017	Final Year B.Sc. (Agri.), Pandit Jawaharlal Nehru College of Agriculture and Research Institute	92
20-Dec-2017	Exposure Visit to ATMA by SAMEITI UAS, Dharwad	40
22-Dec-2017	2nd Year MBA (Agri.), College of Agriculture, Shivajinagar, Pune	35
27-Dec-2017	Final Year B.Tech (Agricultural Information Technology), Tamil Nadu Agricultural University	20
27-Dec-2017	Schools from class Ixth & Xth, United Public High School, Hasan Nagar, Hyderabad	40
4-Jan-2018	Junior Commissioned officers of Army, Navy and Air force of Ministry of Defence, GOI Institute of Cooperative Management	20
1-Jan-2018	Final Year B.Sc. (Horti.), College of Horticulture and Forestry, Himachal Pradesh	49
5-Jan-2018	M.Sc. (Agri.), College of Agriculture, GKVK, Bengaluru	35
19-Jan-2018	B.Sc. (Agri.), Training Programme at NIPHM Participants	25
22-Jan-2018	B.Sc. (Agri.), SKS College of Agriculture and Research Station, Rajnandgaon, IGKV, Raipur,	20
27-Jan-2018	B.Sc. (Agri. Hort.), University of Agricultural Sciences & Technology of Kashmir (SKUAST-K)	83
29-Jan-2018	B.Sc. (Hort.), College, of Horticulture, Kolar	50
1-Feb-2018	M.Sc (Statistics) Student, Pune University	46
8-Feb-2018	Final Year B.Sc (Hort.), SKUAST-Kashmir	48
Total No. Students Visiting NAARM in 2017-18		2239



Chapter - 5

Infrastructure & Facilities



Infrastructure & Facilities

The Academy is established in a green serene, sylvan and sprawling campus of about 50 hectares. It has various facilities like Faculty Block, Academic Block, Administrative Block, Library, Medical and Health Centre, Sports Complex, Guest houses and Trainees living facilities, Housing and Residential Quarters.

5.1 Library



Journal section of library

The NAARM Library has more than 30,813 books, 73 International journals, 18 International online journals, 78 Indian Journals and 12 online databases / software, i.e. EBSCO Business Source Premier, ABI-INFORM, Indiatat.com, Commodities, Economic Outlook, Patent Optimizer, Total Patent, Nexis.com, Patent Search, OECD iLibrary, STATA/IC14, Orbit.com etc. The Library has many of its publications in digital formats like CD, VCD and its own digital repository (EPrints@NAARM). The Academy is also a member of CeRA, and has access to journals and databases through CSIRO, Springer and Open J-Gate etc. Library management has been completely digitized through the open source software "Koha". The Academy also has a rich repository of online patent search providers, GIS software and other statistical software.



Library facility at NAARM

5.2 Health Center



NAARM Health Centre

ICAR-NAARM runs one health centre headed by Dr. (Maj) A. Debnath and have facilities for daily OPD, Pharmacy facilities, Physiotherapy, Referral service and Day care service for emergency patients.

NAARM Health centre provides preventive as well as curative medical service to its staff and their dependents, students, trainees and about 1200 retired ICAR pensioners including their dependents. The pensioners are provided regular medicines for chronic diseases like Diabetes, Hypertension, heart



diseases, Asthma, Arthritis for a period of two months at a time. On an average 40 patients are being attended by the medical officer daily. For the benefit of the patients including pensioners, health camps including facilities for investigations and consultation with specialist doctors are conducted regularly at health center. In the last calendar year as many as eight health camps were conducted with the help of corporate hospitals and Pharma companies. With the help of NAARM PG-Alumni, the centre also organized Blood Donation camp on Sep 15, 2017.

For recent update of knowledge medical officer attends regularly CME (Continuing medical education) programs, conducted by corporate hospitals. Dr A Debnath attended one training program, "Cancer Research and Training" at Indian Institute of Public Health- Delhi during Sept 18-21, 2017.

5.3 Sports Complex

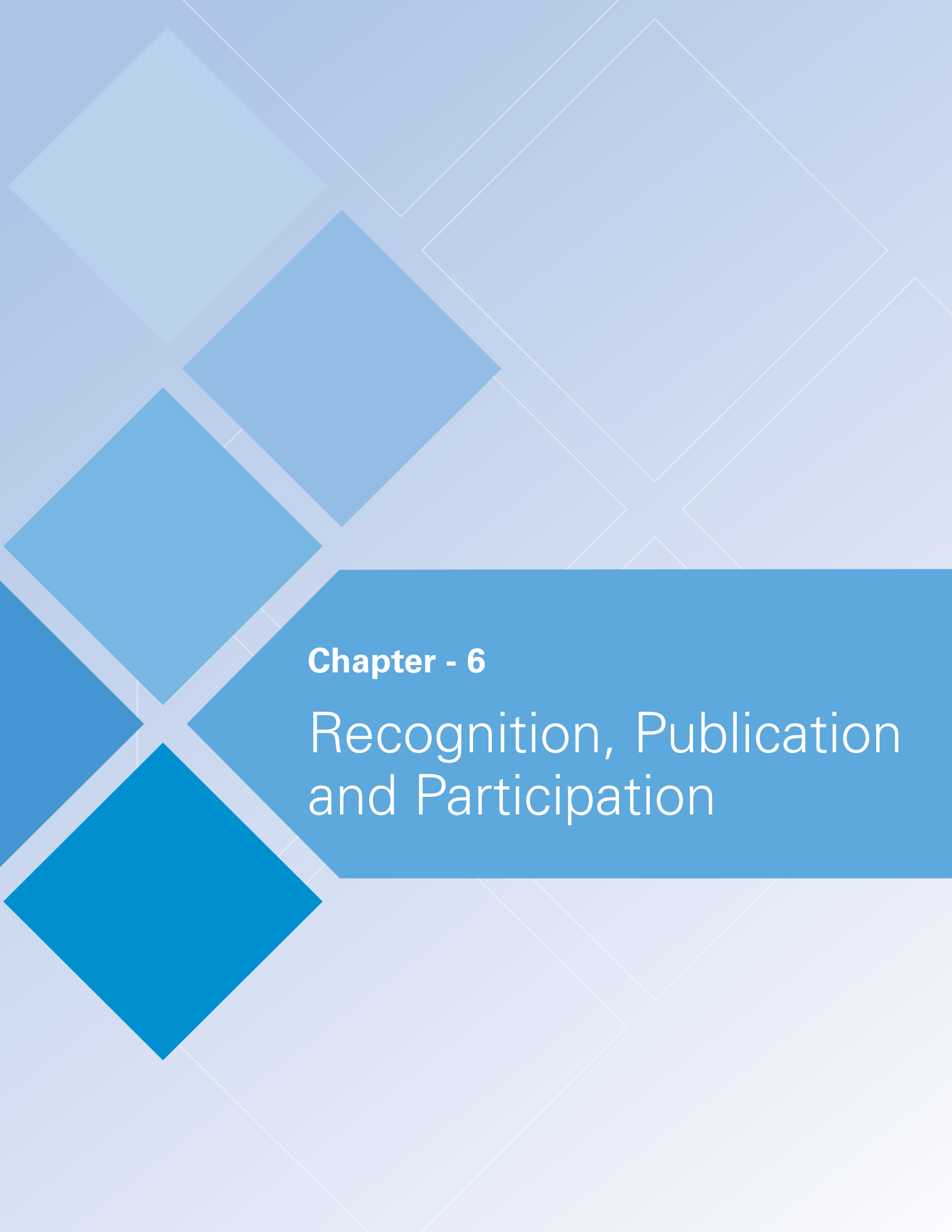
The sports complex at NAARM has several facilities for sports which includes a stadium for outdoor events like cricket, football and various athletic events. One lawn tennis court has also been developed in the year 2017-18. Indoor complex comprises of badminton court, table tennis, billiard/snooker, and gymnasium.



Blood donation camp organized by Red Cross society at NAARM



Various sports facilities at NAARM



Chapter - 6

Recognition, Publication and Participation



Recognition, Publication and Participation

6.1 Recognitions

6.1.1 Visits Abroad



B. Ganesh Kumar presenting in GFRA scientific meeting

B. Ganesh Kumar visited **Zurich, Switzerland** to present the progress made in the project on 'Socio-economic Analysis of Finger Millet in India' under the ISCB network on 'Genetic Enhancement & Bioavailability – Finger Millet (Ragi) organized by Indo-Swiss Collaboration in Biotechnology (ISCB) during Aug 24-25, 2017.

B. Ganesh Kumar visited **Incheon, Seoul, South Korea** to chair the technical session on 'Socio-economics of FMD: Endemic and Non-endemic Settings' and presented a paper on 'Economic Impact of Foot-and-Mouth Disease in India: An Evidence from Andhra Pradesh' organized by Global Foot-and-Mouth Disease (GFRA) and United States Department of Agriculture (USDA) during Oct 25-27, 2017

Ranjit Kumar visited **Bangkok, Thailand** to present paper on 'Agricultural Transformation in Eastern India: Lessons Learnt and Future Opportunities' in International Conference on 'Agriculture and Rural Transformation in Asia: Past Experience and Future Opportunities' organized by IFPRI, New Delhi during Dec 12-14, 2017. The conference was organized by the Regional Strategic Analysis and Knowledge Support System for Asia (ReSAKSS-Asia) facilitated by the International Food Policy Research Institute (IFPRI), the Thailand Development Research Institute (TDRI) and the Leibniz University, Hannover. The event brought together the country and regional stakeholders from research, government, private sector, and civil society to showcase policy research results. There were discussions on policy options to promote cross-country learning and identify emerging policy issues to guide future research.



Ranjit Kumar along with other delegates of conference at Bangkok



6.1.2 Awards Received by the Faculty Members of NAARM

Recipient	Award/Fellowship	Date	Purpose	Awarding body
M. Balakrishnan	Best Paper Presentation Award	10 Mar 2018	Best paper presentation in the conference	INTFES 18, Extension Education Society, TNAU, Coimbatore
N. Sivaramane	Best Paper Presentation Award	31 March 2018	Beat paper presentation in the conference	Endling Conferences, Dhanbad, Jharkhand
P.Venkatesan	Best Presentation Award	12 May 2017	Beat paper presentation in the conference	Himalayan Scientific Society for Fundamental and Applied Research
P.Venkatesan	Outstanding Achievement Award	31 March 2018	For outstanding research achievement	Endling Conferences, Dhanbad, Jharkhand
P.Venkatesan	Best Poster Award	10 March 2018	Best Poster Award in the conference	INTFES 18, Extension Education Society, TNAU, Coimbatore.
Surya Rathore	ISEE Fellow Award	30 November 2017	For contribution in the field of Extension Education	Indian Society of Extension Education, IARI, New Delhi

6.1.3 Honours and Awards Received by the PGDMA Students

Name of Student	Award/ fellowship	Date	Occassion	Awarding body
PGDMA(2016-2018)				
Ms.Meenakshi Chauhan and Ms. Manisha	2nd Prize	02/09/2017	Krishi Chanakya-Case Study Competition	MANAGE, Hyderabad
Mr.Ramagoni Mukesh and Ms.Naushaba Khatoon	National Finalist			
Mr. S.Kavinraaj and Mr.Siddharth Bhattacharya	2nd Prize	13/01/2018	"Madhyam"-Ralationship Marketing event	N.L Dalmia Institute of Management Studies and Research, Mumbai
Ms. Naushaba Khatoon	3 rd Prize	11/03/2018	Best Manager	VAMNICOM, Pune
PGDMA(2017-2019)				
Ms. Sayyeda Salima Asra, Mr. Romit Kumar Singh, Ms. Merin Maria Thomas, Mr. Rahul Kumar Singh, Ms. Sruthy Antony	2 nd Prize	07-12-2017 to 09-12-2017	Game Changer in Agriculture, EIMA Agrimach India 2017, IARI, NEW DELHI	Jointly Organized by FICCI, INDIA & FEDERUNACOMA, ITALY
Ms. Anushree, Mr. Yogesh R, Mr. Dhivagar B, Mr. Shailendra Nageshwar, Ms. Manjushree S N	3 rd prize			
Ms. Usha B, Mr. Mallikarjun G	2 nd Prize	23-01-2018	Agri Innovatia –B Plan Competition	ASPEE Agribusiness Management Institute, Navsari Agricultural University, Gujarat
Mr. Adapa Teja, Ms. Karthika R, Mr. Sheshank Yadav	2 nd Prize	23-01-2018	Agri Innovatia- Case Study Competition	
Mr. Sheshank Yadav	2 nd Prize	23-01-2018	Agri Innovatia-Collage Making	
Mr. Rahul Kumar Singh, Ms. Bhargavi D	2 nd Prize	06-02-2018 to 07-02-2018	Ideators'18 - B Plan Competition	Jansons School of Business, Coimbatore



6.1.4 NAARM in Media

1. Srinivasarao, Ch. 2017. Farm pond based training and capacity building to Agriculture Scientists at National Academy of Agricultural Research Management (ICAR-NAARM). Andhra Jyothi, 14 November 2017
2. Srinivasarao, Ch. 2017. News item about the visit of Parliamentary Standing Committee on Agriculture to ICAR-NAARM on November 11, 2017, Rythunestham, an agricultural family monthly Telugu magazine, Vol.13(5), January 2018.
3. Srinivasarao, Ch. 2017. Press Meet on National Workshop on Farmers' feedback on Doubling Farm Income by 2022 at ICAR-NAARM, Hyderabad on December 22-23, 2017. The Hindu National Newspaper and Namasthe Telangana Telugu Daily Newspaper, December 22, 2017.
4. Srinivasarao, Ch. 2017. News item about the National Workshop on Farmers' feedback on Doubling Farm Income by 2022 at ICAR-NAARM, Hyderabad on December 22-23, 2017. Eenadu, Andhra Jyothi, Andhra Prabha. Namasthe Telangana, Saakshi Telugu Newspaper Hindu Newspapers, dated December 23, 2017.
5. Srinivasarao, Ch. 2017. Video coverage of National Workshop on Farmers' feedback on Doubling Farm Income by 2022 at ICAR-NAARM, Hyderabad on December 22-23, 2017, HMTV Channel <https://youtu.be/GDzRUY7S09c>.
6. Srinivasarao, Ch. 2017. Video coverage of National Workshop on Farmers' feedback on Doubling Farm Income by 2022 at ICAR-NAARM, Hyderabad on December 22-23, 2017, Bigbusinesshub.com, an Online Business Channel, <https://youtu.be/-00Nqc0Kgnc>; <https://youtu.be/6-JffqDoLvU>.
7. Srinivasarao, Ch. 2017. News item about the Agricultural Education Day celebrations held at ICAR-NAARM on December 3, 2017 on the eve of Agricultural Education Day, The Hans India, English daily news paper dated December 4, 2017 and webindia123, an online web news portal, <https://news.webindia123.com/news/articles/india/20171203/3227159.html>.
8. Srinivasarao, Ch. 2017. News item about the 42nd ICAR-NAARM Foundation Day Celebrations at ICAR-NAARM on September 1, 2017, Andhra Jyothi Telugu Daily Newspaper dated September 2, 2017.
9. Srinivasarao, Ch. 2017. News item about the 42nd ICAR-NAARM Foundation Day Celebrations at ICAR-NAARM on September 1, 2017, Rythunestham, an agricultural family monthly telugu magazine, Vol.13(5), January 2018.
10. Srinivasarao, Ch. 2017. News item about the visit of Hon'ble Union Minister of State for Agriculture, Sri Sudarshan Bhagat to ICAR-NAARM, Hyderabad on June 24, 2017, United News of India, dated June 25, 2017; <http://www.uniindia.com/evolve-new-agricultural-technologies-and-transfer-to-farmers-sudarshan-bhagat/states/news/911317.html>.

- సైన్స్ దినోత్సవం సందర్భంగా వ్యాసరచన పోటీలు



ವಿದ್ಯಾರ್ಥಿ ಬಹುಮತಿ ಲಂಡನ್‌ನಲ್ಲಿ ನಾರ್ತ್ ಡೈರೆಕ್ಟರ್ ಕ್ರಿನಿವಾನ್‌ರಾವು

రాజేంద్రనగర్, న్యూనటుడే: రాజేంద్రనగర్‌లోని జాతీయ
వ్యవసాయ మరియు యాజమాన్య సంస్థ(నార్సి)లో సైన్స్
డిగ్రీల్లోని పేరుకలసి బుధవారం మునగా నిర్వహించారు.
ఈ సందర్భంగా పాఠశాల విద్యార్థులకు వ్యాసరచన పోటీలు
ఉర్రవ్వారు. అనంతరం విజ్ఞానరత్న పురస్కారాలు

විද්‍යාඥයන් පර්යේෂණයන් සිදු කරමින් සිටින අතර, මිනිසාගේ ස්වභාවය සහ ජීවිතයේ විවිධ අංශවලට එහි බලපෑම පිළිබඳව අධ්‍යයනයන් සිදු කරමින් සිටිති. මෙම පර්යේෂණයන් මගින්, මිනිසාගේ ස්වභාවය සහ ජීවිතයේ විවිධ අංශවලට එහි බලපෑම පිළිබඳව අධ්‍යයනයන් සිදු කරමින් සිටිති. මෙම පර්යේෂණයන් මගින්, මිනිසාගේ ස්වභාවය සහ ජීවිතයේ විවිධ අංශවලට එහි බලපෑම පිළිබඳව අධ්‍යයනයන් සිදු කරමින් සිටිති.



ನಿಮ್ಮ ಕೆಲಸವನ್ನು ಮುಂದುವರಿಸುವಲ್ಲಿ
ಪ್ರಾಜೆಕ್ಟ್ ನಿರ್ವಹಣೆಯು ಸಹಾಯಕವಾಗಿದೆ.



ప్రేమంగీ శాస్త్రవేత్తలను చివరీ
రావీలానె.

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ನೀವು ಈ ಹೊತ್ತಿನಲ್ಲಿ ಒಂದು ಸರಿಯಾದ ಆರೋಗ್ಯಕರ ಆಹಾರವನ್ನು ತಿಳಿದುಕೊಳ್ಳುವುದು ಅತ್ಯಗತ್ಯ. ಇದು ನಿಮ್ಮ ಆರೋಗ್ಯವನ್ನು ಸುರಕ್ಷಿತವಾಗಿಡುತ್ತದೆ.

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Reviews, 12 figures

Reviews, 12 figures

(विद्यमान मन्त्री)
यह विधायक कोटावली
का है। भविष्य में
कोटावली राष्ट्रीय युवासेवा
मंडल संस्थापन अधिनियम के
अनुसार कोटावली राष्ट्रीय
सेवा समिति स्थापनी की
जाएगी।

आई नारी ऐम बिजनेस
कानूनी, वित्तियारूपी
द्वितीय प्रतीति अधिनियम
एक विशेषता धरोहर के रूप
में प्रकाश अर्पित अन्वय



संस्थानकी अधिष्ठाता जी, अधिका-
र्यम विज्ञानिक डॉ. सुभाष कुमारी
जी, सती रायू जी आदि-आदि
प्राचार्य जी स्वयं संस्थान की जी
अतिथिगत रूप से प्रथम विज्ञान
अनुष्ठान पर मेरी बहुत अतिथि
सम्मानना की जा रही थी।
मेरे साथ विज्ञान अधिष्ठाता जी की
आधीन की विज्ञान पर संस्थानिक
की प्रशंसा, विज्ञान क्षेत्र में मेरी
विशेषताएँ तथा प्राचार्य संस्थान की
पदाविज्ञान। अधिष्ठाता जी विज्ञान
में प्राचार्य 200 की जी अधिका-
र्यम विज्ञानिकों में प्राचार्य विज्ञान।

పంజాగుట్ట: నేషనల్
అకాడమీ ఆఫ్
అగ్రికల్చర్ రిసెర్చ్
మేనేజిమెంట్
(నారమ్) ఆర్గ్
ర్యంలో శుక్ర, శని
వారాలలో



మాట్లాడుతున్న నారమ్ డైరెక్టర్ శ్రీనివాసరావు

(ఈనెం 22, 23వ తేదీలో) రైతుల ఆదాయం రెట్టింపు చేసేందుకు రైతుల ఫీడ్బ్యాక్ అనే అంశంపై జాతీయ సదస్సు నిర్వహిస్తున్నట్లు నారమ్ డైరెక్టర్ తీసి వాసరాయ్ తెలిపారు. గురువారం సోమూతిగూడ ప్రసిక్షన్లో విశేషరం సమావేశంలో మాట్లాడుతూ.. 2022 లోపు రైతుల ఆదాయం రెట్టింపు కావాంని ప్రధాని చిట్టిన పిలుపుమేరకు తీసుకొచ్చాల్సిన ప్రణాళికల రూపొందించేందుకు ఈ సదస్సు నిర్వహిస్తున్నట్లు తెలిపారు. 23వ తేదీన కేంద్ర వ్యవసాయ సన్నాహ కార్యమంత్రి గణేంద్ర సింగ్ మోహన్ హాజరుకానున్నట్లు తెలిపారు.

366 ರನ್ ಪ್ರಾರಂಭಿಸುತ್ತಿ, ಸಾರ್, ಧರ್ಮರ ಕ್ರೀಡಾವಾಸರಾಢು

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విద్యార్థిని పై ప్రశంసించడం కొరతలేదు, శాస్త్రవేత్తలు

[illegible]

2017年2月24日

2017年2月24日



ရွှေတိဂုံမြို့ ငါးမင်္ဂလာ အောင် ဥပဇ္ဈိတ အထကတံ ဘုရား တစ်ဆူတည်တော်မူ

[illegible]

გარდა, ასევე და გრძელდება მუშაობა



कक्षा 10 के छात्रों के लिए, यह पुस्तक एक अच्छा संदर्भ है।

[illegible]

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6.1.5 TV/Radio Talk delivered

S. No.	Name of the person who delivered	Topic of the talk	Date	Name of the Channel and the Programme
1	N. Sandhya Shenoy	Kisan 2017	01 August 2017	Developed for "Siddhi Se Sankalp" programme for the ZPD 5 (ATARI, Hyderabad) which was screened in all the KVKs

6.1.6 Invited Lectures Delivered by the Faculty

Name of Staff	Lecture	Programme/Organized by
A. Dhandapani	Multivariate Analysis	CAFT training on "Quantitative Methods for Agricultural Policy Analysis" by IARI, New Delhi on 05 Feb, 2018
B. S. Sontakki	Content writing	Refresher Course by MANUU, Hyderabad on 30 Jun, 2017
	Communication as a tool for extension professionals	ICAR Sponsored Summer School by BHU, Varanasi on 10 Jul, 2017
	Introduction to concept note preparation	Off-campus Programme on "Competency Enhancement for efficiency and Effectiveness at Workplace" for Faculty and Scientists by VNMAU, Parbhani on 07 Sep, 2017
	Research Paper writing	
	Winning research proposal writing	
	Content writing	Refresher Course by MANUU, Hyderabad on 15 Sep, 2017
	Business writing skills	Orientation Programme by NFCL, Hyderabad on 24 Sep, 2017 and 17 Feb, 2018
	Presentation Skills	
	Addressing gender issues through effective communication using ICTs	Model Training Course by UAS, Raichur on 25 Nov, 2017
	Communication skills (Presentation and writing)	Early Career Motivation Course" for Faculty (Assistant Professor) of UAS, Dharwad by UAS, Dharwad on 28 Nov, 2017
	Communication Skills for Effective Teaching	
	Student Centred Learning	
	Oral Communication Exercise	
Ch. Srinivasa Rao	Adaptation of Agriculture to Climate Change	National Consultation on SDG2 : Zero Hunger by Niti Aayog, New Delhi on 13 Apr, 2017
	Climate Risks	National Conference on "Sustainable Development Goals: Preparedness and Role of Indian Agriculture" by TAAS-IFPRI in collaboration with ICAR, at NASC Complex, New Delhi on 11-12 May, 2017
	Doubling of Farmers' Income	NAAS National Symposium by NAAS, at BCKV, Kalyani on 08-09 Jun, 2017



	Emerging Challenges and Good Agricultural Practices for Andhra Pradesh Agriculture	Three Day Residential Training Programme on “Best Practices in PPP” by APHRD Institute, Bapatla, Andhra Pradesh on 03 Jul, 2017
	Addressed the students of new academic year	Inauguration of new academic year by TISS, Hyderabad on 11 Jul, 2017
	Keynote address	Brainstorming Workshop on ‘National Soil Management Policy’ by MANAGE, Hyderabad on 20 Jul, 2017
	Lecture on Disaster Management	Students of TISS by TISS, Hyderabad on 07 Aug, 2017
	Presentation on ‘Potassium for mitigation of biotic and abiotic stresses in plants’	International Conference on ‘Advances in Potassium Research for efficient Soil and Crop Management’ by IPNI, FAI, TAAS, CIMMYT, ICAR and IARI, at New Delhi on 28 Aug, 2017
	Invited speaker on the theme: Preparing the Producer for Climate Change: Role of Adaptation Technologies in the Fight for Food Security	Conference on Technologies and Innovations in Climate Smart Agriculture (CSA): Practice or Perish by National Skills Foundation of India, at New Delhi on 12 Oct, 2017
	Presented a paper on ‘Soil Information System’	AP AgTech Summit by CII and Bill & Melinda Gates Foundation, at FinTech Capital and Smart City, Visakhapatnam on 16-17 Nov, 2017
	Dr M.V.Shantaram Memorial Lecture on Soil Organic Carbon Sequestration Technologies and Implementation Issues	On the eve of World Soil Day organized Indian Society of Soil Science, Hyderabad Chapter by PJTSAU, Hyderabad on 05 Dec, 2017
	Keynote address on Strategies for Climate resilient technologies for sustainable vegetable production in India.	National Conference on Food and Nutritional Security through Vegetable Crops in relation to Climate Change by Indian Institute of Vegetable Research, Varanasi on 11 Dec, 2017
	Climate Change Adaptation Planning for Indian Agriculture – Experiences and Way Forward	Livelihood Asia Summit on Climate Related Risks in Indian Agriculture: Challenges and Opportunities by New Delhi on 13 Dec, 2017
	Presentation on Opportunities for Carbon Sequestration in Rainfed Production System	International Seminar on Global Climate Change, Implications for Agriculture and Water Sectors by Four agricultural universities in Maharashtra and WALMI, Aurangabad, at Mahatma Gandhi Mission College Complex, Aurangabad on 14-15 Dec, 2017
D. Thammi Raju	Overview of Educational Technology	Off-campus Programme on “Competency Enhancement of Faculty and Scientists of ICAR-NDRI by ICAR - National Dairy Research Institute , Karnal on 25-26 May, 2017
	Innovative Teaching Methods	
	ICTs in Teaching – Learning – Experience Sharing	
	Curriculum Design and Lesson Planning	



	Innovative Teaching Methods	Off-Campus Programme on Competency Enhancement for Efficiency and Effectiveness at Workplace for Faculty and Scientists of VNMAU, Parbhani by Vasantrya Naik Marathwada Krishi Vidyapeeth , Parbhani on 04-05 Sep, 2017
	Overview of Educational Technology	
	Curriculum Design and Lesson Planning	
	Innovative ICT initiative in agriculture and allied sectors	ICTs for Effective Knowledge and Extension Delivery by Extension Education Institute , Hyderabad on 07 Oct, 2017
	ICT in Livestock Extension – Research Evidences	Model Training Course on Improving the core competencies of Women Livestock Extension Professionals, 22-30, October, 2017 by College of Veterinary Science , Gannavaram, Andhra Pradesh on 24 Oct, 2017
	Online Education in Agriculture	Model Training Course on “Capacity building of Extension Professionals on Emerging Digital Extension Strategies” by Professor Jaya Shankar Telangana State Agricultural University, Hyderabad on 21 Nov, 2017
	Innovative Teaching Methods	Early Career Motivation for faculty and Scientists of UAS, Dharwad by University of Agricultural Sciences, Dharwad on 28 Nov, 2017
	Oral Communication	
	Overview of Educational Technology	Faculty Development Programme on Enhancing Efficiency and Effectiveness at Workplace by Dhanwate National College, Nagpur of Shri Shivaji Educational Society, Amaravati, Nagpur on 06-8 Feb, 2018
	Innovative Teaching Methods	
	Micro-Teaching Exercise	
	Curriculum Design and Lesson Planning	
	Teaching Effectiveness	
	Innovative Teaching Methods	Orientation Training programme for the newly recruited Assistant Professor of TANUVAS by Tamil Nadu Veterinary and Animal Sciences University , Chennai on 13 Mar, 2018
	Curriculum Design and lesson Plan	
	Innovative Teaching Methods	Short term training on Capacity Building and Stress Management by Veterinary College, Karnataka Veterinary and Animal Sciences University, Bidar on 24 Mar, 2018
	Curriculum Design and lesson Plan	
	Oral Communication	
G. R. K. Murthy	ICT in Agriculture Education	ICT Application in Changing Face of Agriculture by Birsa Agricultural University, Ranchi, Ranchi on 19 Jan, 2018
	Education and Training in the Digital Era: Online education platforms for extension education	National Workshop on Agricultural Extension: Time to Change by MANAGE, Hyderabad on 16 Feb, 2018
	E-learning	National Conference on E-learning and Technologies: Prospects and Challenges by MANUU , Hyderabad on 20-21 Feb, 2018
	ICTs for agri entrepreneurship Education	National Conference on Digital and Engineering Technologies for Precision Agriculture by College of Agricultural Engineering, ANGRAU, Bapatla (AP) on 26-28 Feb, 2018
K. H. Rao	Understanding Roles and Responsibilities for Personal and Organizational Effectiveness	HRD Training Program on Personal and Work Excellence by ICAR-IIOR, Hyderabad on 15-21 Nov, 2017



K. Kareemulla	Role of Animal Husbandry and Dairying in Doubling farmers' income – Challenges and Strategies	Developing Competency for Harnessing New Agribusiness Opportunities for Doubling Farmers' Income by School of Agribusiness Management, College of Agriculture, PJTSAU, Hyderabad on 21-27 Nov, 2017
	Personality for teamwork	Workshop by NFCL, Kakinada on 08 Feb, 2018
	Role perception	Bloom lectures by Netafim, Hyderabad on 24 Feb, 2018
	Developing Indicators for Monitoring and Evaluation	Project Management by MANAGE, Hyderabad on 16 Jun, 2017
	"Concept of producer organizations and their success"	Marketing Extension in Agriculture and Allied Sectors by EEI, Hyderabad on 29 Aug, 2017
	Developing Indicators for Monitoring and Evaluation	Tools on Monitoring, Evaluation and Impact Assessment of Rainfed Technologies and Agricultural Development programmes by ICAR-CRIDA, Hyderabad on 06 Nov, 2017
K. Srinivas	Augmenting Community Infrastructure for Ensuring Vibrancy and Income Stability of farmers	Training Programme on Doubling Farmers Income by MANAGE, Hyderabad on 08 Dec, 2017
	Agri Startups, Agri Fintechs and Agri Finance: Shape of things to come	National Conference on "Future of Agrifinance 2030 during April 24-25, 2017 at College of Agricultural Banking (CAB), Pune, Maharashtra
M. B. Dastagiri	WTO & Oilseeds: World markets, Trade and Policies	Guest Lecture by Indian Institute of Oilseeds Research, Rajendranagar, Hyderabad on 14 Feb, 2018
M. Krishnan	Suitable models for rainfed aquaculture	Entrepreneurship Development in Rainfed Aquaculture by MANAGE, Hyderabad on 15 Dec, 2017
Manju Gerard	Role of Intellectual Property Rights in leveraging Public Private Partnerships	National Conference on "Food and Nutritional Security through Vegetable Crops in Relation to Climate Change" by Indian Society of Vegetable Science, ICAR-IIVR, Varanasi on 11 Dec, 2017
N. Sandhya Shenoy	ICTs for empowering farm women	Model Training Course on "Capacity building of Extension Professionals on Emerging Digital Extension Strategies" for the officers of development departments of all states of India by EEI, Rajendranagar, on 25 Nov, 2017
	ICTs for empowering farm women	Training programme on ICTs for Effective Knowledge and Extension Delivery for the Officers of Development Departments of Southern states i.e. Andhra Pradesh, Telangana, Karnataka, Tamil Nadu, Kerala, Odisha, Puducherry, Andaman & Nicobar Islands and Lakshadweep by EEI, Rajendranagar, on 06 Dec, 2017
N. Sivaramane	Modelling human decisions in adoption of agricultural technologies – LOGIT, PROBIT and TOBIT" and "Conjoint Analysis"	CTCRI-AESA Workshop on "Good Practices in Quantitative Social Science Research: A journey from conceptualization to research utilization" during August 7-12, 2017 by ICAR-Central Tuber Crops Research Institute, Thiruvananthapuram, Kerala on 10 Aug, 2017
	Logistic Regression	Qualitative and Quantitative techniques in analyzing of Agriculture Research Data by MANAGE, Hyderabad on 27 Feb, 2018
P. D. Sreekanth	GIS Applications on Mobiles for Field Workers	Workshop on Digital Field Book by ICAR-Indian Institute of Millets Research, Hyderabad on 30 Nov, 2017
P.Venkatesan	Tools for facilitation skills for community mobilization	Community Forestry and JFM for the in-service SFS officers of State Forest Department by Central Academy for State Forest Services, GOI, Ministry of Environment and Forest, Coimbatore, on 31 Oct, 2017



S. K. Soam	Agriculture and TRIPs	Exploring opportunities for agriculture under WTO by National Institute of Agricultural Extension Management (MANAGE), Hyderabad during 12-13 Jun, 2017
	GI and Implications for Indian farmers	
	Technical barriers to trade in agricultural commodities	
	Logical Framework Approach	Project Management Skill for officers of agriculture and allied sector by National Institute of Agricultural Extension Management (MANAGE), Hyderabad on 14 Jun, 2017
	Writing winning research proposals	Skill and competency enhancement week by ICAR-National Bureau of Fish Genetic Resources (NBFGR), Lucknow on 02 Nov, 2017
	Protection of Plant Varieties and Farmers Rights	One-day training programme on Protection of Plant Varieties and Farmers Rights by ICAR-National Institute of Abiotic Stress Management (NIASM), Baramati on 06 Dec, 2017
	Writing winning research proposals	Competency enhancement of NIASM scientists by ICAR-National Institute of Abiotic Stress Management (NIASM), Baramati on 07 Dec, 2017
S. Senthil Vinayagam	Market Led Extension	Training Programme on “Linking Farmers to Market-Extension Strategies” by MANAGE, Hyderabad on 02 May, 2017
	Technology Enhanced Learning	Summer School on Dynamics of In-Tension of Ex-Tension by BHU, Varanasi on 20 Jul, 2017
	Market led Extension	Training Programme on “Linking Farmers to Market-Extension Strategies” by MANAGE, Hyderabad on 02 Aug, 2017
	ICT in Agriculture and Rural Development	National Training Programme on “Strategies for E-Governance for Better Service Delivery by NIRD&PR, Hyderabad on 11 Sep, 2017
	Competency enhancement for technology transfer	Model Training Course (MTC) on “Improving the core competencies of women livestock extension professionals” by College of Veterinary Science, Gannavaram, Vijayawada on 27 Oct, 2017
	Grassroots level Innovations for rural entrepreneurship and Strategies	Winter School on Farmers Empowerment and Entrepreneurship Development in Agriculture by UAS, Bangalore on 06 Nov, 2017
	Appraisal Techniques in Project Planning	International Training programme on Planning and Management of RD Programmes by NIRD&PR, Hyderabad on 01 Jan, 2018
	Extension Programme Planning, Need and importance - An overview	Training Programme on Extension Programme Planning and Implementation for Effective Extension Delivery by EEI, Hyderabad on 08 Jan, 2018
	Extension for Supply Chains in Agriculture	Training Programme on “Supply Chain Management of Agricultural Produce” by MANAGE, Hyderabad on 10 Jan, 2018
	Extension for Supply Chains in Agriculture	Training Programme on “Propelling Farming to Business - Shifting the paradigms” by MANAGE, Hyderabad on 02 Feb, 2018
	Entrepreneurship Development and Capacity Building: Participatory Rural Development in India	National Workshop on Agricultural Extension: Time to Change by MANAGE, Hyderabad on 16 Feb, 2018
	Overview of Educational Technology and Classroom Communication	Orientation Training programme for the newly recruited Assistant Professor of TANUVAS by TANUVAS, Chennai on 13 Mar, 2018



Sanjiv Kumar	Marketing Strategies for livestock products to increase farmers' income	Model Training course-Developing Competency for Harnessing New Agribusiness Opportunities for Doubling Farmers' Income by SABM, PJTSAU, Hyderabad on 24 Nov, 2017
	Advanced features of MS-Excel	Qualitative and Quantitative techniques in analyzing of Agriculture Research Data by MANAGE, Hyderabad on 26-28 Feb, 2018
	Correlation and Regression techniques	
	Report Writing using MS-Word	
Surya Rathore	Gender Concerns and Strategies for Empowerment of Women – International perspectives	International Programme on “Empowerment of Women for Rural Development” by Ministry of External Affairs, Govt. of India at National Institute of Rural Development and Panchayati Raj (NIRD & PR), Hyderabad on 04 July, 2017
	Doubling Farmers' Income: Contribution of Women Farmers”	“Sankalp Se Siddhi” by Central Arid Zone Research Institute, Jodhpur on 30 Aug, 2017
	Impact Assessment of Extension Programmes	Department of Agricultural Extension by College of Agriculture, Bapatla, ANGRAU on 17 Feb, 2018
V. K. J. Rao	Process Documentation Skills	Department of Agriculture & Allied departments by SAMETI, Hyderabad on 12 Jul, 2017
	Agri-business: An engine of growth for the reorientation and capacity building	Capacity building Programme for the district Officers by MCHARDI, Hyderabad on 18 Jul, 2017
	Frontline and First Line Extension	Extension Education Management by UAHS, Shimoga on 29 Jan, 2018
	Role of Frontline extension systems of Horticultural Universities	Annual technical meet of University of Horticultural sciences by College of Horticulture, Bidar on 26 Mar, 2018
	Role of Higher Education Institutions in Rural Development	National Seminar by Kulbhaskar Ashram Post graduate college, Allahabad on 29 Mar, 2018
V. V. Sumanth kumar	ICT Applications in Agribusiness	PJTSAU Training programme on “Developing Competency for Harnessing New Agribusiness Opportunities for Doubling Farmers' Income”. by School of Agribusiness Management, PJTSAU, Hyderabad on 4 Nov, 2017



6.2 Publications

NAARM faculty contributed 245 publications in the year 2017-18 in various forms. Research papers formed more than 43 percent of the total publications followed by seminar/conference proceedings. Thirty book chapters were authored by the NAARM faculty and eight books were written or edited by them. The faculty contributed in developing 22 working papers or project reports

Publication	Total
Research Papers	110
Books Authored	8
Book Chapters	29
Popular Article/newspaper article/newsletter/blogs	10
Souvenir	7
Training Manuals /Technical bulletins	24
Papers in proceedings of Conferences / Seminars/ workshops	47
Research Projects/Working Papers/Policy Papers	10

6.2.1 Publications in Refereed Journals

- Ahire, L. M., Rao, K. H. & Kumar, A. (2017). Evaluation of competence enhancement programme on motivation and positive thinking for technical officers of ICAR. *Indian Research Journal of Extension Education*, 17 (4), 48-55.
- Ahire, L. M., Rao, K. H. & Sontakki, B. S. (2017). Perception of horticultural students towards awareness and preference of self-employment avenues. *Indian Research Journal of Extension Education*, 17 (3), 47-52.
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6.2.2 Popular Articles

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- Kareemulla, K. & Sumanth Kumar, V. V. (2018). *Fifth foundation course for faculty of agricultural universities*. vol I & II. p. 409.
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- Sivakumar Sethuraman, P., Sontakki, B. S., Rasheed Sulaiman, V., Saravanan, R., Mittal, N. (Eds.), (2017). *Manual on good practices in extension research and evaluation*, AESA, C/o CRISP, Hyderabad, p.255.
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- Thammi Raju D. & Rao, K. H. (2017). *Competency enhancement for efficiency and effectiveness at workplace for faculty and scientists of Vasantrao Naik Marathwada Krishi Vidyapeeth, Parbhani*, ICAR –NAARM, Hyderabad.
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6.2.7 Book Chapters

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- Chakravarty, A., Sumanthkumar, V. V. & Patil, M. D. (2017). *Customised information delivery for dryland farmers- In Mobile technology for agricultural and rural development* in the Global South Centre for Development Informatics (CDI) at the Global Development Institute (GDI), University of Manchester, UK and CABI (Centre for Agriculture and Biosciences International).
- Dam Roy, S., Krishnan, P., Velmurugan, A., Anand, A., Grinson George, Kiruba Sankar, R. and Swarnam, T.P. 2017. *Wetlands of tropical islands under changing climate: A case from Nicobar Group of Islands, India*. In: Prusty et al., (Eds), *Wetland Science: Perspectives from South Asia*, Springer. p205-224. ISBN: 978-81-322-3713-6; DOI 10.1007/978-81-322-3715-0_11;
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6.2.8 Blogs

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6.2.9 Papers presented in Conferences / Seminars

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- Balakrishnan, M., Venkatesan, P. & Soam, S. K. (2018, March). *Designing of database for indigenous technical knowledge in agriculture*. Proceeding of the International Conference on Invigorating Transformation of Farm Extension towards Sustainable Development (INTFES-18): Futuristic Challenges and Prospects, TNAU, Coimbatore, p.315.
- Dhandapani, A. (2018, January). *Statistical packages – Use or misuse?* Invited talk in International Conference on Changing Paradigms and Emerging Challenges in Statistical Sciences (IPECS 2018), Pondicherry.
- Dhandapani, A. (2017, December). *Planning and analysis of coordinated crop improvement trials*. Invited talk in the session on Advances in Design of Experiments, International Conference of IISA-2017 held at Hyderabad.
- Ganesh Kumar, B. (2017, October). *Economic Impact of Foot-and-Mouth Disease in India: An Evidence from Andhra Pradesh*. GFRA Scientific Meeting, Incheon, Seoul, South Korea.
- Ganesh Kumar, B., Sivaramane, N., Umesh K. B. & Sheshshayee M. (2017, November). *Finger Millet, the Poor men's Crop and Rich people's Diet: An Economic Analysis on the Status, Scope and Policy*. In Silver Jubilee Conference of Agricultural Economics Research Association (India) on 'Doubling Farmers Income: Options and Strategies'. ICAR-NAARM, Hyderabad.
- Gaurav, N., Srinivas, K., Subash S.P and Sreekanth, P.D. (2017, November). *Assessing nutrient status and market potential for Fertilizers at district level using geospatial analysis*, Silver Jubilee Conference of Agricultural Economics Research Association at ICAR-NAARM, Hyderabad.
- Gomathi, P., Krishnan, M., Ramasubramanian, V., Saravanakumar, V., Prakash, S. & Qureshi, N. W. (2017, November). *Marine ornamental fisheries – A study on marketing in ramanathapuram district of Tamil Nadu*. Paper Presented at 11th Indian Fisheries and Aquaculture Forum (11th IFAF), Asian Fisheries Society, Kochi.
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- Kumar, R. (2017, December). *Agricultural transformation in Eastern India: Lessons learnt and future opportunities*. International Conference on "Agriculture and Rural Transformation in Asia: Past Experiences and Future Opportunities". Bangkok, Thailand.
- Kumar, R. & Kumar, S., (2018, March). *Innovations in ensuring remunerative prices (MSP) to Farmers: Challenges and strategies*. In Policy Dialogue on Innovations in Ensuring Remunerative Prices (MSP) to Farmers: Challenges and Strategies held at NASC Complex, Pusa, New Delhi.
- Manju, G., Gupta, M., Subash, S. P. & Kalpana Sastry R., (2018, February). *Assessment of CRISPR technology as a potential application in agriculture* at Second Asia Pacific Workshop on Innovation, IP, and Competition, Centre for Innovation, Intellectual Property and Competition (CIIPC), National Law University, Delhi.
- Manju, G., Gupta, M., Subash, S. P. & Kalpana Sastry R., (2018, March). *"Patent landscaping of CRISPR technology in agriculture"* at International Conference "Integrative Biology & Applied Genetics" ICBAG-2018, Dept. of Genetics, Osmania University, Hyderabad.
- Mooventhana, P., Singh, S. R. K., Venkatesan. P., Dixit Anil, Sharma, K. C., Sivalingam, Gupta, P. N., Kumar, A., Singh, U. and Dixit, A. (2018, March). Poster presentation on *Socio-economic upliftment of tribal farmers through suitable agricultural enterprises integration in Rice fallow pulse cropping system- A farmer participatory approach* in the International Conference on Invigorating Transformation of Farm Extension towards Sustainable Development: Futuristic Challenges and Prospects - INTFES, Extension Education Society, TNAU, Coimbatore.
- Murthy, G. R. K. (2018, February). *ICTs for Agri entrepreneurship education*. Lead Paper presented at National Seminar on Digital and Engineering Technologies held at College of Agricultural Engineering, Bapatla.



- Murthy, G. R. K. (2018, January). *ICTs in Agricultural education*. Lead Paper presented at National Workshop on ICTS in Agriculture held at Birsa Agricultural University, Ranchi.
- Ramesh, P. (2017, November). *Organic farming towards climate change adaptation and mitigation*. Invited paper presented in the International Workshop on Building Perspective capacity to measure climate change impacts due to changes in Agricultural practices. Held at the Gateway Hotel, Vijayawada, Andhra Pradesh.
- Rao, V. K. J., (2018, March). Key note address for the national seminar on *Role of higher education institutions in rural development* at Kulbhaskar Ashram Post graduate college Allahabad, pp 1-4, ISBN-978-81-86100-99-8
- Rathore, S. (2017, November). *Geographical Indications in Agriculture: A single answer to doubling farmers' income and ARYA*. Paper presented in ISEE National Seminar on Doubling Farmers' Income & Farm production through Skill development and Technology Application at BAU, Sabour (Bihar).
- Ravichandran, S. (2018, January). *Big Data Analytics in Agriculture*. Invited paper presented in International Conference on Changing Paradigms and Emerging Challenges in Statistical Sciences, Society of Statistics, Computer and Applications and Department of Statistics, Puducherry.
- Sandhya Shenoy N. (2017, September). *Gender issues in Horticulture*. In *International symposium on Horticulture: Priorities and Trends*. Jointly organised by ICAR, New Delhi and International Society of Horticulture Science (ISHS), Belgium at ICAR- IIHR, Bengaluru.
- Sandhya Shenoy N. (2018, February). *ICTs for empowering farmwomen*. Lead paper presentation in National Conference on Digital and Engineering technologies for precision agriculture and value addition, College of Agricultural Engineering, ANGRAU.
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- Sontakki, B. S. (2018, February). *Mainstreaming farmer innovations: implications for extension with focus on partnerships*. Presentation as Invited speaker in the 1st International Extension Congress on New Horizons of Extension-Challenges and Opportunities at ICAR-CIWA, Bhubaneswar.
- Sreekanth, P. D., Kumar, K. V., Soam S. K. & Srinivasarao, Ch. (2018). *Spatial decision support systems for smart farming using geo-spatial technologies*. Proceedings of the National Conference on Application of Geo-Spatial Technologies and ICT's in Smart Agriculture (SMARTAGRI-2018), pp. 118-122. ISBN:978-81-88367-42-9
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6.2.11 Policy Papers

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6.3 Participations

6.3.1 Participation in Workshop/Conferences/Symposia/ Meetings

Month/ Year	Employee Name	Programme Name	Venue	Period
April '17	Ch. Srinivasa Rao	Meeting about Drought Monitoring Project.	Regional Office, IWMI, New Delhi	10 Apr, 2017
	I. Sekar M. B. Dastagiri	Scoping Consultation on Framework for Assessment of Socio-economic Vulnerability of Coastal Community	ICAR-NAARM, Hyderabad	06-07 Apr, 2017
	K. Srinivas P. C. Meena	National Conference on Future of Agrifinance 2030	College of Agricultural Banking (CAB), Pune, Maharashtra	24-25 April, 2017
	S. Senthil Vinayagam	National Conference on "Revisiting Agricultural Extension Strategies,"	PJTSAU, Hyderabad	22-24 Apr, 2017
	V. V. Sumanthkumar	One day workshop on Digitization for the Department of Agriculture, Telangana	ICRISAT, Hyderabad	12 Apr, 2017
May '17	Alok Kumar	8 th Conference on Excellence in Research & Education (CERE)	IIM, Indore	04-07 May, 2017
	Manju Gerard	IP-Ideas to Innovation-WIPS	Hyderabad	05 May, 2017
	Ch. Srinivasa Rao	Meeting on Forecasting Models	BHU, Varanasi	17 May, 2017
	Ch. Srinivasa Rao	1 st Meeting of TIFAC-Global Technology Watch Group (GTWG) – Sustainable Agricultural Sector	TIFAC, New Delhi	19 May, 2017
	P. Venkatesan	International Conference on "Contemporary Issues in Integrating Climate- The Emerging Areas of Agriculture, Horticulture, Biodiversity, Forestry, Engineering Technology, Fundamental/ Applied Science and Business Management for Sustainable Development (AGROTECH-2017) "	Kalimpong Science Centre, Kalimpong, West Bengal, India	11-12 May, 2017
	V. V. Sumanthkumar	ICT4D Data Jam event 2017	iHub, ICRISAT	12-13 May, 2017
	V. V. Sumanthkumar	ICT4D Conference – From Innovation to Impact	Hitex	15-18 May, 2017
June '17	B. S. Sontakki	Brainstorming Workshop on Collaboration in Research, Education, Extension and Management	ICAR-NAARM, Hyderabad	23 Jun, 2017
	B. S. Sontakki	8 th Meeting of the Extension Education Council of the University of Horticultural Sciences, Bagalkot (as nominated member)	University of Horticultural Sciences, Bagalkot	29 Jun, 2017
July '17	D. Thammi Raju	Consultative meeting on capacity building of stakeholders for effective delivery of Livestock extension services	Sri Venkateswara Veterinary University, Tirupati	04 Jul, 2017
	S. K. Soam D. Thammi Raju S. Senthil Vinayagam Surya Rathore	Global Technology Watch Group (GTWG) under Climate Change programme of Dept. of Science & technology, Govt. of India	ICAR-NAARM, Hyderabad	13-14 Jul, 2017



August '17	B. S. Sontakki	Meeting of the Selection Board for the posts of Assistant and Associate Professors in the subject of Agricultural Extension	UAS, Raichur	19 Aug, 2017
	B. S. Sontakki	1 st Meeting of the VIII RAC of NAARM	ICAR-NAARM, Hyderabad	24 Aug, 2017
	D. Thammi Raju K. H. Rao	Reshaping Agricultural Research Education and Extension Systems Management for 2030	ICAR-NAARM, Hyderabad	31 Aug, 2017
	I. Sekar P. Krishnan	Workshop on Agri-Innovations	ICAR-NAARM, Hyderabad	17-18 Aug, 2017
	Ranjit Kumar Manju Gerard	AGRI-UDAAN: Food & Agribusiness Accelerator 2.0	NAAS Complex, New Delhi	04 Aug, 2017
	Surya Rathore	<i>Sankalp Se Siddhi</i>	KVK, CAZRI, Pali	30 Aug, 2017
September '17	All Faculty	Foundation Day of NAARM	ICAR-NAARM, Hyderabad	01 Sep, 2017
	N. Sandhya Shenoy	International symposium on "Horticulture: Priorities and Trends" jointly organised by ICAR, New Delhi and International Society of Horticulture Science (ISHS), Belgium.	ICAR- IIHR, Bengaluru	04-08 Sep, 2017
	P. Krishnan	Review meeting on, "Assessing socio-economic vulnerability of coastal community from Sidhudurg District, Maharashtra	National Centre for Sustainable Coastal Management, Chennai	13-14 Sep, 2-17
	Manju Gerard	Society for Technology Management (STEM)- Annual Summit	Hyderabad	14-15 Sep, 2017
October '17	B. S. Sontakki D. Thammi Raju P.Venkatesan	National Consultation Meet on Fostering Faster reach of innovations from Aquaculture Research through Media: A science Communication Perspective	ICAR-CIFA, Bhubaneshwar	27 Oct, 2017
	Ch. Srinivasa Rao	Meeting with Hon'ble Chief Minister of A.P and participated in the event for Exchange of Memorandum of Cooperation (MoC) between AP Food Processing Society and ICAR-NAAM.	Vijayawada	16 Oct, 2017
	M. Krishnan	77 th Annual conference of the Indian Society of Agricultural Economics	Central Agricultural University, Umiam, Barapani, Meghalaya,	12-14 Oct, 2017
	Ranjit Kumar	Workshop on 'Green Revolution in Eastern India: Constraints, Opportunities and Way Forward'	IFPRI, New Delhi	9-10 Oct, 2017
	Ranjit Kumar	Workshop on 'Promoting Pulses Cultivation in Bihar'	Motihari, Bihar	12 Oct, 2017



November '17	All Faculty	44 th Foundation Day of ASRB & ASRB Review Committee Meeting	ICAR-NAARM, Hyderabad	01 Nov, 2017
	All Faculty	Silver Jubilee Conference of Agricultural Economics Research Association on Doubling Farmers' Income: Options and Strategies	ICAR-NAARM, Hyderabad	7-9 Nov, 2017
	All Faculty	Visit of Parliament Standing Committee	ICAR-NAARM, Hyderabad	11 Nov, 2017
	B. S. Sontakki	4 th International Conference on Life Skills & Livelihood Skills - Leveraging Digital India for Rural Empowerment	Dr. MCR HRD Institute of Telangana, Hyderabad	06-07 Nov, 2017
	K. Kareemulla	Annual Conference of Indian Society of Agricultural Marketing	Nabhakrishna Chaudhary Centre for Development Studies, Bhubaneswar	23-25 Nov, 2017
	Manju Gerard	AGRI UDAAN - Capacity Building Workshop	ICAR-NAARM, Hyderabad	13-16 Nov, 2017
	P. Ramesh	International workshop on Building Perspective capacity to measure climate change impacts due to changes in Agricultural practices.	Vijayawada, Andhra Pradesh	01-02 Nov, 2017
	P. Krishnan	11 th Indian Fisheries and Aquaculture Forum (IFAF)	Le Meridien Hotel, Kochi Jointly organized by Asian Fisheries Society (Indian Branch) and ICAR-Central Institute of Fisheries Technology (CIFT), Kochi.	21-24 Nov, 2017
	Ranjit Kumar	Capacity building workshop for agri-start ups	ICAR-NAARM, Hyderabad	15-16 Nov, 2017
	S. K. Soam	Policy Dialogue on Mainstreaming Biodiversity into Inland Fisheries Sector	ICAR-NAARM, Hyderabad	28 Nov, 2017
December '17	All Faculty	Agriculture Education Day	ICAR-NAARM, Hyderabad	03 Dec, 2017
	All Faculty	Let's Listen to farmers: National Workshop on Farmers Feedback on Doubling of Farmers Income	ICAR-NAARM, Hyderabad	22-23 Dec, 2017
	A. Dhandapani	IISA 2017	Hyderabad	28-30 Dec, 2017
	B. S. Sontakki N. Sandhya Shenoy V. K. J. Rao Surya Rathore P. Venkatesan	World Soil Day Celebration (Chief Guest – Shri Anjaiah Yadav Yalganamoni, MLA, Shadnagar Constituency, Telangana)	Vemulanarva Village, Kesampet Mandal, Ranga Reddy Dist	05 Dec, 2017
	Ch. Srinivasa Rao	FAI Annual Seminar on Fertilizers and Farm Income	Delhi	06 Dec, 2017
	D. Thammi Raju VV Sumanth Kumar G. Aneeja	Consultative Workshop on Agri Journalism	MANAGE, Hyderabad	05 Dec, 2017
	D. Thammi Raju G. Aneeja	39 th Public Relations Society of India Annual Convention	Vizag	15 Dec, 2017
	Ranjit Kumar	International Conference on 'Agriculture and Rural Transformation in Asia: Past Experience and Future Opportunities'	Bangkok, Thailand	12-14 Dec, 2017
	S. Ravichandran	Brainstorming Session on "Emerging issues for socio-economic and policy research in Asian Rice base Systems".	PJTSAU, Hyderabad	26 Dec, 2017
	S. Senthil Vinayagam	TELAGe workshop	TANUVAS, Chennai	11 Dec, 2017



January '18	Ch. Srinivasa Rao	SOC, ICAR	Krishi Bhavan, New Delhi	7 Jan, 2018
	D. Thammi Raju	7th National Conference on Shaping Livestock Extension Advisory Services (LEAS) for Doubling of Farmers Income	Veterinary College, Shivamogga, Karnataka	29-31 Jan, 2018
	K. H. Rao	National Workshop on Doubling Farmers' Income by 2022	CGGG & NABARD, Hyderabad	17-18 Jan, 2018
	N. Sandhya Shenoy D. Thammi Raju Surya Rathore	Consultative Meeting on Digital Farming Initiatives of ICAR in collaboration with TCS	NASC, New Delhi	22 Jan, 2018
	N. Sivaramane	Changing Paradigms and Emerging Challenges in Statistical Sciences (IPECS-2018)	Ponducherry University, Puducherry	28-31 Jan, 2018
	P. D. Sreekanth	GeoAgri Conference in GE04SDGS of Geospatial World Forum 2018	Hyderabad International Convention Centre (HICC), Hyderabad	18-19 Jan, 2018
	P. D. Sreekanth	National Conference on Applications of Geo-spatial Technologies and ICT in Smart Agriculture (SMART AGRI- 2018)	UAS, Dharward, Karnataka	23-24 Jan, 2018
	P. Krishnan	Scoping Consultation on Coastal vulnerability assessment in Kerala	KILA Office, Thiruvanthapuram Jointly organized by Kerala Institute for Local Administration (KILA), Thiruvananthapuram and Action on Climate Today (ACT), Climate Change Innovation Programme (CCIP) of DFID	12 Jan, 2018
	S. K. Soam B. S. Sontakki D. Thammi Raju Surya Rathore P.Venkatesan	Policy cum Concluding Workshop on "Technology Enhanced Learning in Agricultural Education"	ICAR-NAARM, Hyderabad	24-25 Jan, 2018
	S. Ravichandran A. Dhandapani	International Conference on Changing Paradigms and Emerging Challenges in Statistical Sciences (IPECS 2018)	Pondicherry	29-31 Jan, 2018
	S. Senthil Vinayagam	International Conference on Biocontrol for sustainable insect Pest Management (ICBS 2018)	Agricultural College & Research Institute, TNAU, Killikulam, Vallanad, Tamil Nadu	29-31 Jan, 2018
February '18	B. S. Sontakki P.Venkatesan	1 st International Extension Congress on New Horizons of Extension-Challenges and Opportunities	ICAR-CIWA, Bhubaneswar	01-03 Feb, 2018
	B. S. Sontakki P.Venkatesan	Annual Review Workshop of Farmer FIRST programme	ICAR-IARI, New Delhi	21-22 Feb, 2018
	G R K Murthy	National Conference on Digital and Engineering Technologies for Precision Agriculture	College of Agricultural Engineering, Bapatla, ANGRAU	26-28 Feb, 2018
	G R K Murthy	National Conference on E-learning and Technologies: Prospects and Challenges	MANUU, Hyderabad	20-21 Feb, 2018
	M. Balakrishnan	National Workshop on National Academic Depository (NAD)	AICTE, New Delhi	5 Feb, 2018



	M. Balakrishnan N. Srinivasa Rao P. D. Sreekanth	Policy Workshop on Big Data Analytics in Agriculture	ICAR-NAARM, Hyderabad	08-09 Feb, 2018
	Manju Gerard	Second Asia Pacific Workshop on Innovation, IP, and Competition	National Law University, New Delhi	05-10 Feb, 2018
	Manju Gerard	Demo Day (Investors' Meet) of AGRI UDAAN, India's unique Food & Agribusiness Accelerator program 2.0.	ICAR-NAARM, Hyderabad	12 Feb, 2018
	P. D. Sreekanth	Workshop on Google Earth Engine	Google, Hyderabad	19 Feb, 2018
	P. Krishnan	Consultative Workshop for synchronizing data collection of Agricultural Research and Technology Indicators	NSTMIS-DST, MoST, New Delhi	09 Feb, 2018
	Ranjit Kumar	National Conference on 'Agriculture 2022- Doubling Farmers' Income'	NASC Complex, New Delhi	19-20 Feb, 2018
	S. Ravichandran	Screening Committee Meeting at AICTE, Trivandrum as Member, Academics	AICTE, New Delhi	25 Feb, 2018
March '18	All Faculty	National Workshop on Revisiting Foundation Course for Agricultural Research Service (FOCARs): Reflections and Feedback of Trained Scientists	ICAR-NAARM, Hyderabad	15-16 Mar, 2018
	All Faculty	Interactive Session on "Doubling Farmers Income" with Hon'ble Vice President of India	ICAR – IIRR, Hyderabad	31 Mar, 2018
	B. S. Sontakki	7 th Executive Committee/GB meeting of the a-IDEA	ICAR-NAARM, Hyderabad	07 Mar, 2018
	B. S. Sontakki	Institute Management Committee Meeting of ATARI, Hyderabad	ATARI, Hyderabad	26 Mar, 2018
	B. S. Sontakki	Meeting of the Selection Board for the posts of Assistant and Associate Professors in the subject of Agricultural Extension	PDKV, Akola	27 Mar, 2018
	M. Balakrishnan P. D. Sreekanth	105 th Indian Science Congress	Manipur University, Imphal	16-20 Mar, 2018
	N. Sivaramane	International training session on "Climate-Smart Agriculture and Soils" held at Dhanbad- Jharkhand	Dhanbad, Jharkhand	29 Mar, 2018
	N. Sivaramane	International Conference on Food & Agriculture	Dhanbad, Jharkhand	29-31 Mar, 2018
	P. Krishnan	Expert Consultation for strengthening the Agricultural Science & Technology Indicators in India.	Bill and Melinda Gates Foundation	23 Mar, 2018
	Ranjit Kumar Sanjiv Kumar	Policy Dialogue on "Innovations in Ensuring Remunerative Prices (MSP) to Farmers: Challenges and Strategies	NASC Complex, New Delhi	23 Mar, 2018
	Ch. Srinivasa Rao S. K. Soam	ICAR Directors Conference	NASC Complex, New Delhi	08-09 Mar, 2018
	S. Ravichandran	STI and Make in India: Promoting Manufacturing at District Level – Workshop as expert.	Osmania University	29 Mar, 2018
	S. Senthil Vinayagam M. Balakrishnan	International Conference on Invigorating Transformation of Farm Extension Towards Sustainable Development: Futuristic Challenges and Prospects.	TNAU, Coimbatore	09-10 Mar, 2018



6.3.2 Training Programmes Attended by the Staff of the Academy

Employee Name	Programme Name	Venue	Period
Alok Kumar	Faculty Development Programme	IIM, Indore	24 Apr-30 May, 2017
	Priority setting, Monitoring & Evaluation (PME) of Agricultural Research Projects	ICAR-NAARM, Hyderabad	6-11 Oct, 2017
B. S. Yashavanth	Emerging tools and techniques for Agricultural Knowledge Communication	ICAR-NAARM, Hyderabad	01-10 Nov, 2017
	Advances in Simulation Modelling and Climate Change Research towards Knowledge Based Agriculture	CESCRA, IARI, New Delhi	16 Nov-06 Dec, 2017
	Google Earth Engine	Google, Hyderabad	19 Feb, 2018
Ch. Srinivasa Rao	Leadership and Change Management	IIM, Ahmedabad	11-15 Sep, 2017
D. Thammi Raju	Elsevier Authors Workshop	ICAR-NAARM, Hyderabad	04 Aug, 2017
Manju Gerard	Priority Setting, Monitoring & Evaluation(PME) of Agricultural Research Projects	ICAR-NAARM, Hyderabad	6-11 Oct, 2017
	Science for progress in India: Focus on sustainability	National Institute of Advanced Sciences, Bengaluru	22-26 Jan, 2018
N. Srinivasa Rao	Training programme on Priority Setting, Monitoring and Evaluation (PME) of Agricultural Research Projects	ICAR-NAARM, Hyderabad	06-11 Oct, 2017
P. D. Sreekanth	Geo-spatial Data Analysis using R Open Source Software	International Institute of SAS Training and Research (IISASTR), New Delhi	21-24 Mar, 2018
P. Venkatesan	Global Technology watch Group (GTWG) – Sustainable Agriculture Stakeholders 2nd Meeting	ICAR-NAARM, Hyderabad	13 Mar, 2017
	Climate-Smart Agriculture & Soils	Dhanbad, Jharkhand	29-30 Mar, 2018
	Brainstorming session on role of SMS Extension in KVKs	ATARI-Hyderabad	08 Apr, 2017
S. K. Soam	The Winning Edge: Communication Strategies for Leaders	IIM, Ahmedabad	18-23 Sep, 2017
S. Ravichandran	MDP on “Business Analytics for Strategic and Tactical Level Decision Making”	IIM, Calcutta	18-22 Dec, 2017
V. K. J. Rao	MDP on ‘What Every Teacher Should Know for Bringing Out The Best From Their Students’	IRMA, Anand	23-25 Nov, 2017
	Android and IOS Mobile App development	C-DAC, Hyderabad	19-24 Mar, 2018



6.3.3 Other Responsibilities and Assignments

Scientist	Responsibilities/Assignments
Bharat S. Sontakki	• Member, General Body of ICAR Society
	• Member, Sub-group on Empowering the Farmers through Extension and Knowledge Diffusion of the Inter-Ministerial Working Group on Doubling of Farmers' Income.
	• Member, Extension Education Council, University of Horticultural Sciences, Bagalkot, Karnataka
	• Co-Chair & Chair in National Consultation Meet on Fostering Faster reach of innovations from Aquaculture Research through Media: A science Communication Perspective at ICAR-CIFA, Bhubaneswar on Oct 27, 2017
	• Panelist on the theme – Innovative approaches in Agricultural Extension in the 1st International Extension Congress 2018 on New Horizons of Extension –Challenges and Opportunities
	• Member of the Organizing Committee and Convener of Committee on Awards in National Conference on “Revisiting Agricultural Extension Strategies for Enhancing Food and Nutritional Security, Sustainable Livelihoods and Resilience to Climate Change - Towards Transforming Agriculture” held in PJTSAU, Hyderabad on 23-04-2017
	• Co-Chair & Rapporteur on the theme – Agricultural Extension & SDGs in the National Conference on “Revisiting Agricultural Extension Strategies for Enhancing Food and Nutritional Security, Sustainable Livelihoods and Resilience to Climate Change - Towards Transforming Agriculture” held in PJTSAU, Hyderabad on 23-04-2017
	• Member of the International Organizing Committee in 4th International Conference on Life Skills & Livelihood Skills - Leveraging Digital India for Rural Empowerment at Dr. MCR HRD Institute of Telangana, Hyderabad during Nov 16-17, 2017
	• Member, Editorial Advisory Board, Amity Journal of Agribusiness, New Delhi
	• External Examiner to evaluate PhD thesis and External Expert in the Thesis Viva Voce of a PhD student in the subject of Agricultural Extension at PJTSAU, Hyderabad
M.Balakrishnan	• Honorary Member for the Society for Bioinformatics and Biological Sciences (SBBS), Allahabad
	• External examiner for evaluation of Ph.D. thesis by a students of Bharathidasan University, Chennai, TN
	• Member for the for the ARS online facility at ASRB, New Delhi
	• External examiner for evaluation of Ph.D. thesis by a students of Karpagam University, Coimbatore, TN
	• External examiner for evaluation of Ph.D. thesis by a students of Anna University, Chennai, TN
	• External examiner for evaluation of Ph.D. thesis by a students of PRIST University, Thanjavur, TN
	• Centre Supervisor- ARS-2016(Prelims) and NET(I)-2017 online Examination during May 16-21, 2017.
	• Treasurer, NAARM Alumni Association, NAARM



	- Member Selection committee for the selection of SRF (Crop Protection, Computer and Statistics) post at CRIDA (05.08.2017) - Vice president, NAARM PG Alumni Association, NAARM
N Sivaramane	- Chaired a session on “Digital Agriculture, Big Data Analytics, Bridging the digital divide and New Age Startups” at International Conference on Food & Agriculture Organized by Endling Conferences at Dhanbad, Jharkhand - Conducted Oral qualifying examination of a Ph.D. student on Oct 28, 2017 at Dept. of agricultural economics, UAS, Raichur - External Examiner for a M.Sc.(Ag.) student in the Dept. of Agricultural Economics, College of Agriculture, Vellayani, Kerala on Oct 13, 2017
N. Sandhya Shenoy	- Chairperson for the technical session on ‘Participatory Technology Development, Adoption and Gender issues’ International symposium on “Horticulture: Priorities and Trends” jointly organised by ICAR, New Delhi and International Society of Horticulture Science (ISHS), Belgium during Sep 4-8, 2017 - External Examiner for comprehensive viva of M. Sc. Home Science Extension Communication Management (HECM) at ANGRAU, Guntur on 15.12.2017. - External Examiner for Ph.D.thesis of Home Science Extension & Communication Management at ANGRAU, Guntur
P. Krishnan	- Resource Person for the Mid-Career Training (MCT) for IFS officers of 14-16 years of service held at Indira Gandhi National Forest Academy (IGNFA), Dehradun on 17 Apr 2017 and delivered a lecture on “Environmental Governance and EIA Studies”. - Invited as an Expert for the meeting held at MoEFCC, New Delhi on 26th May, 2017 to discuss proposal to develop national biodiversity portal, and India’s membership of Global Biodiversity Information Facility (GBIF) and presented a conceptual framework for the National Biodiversity Portal. - Invited Speaker in the Workshop on Climate Resilient Aquaculture organized by Tamilnadu Fisheries University at Fisheries College and Research Institute, Thoothukkudi on 05 Jun. 2017 and delivered a talk on, “Climate Smart Aquaculture: Trends and Opportunities”. - Invited lecture on “Environmental Governance and EIA Studies” for the IFS officers of 14-16 years of service, as a part of the Mid-Career Training (MCT) held at Indira Gandhi National Forest Academy (IGNFA), Dehradun on 24 Jul 2017. - Invited as an Expert for the Scoping Workshop on IT-based Monitoring of Indian Bio-resources organized by Ministry of Environment, Forests, and Climate Change (MOEF&CC), National Biodiversity Authority (NBA) in partnership with Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH as a part of their project on Access and Benefit Sharing (ABS) at NBA, Chennai, during 13-14th February 2018. Presented a case for building a “National Digital Repository for India – Issues and Strategies - Invited to deliver a Keynote Address on, “Dealing with Climate Change- Researchers’ Perspective” during the National Conference on “Climate Change and Global Warming”, organized by the P.G. & Research Department of Zoology at Justice Basheer Ahmed Sayeed College for Women (Autonomous), Chennai, on 22 Feb 2018.
P.Venkatesan	Co-Chair in National Consultation Meet on Fostering Faster reach of innovations from Aquaculture Research through Media: A science Communication Perspective at ICAR-CIFA, Bhubaneswar on Oct 27, 2017



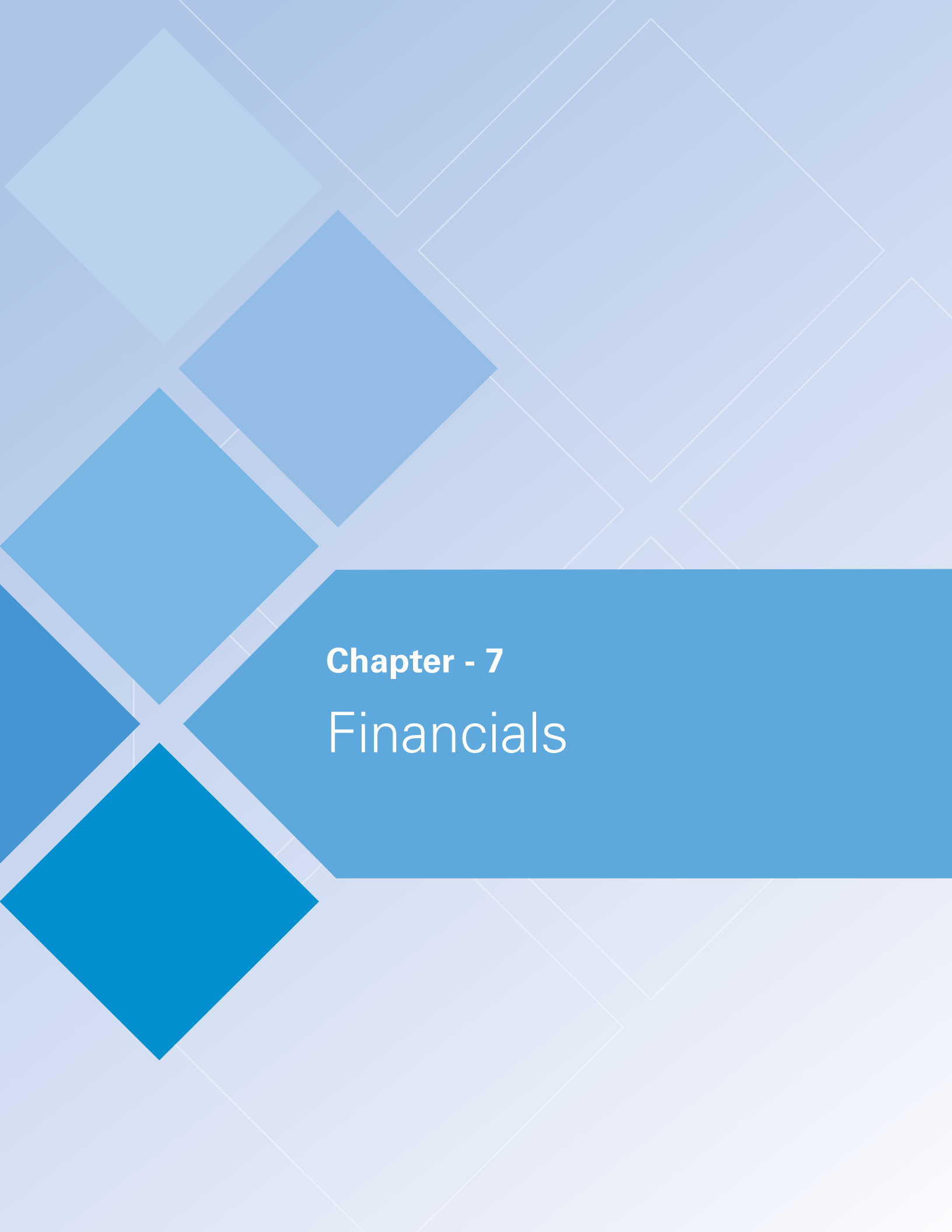
	• Panelist in the topic- Interface among stakeholders in the 1st International Extension Congress 2018 on New Horizons of Extension –Challenges and Opportunities held in ICAR-CIWA, Bhubaneswar on Feb 02, 2018 • Co-chair in the thematic area “Stocktaking of Extension Research” in the International Conference on Invigorating Transformation of Farm Extension towards Sustainable Development: Futuristic Challenges and Prospects - INTFES – 18 at TNAU, Coimbatore during March 9-10, 2018 • Co-chair in the thematic area “KVK in action: Potential to revolutionize agricultural transformation” in the 2nd International Conference on Food and Agriculture at Dhanbad, Jharkhand during March 29-31, 2018 • External Examiner for M. Sc. (Agricultural Extension) thesis at TNAU • External Examiner for M. V. Sc. (Veterinary Extension) thesis at TANUVAS
Ranjit Kumar	• Co-organized Silver Jubilee Conference of Agricultural Economics Research Association (AERA) during Nov 7-9, 2017 • Co-chaired Technical Session on Agribusiness and Market Reforms in the AERA Conference during Nov 7-9, 2017 • Chairperson of Technical Committee of Fifth Meeting of the ASRB Revamping and Restructuring Committee • Chairman, PGDMA Admission Committee (2017-19) of NAARM • Member, Institute Management Committee (IMC) of CTCRI, Rajamundry • Member, Interview Board for selection of Assistant/Associate Professor for ANGRAU Guntur and Banda Agricultural University • Coordinated 2nd Annual Business Competition Event “Sanikalp” at the Academy during Aug 28-29, 2017 • President, NAARM PG Alumai Association • Coordinator for accreditation of PGDMA programme visit team of National Board of Accreditation during Sep 16-17, 2017
Sanjiv Kumar	• Rapporteur in Deans committee review meeting held during June 29-30, 2017 in ICAR-NAARM • Rapporteur in 25th Agricultural Economics Research Association conference which is organized at ICAR- NAARM Hyderabad during November 7-9, 2017 • Rapporteur in ASRB review meeting held during Nov 1-3, 2017 in ICAR-NAARM. • External Examiner for the subject of Operations Research of College of Dairy Science & Technology, GADVASU, Ludhiana, Punjab
S. K. Soam	• IPR Expert member of Institute Technology Management Committee (ITMC) of ICAR-Indian Institute of Millets Research, Hyderabad. Attended the meetings on 01.09.2017, 27.04.2017, 20.02.2018 • Chairman, Selection Board for the post of ‘Executive Director’, National Fisheries Development Board, Hyderabad • Member, Selection Board for the post of ‘Senior Executive’, National Fisheries Development Board, Hyderabad



	- Chairman, Session-I: Linking product, places and people-Exploring Telangana, in Geographical Indication Festival of Telangana (GIFT-2017) held at Hotel Taj Deccan, Hyderabad on 9th June 2017 organized by Confederation of Indian Industry (CII) in association with Tourism Department, Govt. of Telangana - Chief Guest at 'Two-days learning workshop on Digital Field Book', on 29th November 2017 organized by ICAR- Indian Institute of Millet Research, Hyderabad. - Chief Guest in inaugural function of one-day sorghum germplasm field day and one-day learning workshop on Apps for agriculture (21-22 March 2018) organized by ICAR- Indian Institute of Millet Research, Hyderabad. - Expert for 28th meeting of the Board of Faculty for PG Studies, 19.12.2017 at Lam, Guntur, Acharya NG Ranga Agricultural University, Guntur - As AHP expert, helped ICAR-VPKAS scientist for project data analysis - Developed questionnaire for GI studies and provided to Gujarat Institute of Development Research, Ahmedabad for their research use - As project expert, reviewed research proposal from Sri Venkateswara Veterinary University (SVVU), Tirupathi - As project expert, helped in development of 'Swatcha bharat mission-rural manual' of Administrative Staff College of India (ASCI), Hyderabad
S.Ravichandran	Chairman of Technical Session on Data Analytics in International Conference organized by Pondicherry University and Society of Statistics, Computer and Applications during January 29-31, 2018
S. Senthil Vinayagam	- Organised PGDTMA Examination at IASRI, New Delhi from 12-15, September, 2017 - Participated as an expert in Training Quality Improvement Measures Committee (TQIMC) at NIRD & PR on 26-5-2017 - Conducted ICAR-AIEEA examination as exam superintendent on 10-6-2017 - Acted as an expert for CAS for the position from Associate Professor to Professor (Agril. Extn), ANGRAU, Guntur on 28-6-2017 - Acted as an expert for Selection of Assistant Professor (Agrl. Extn.), ANGRAU, Guntur on 28-6-2017 - Acted as an Expert for Selection of Best Scientist award PJTSAU, Hyderabad on 26-08-2017 - Acted as an Expert for selection of Sr. Training Project Manager (SPTM), NIRD&PR, Hyderabad on 07-02-2018 - Acted as referee for Journal of Rural Development, NIRD, Hyderabad. - Acted as expert for thesis evaluation of TNAU, Coimbatore
Surya Rathore	- Member, Editorial Advisory Board in Amity Journal of Agribusiness, New Delhi - Member, Personal Interview Board in Selection of candidates for PGDMA – 2017-19 - Member, Expert Panel in Selection of candidates for 14th & 15th Batch of NIRD & PR's P. G. Diploma in Rural Development Management (PGDRDM) 2017-18 & 2018-19 - Centre Superintendent in AIEEA – UG - 2017



	• Expert in Advice on Training Needs Assessment for On-campus & Off Campus programmes to prepare training calendar for 2018-19
	• Subject Expert in Selection Committee for the post of Subject Matter Specialist (Home Science) in Agricultural University, Jodhpur during Feb 8-9, 2018.
	• External Examiner to evaluate the outline of the research work (Synopsis) of a PhD scholar(Veterinary Extension) from Indian Veterinary Research Institute, Bareilly (U. P.)
	• External Examiner for Ph.D. (Agricultural Extension) thesis evaluation and viva Voce at College of Agriculture, Bapatla, Acharya N. G. Ranga Agricultural University, Andhra Pradesh
	• Rapporteur for Brainstorming Session of 5th Dean's Committee recommendations for Home Science & Fisheries held on June 29, 2017 at ICAR-NAARM
	• Rapporteur for theme "Extension System and Policy imperatives" during Nov 28-30, 2017 in ISEE National Seminar on Doubling Farmers' Income & Farm production through Skill development and Technology Application at BAU, Sabour (Bihar)
	• Paper reviewer in International Journal of Basic and Applied Agricultural Research (IJBAAR)
V. V. Sumanth Kumar	• Awarded "First grant from a new Donor" for the project funded by AP State Skill Development Corporation, during ICRISAT Annual Day Celebrations



Chapter - 7

Financials



Financials

7.1 Budget Allocation and Expenditure during 2017-18 (₹ in lakh)

Head of Account	Revised Estimates	Expenditure
Grants in Aid - Capital	0.00	0.00
Grants-in-Aid - Salaries (REVENUE)		
Establishment Expenses		
i.Establishment charges	1889.77	1889.77
ii.Wages	144.00	143.98
iii.OTA	0.23	0.23
Total Salaries	2034.00	2033.98
Grants-in-Aid-General (REVENUE)		
Pension & Other Benefits	2700.00	2700.00
Travelling Allowance	55.00	55.00
Research & Operational Expenses	190.00	189.98
Administrative Expenses	1791.00	1791.00
H.R.D.	20.00	20.00
Misc.Expenses	123.00	123.00
Total Grants in Aid-General	4879.00	4878.98
Grand Total (Capital+Salaries+General)	6913.00	6912.96
Loans & Advances	5.00	0



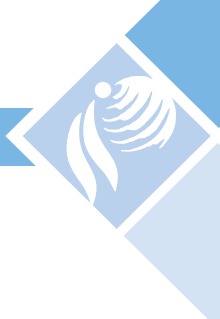
7.2 Resource Generation

7.2.1 Sponsored Research Projects Budget (₹ in Lakh)

S. No.	Name of the Project	Budget outlay	Fund available/ received including Opening Balance for 2017-18	Expenditure during 2017-18
1	RKVY - Telangana State Govt. Developing SAIDP Project	24.72	13.46	8.80
2	Developing Guidelines & Methodologies for IP Policies for PJTSAU	5.00	5.00	1.74
3	PJTSAU Food Consumption Project	10.17	10.17	5.54
4	PJTSAU Medicinal Plants Project	0.75	0.75	0.75
5	Environmental Impact Assessment of Tobacco Curing	20.87	10.40	5.76
6	NAARM-MANAGE Collaboration Project	3.30	1.00	1.00
7	DBT Project - Socio Economic Analysis of Finger Millet in India (Ragi)	13.58	14.96	13.31
8	R&D Strategies and Policies under SERB, DST	12.42	5.71	3.71
9	Agril.Res.&Dev.Infrastructure in Select Indian States under NSTMIS Scheme of DST	12.70	12.70	7.62
10	ICAR NAARM-MANAGE Colloborative Proj.Certified Livestock Advisor	4.68	4.68	4.68
11	Mapping Coastal Vulnerability & Hazard, NCSCM	8.00	8.00	8.00
12	RGMVP, Raibareli	1.35	1.35	0.14
13	IFPRI sponsored Project- ASTI in Indian NARES	28.35	11.35	0.00
14	Consultancy Project on Adoption of Insect Resistance Management	26.00	23.40	3.05
Total		171.89	122.94	64.09

7.2.2 Resource Generation from Off-Campus and Sponsored Programmes (₹ in Lakh)

Name of the Programme	Amount (Gross)
Admn.Training Programme for faculty of Veterinary Colleges (24-26 th Apr, 2017)	1.50
4th FOCAU Programme (2-31 st May, 2017)	5.94
Stakeholders Conclutation Workshop on Climate Change	1.54
Competency Enhancement Programme at Parbhani(4-8th Sep, 2017)	2.18
DWRP at SVVU, Tirupathi (4-6 th Oct, 2017)	3.74
DWRP at NDRI, Karnal (13-16 th Dec, 2017)	3.17
5th FOCFU Programme (17 th Jan to 15 th Feb, 2018)	21.60
FDP at UoA&HS, Shivamogga, Karnataka (29 th Jan to 2 nd Feb, 2018)	2.02
FDP at DN College, Nagpur (6-10 th Feb, 2018)	1.45
Training Programme on ARM for faculty of JNKVV, MP from (14-20 th Mar, 2018)	2.26
Knowledge W/S on Res.Proj.Mgmt. for PG Students JNKVV, Jabalpur from (24-25 th Mar, 2018)	0.54
6th FOCFU Programme (12 th Mar to 10 th Apr, 2018)	22.05
Total	67.98



7.2.3 Resource Generation from Educational Programmes (₹ in Lakh)

Name of the Programme	Amount (Gross)
PGDMA 2016-18	89.65
PGDMA 2017-19	106.50
PGDTMA 2017	13.92
Total	210.06

7.2.4 Resource Generation from Other Activities (₹ in Lakh)

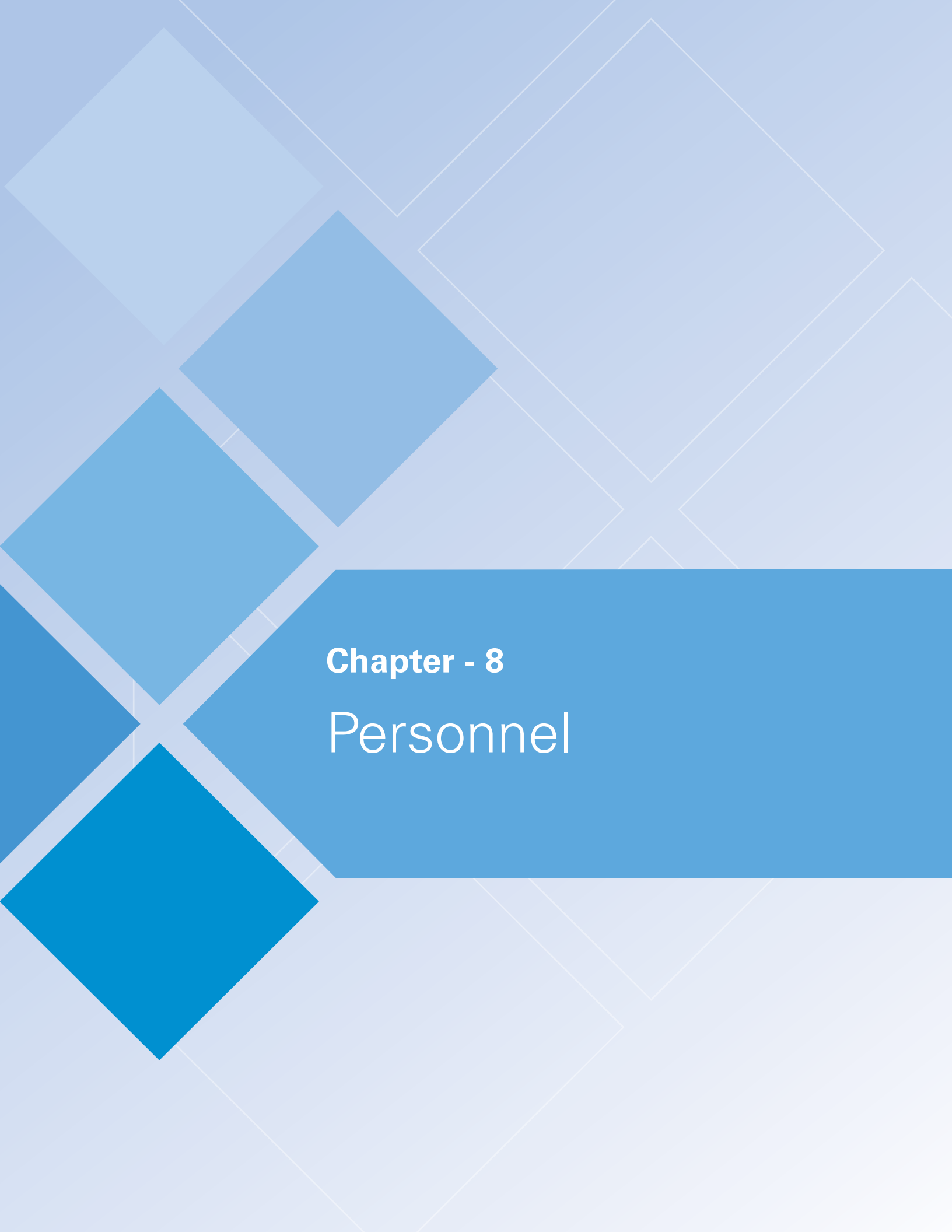
Activities	Amount (Gross)
Sale of Farm Produce	1.41
Sale of Publications	0.03
Guest House and Quarters	34.98
Institute Training Programmes	10.37
Sale of Application Fee	3.15
Services Rendered	7.02
Miscellaneous Receipts	126.83
Total	183.78

7.2.5 Net receipts from Sponsored Research Projects (₹ in Lakh)

Name of Project	Amount (Gross)
RKVY (IS) Project SAIDP Project	1.80
MONSANTO Project, Maize	2.36
ICAR NAARM-MANAGE Colaborative Project	2.42
PJTSAU Medicinal Plants	0.68
MGNREA, MANAGE Programme, NCSCM	0.54
Revenue from Consultancy Projects	7.98
Receipts from Schemes	0.72
Net receipts from Educational Programmes PGDMA 2014-16	0.22
Net receipts from Educational Programmes PGDMA 2015-17	79.43
Net receipts from Sponsored Programmes	40.80
Total	136.94

7.2.6 Plan Schemes (₹ in Lakh)

S. No.	Name of the Scheme	Budget outlay	Fund available/received including Opening Balance for 2017-18	Expenditure during 2017-18
1	NAIF-ZTMC	7.50	7.50	7.47
2	NAIF-ABI	3.00	22.50	13.95
3	Summer/Winter short Courses (4)	13.55	10.84	6.77
4	NICHE - TELAgE Project	56.30	56.30	40.58
5	KRISHI Project	24.16	22.95	6.82
6	Extra Mural Research Project "Documentation of Success stories of Agri/ Graduates turned Entrepreneurs"	2.90	2.90	2.64
7	Extra Mural Res.Project "Rising Start Ups of Agri-entrepreneurs:A Success Story"	3.96	3.96	3.73
8	FFP Project	24.80	24.15	24.14
	Total	136.17	151.09	106.11



Chapter - 8

Personnel



Personnel

8.1 Human Resources at the Academy (as on 31.03.2018)

Category	Sanctioned Strength	In position	Vacant
Scientific – RMP	02	01	01
Scientific – Faculty	60	32	28
Technical	40	36	04
Administrative	49	35	14
Skilled Support Staff	39	33	06
Total	190	137	53

Research Management Positions

1. Dr.Ch. Srinivasa Rao, Director (28-04-2017 onwards)
2. Dr. R. Kalpana Sastry, (Acting Director till 27-04-2017) and Joint Director (19-11 2017)
3. Dr. SK. Soam (Incharge Joint Director, 20-11-2017 onwards)

Scientific Staff

Agribusiness Management Division

1. Dr. Ranjit Kumar, Head
2. Dr. B Ganesh Kumar, Principal Scientist
3. Dr. N. Sivaramane, Senior Scientist
4. Dr. Prem Chand Meena, Senior Scientist
5. Dr. Sanjiv Kumar, Scientist

Education Systems Management Division

1. Dr. M Krishnan, Head
2. Dr. G. R. Ramakrishna Murthy, Principal Scientist
3. Dr. D. Thammi Raju, Principal Scientist
4. Dr. S. Senthil Vinayagam, Principal Scientist
5. Dr. V. V. Sumanth Kumar, Scientist

Extension Systems Management Division

1. Dr. Bharat S. Sontakki, Head

2. Dr. (Smt.) N. Sandhya Shenoy, Principal Scientist
3. Dr. V. K. Jayaraghavendra Rao, Principal Scientist
4. Dr. (Smt.) Surya Rathore, Principal Scientist
5. Dr. P Venkatesan, Senior Scientist

Human Resources Management Division

1. Dr. R. V. S Rao, I/c Head, HRM Division
2. Dr. P. Ramesh, Principal Scientist
3. Dr. K. H. Rao, Principal Scientist
4. Dr. Alok Kumar, Principal Scientist

Information and Communication Management Division

1. Dr. SK Soam, Head
2. Dr. A. Dhandapani, Principal Scientist
3. Dr. S. Ravichandran, Principal Scientist

4. Dr. M. Balakrishnan, Principal Scientist
5. Dr. N. Srinivasa Rao, Principal Scientist
6. Dr. P. D. Sreekanth, Senior Scientist
7. Dr. B. S. Yashavanth, Scientist

Research Systems Management Division

1. Dr. I. Sekar, Head
2. Dr. K. Srinivas, Principal Scientist
3. Dr. K. Kareemulla, Principal Scientist
4. Dr. M. B. Dastagiri, Principal Scientist
5. Dr. P. Krishnan, Principal Scientist
6. Dr. (Smt.) Manju Gerard, Principal Scientist

Administrative

Joint Director (Admn.) & Registrar

Shri Ashish Roy

Chief Finance & Accounts Officer

1. Shri S.George



Senior Administrative Officer

1. Shri J.N.L.Das

Administrative Officer

1. Shri W. Sreenivasa Bhat

Asst. Admn. Officer

1. Sh.P.G.Kohad
2. Sh.K.R.Ghanshyam
3. Sh.M.Dinesh

Private Secretary

1. Sh.N.Raghunath
2. Smt.A.Mercy

Junior Accounts Officer

1. Mrs.N.Vijayalakshmi

Assistants

1. Sh.T.Srinivas
2. Sh.G.Raj Reddy
3. Sh.C.Phani Raj
4. Ms.B.Padma Saroja
5. Sh.P.Srinivasu
6. Ms.G.Jessie Ecclicia
7. Sh.R.Chandra Babu
8. Sh.C.Julius Samuel
9. Sh.M.Sridhar

Personal Assistants

1. Sh.M.Venkatesh
2. Shri T.V.Ramadas
3. Ms.T.Vanisri
4. Ms.Y.Anuradha
5. Ms.S.Sesha Sai
6. Ms.K.K.Rukmani Ammal

Upper Division Clerks

1. Ms.Y.Gayathri
2. Sh.M.K.Samson
3. Ms.Rajashri Bokde
4. Sh.K.Suryanarayana

Stenographers Grade III

1. Ms.S.Shanthi
2. Ms.V.Shailaja

Lower Division Clerk

1. Sh.P.Swamy
2. Sh.M.Narsing Rao
3. Sh.M.Ashok
4. Sh.C.Bikshapathi

TECHNICAL

Grade T-9 (Category-III)

1. Dr.V.Murali, Chief Technical Officer (Garden Superintendent)
2. Dr.A.Debnath, Chief Technical Officer (Medical Officer)
3. Sh.Zameer Ahmed, Chief Technical Officer (Manager {HS})
4. Dr.Ch.Janardhan Rao, Chief Technical Officer (Video cameraman) *(Relieved on 10.04.2017 (AN) with lien for two (2) years to join the post of Reader at UoH.)*
5. Dr. M.A.Basith, Chief Technical Officer (Jr. Farm Superintendent)
6. Sh.K.V.Kumar, Chief Technical Officer (Statistical Assistant)
7. Sh.P.Vijender Reddy, Assistant Chief Technical Officer (Research Assistant)

Grade T(7-8) (Category-III)

1. Sh.P.Namdev, Assistant Chief Technical Officer (Graphic-cum-Visualizer)
2. Sh.Sohail Ahmad Khan, Assistant Chief Technical Officer (Junior Engineer-Civil)
3. Ms.G.Aneeja, Assistant Chief Technical Officer (Assistant Editor)
4. Sh.M.Shekhar Reddy, Assistant Chief Tech. Officer (Dark Room Assistant)
5. Dr.Ahire Laxman Maharu, Assistant Chief Technical Officer (Horticultural Technical Assistant)
6. Sh.M.Ravi, Sr.Technical Officer (Photographer-cum-Artist)
7. Sh.Sham Bahadur, Asst.Chief Technical Officer (Catering in-charge)

Grade T-6 (Category-III)

1. Sh.P.Mohan Singh, Sr.Technical Officer (Computer Assistant)
2. Sh.Ravi Viswanathan, Sr.Technical Officer (Editor-cum-Information Officer)
3. Ms.Savithri Murali, Sr.Technical Officer (Catering in-charge)

Grade T-5 (Category-II)

1. Sh.N.Naresh Kumar, Technical Officer (VCR-cum-Tele cine Operator)

2. Sh.B.S.N.Murthy, Technical Officer (Technician Electrical)
3. Sh.K.Obulapathi, Technical Officer (Technician Electrical)
4. Sh.M.K.Shamshuddin, Technical Officer (Technician Electrical)
5. Sh.M.K.Sonkusare, Technical Officer (Technician Electrical)
6. Sh.K.Shivaiah, Technical Officer (Technician)
7. Sh.D.Rajagopal Rao, Technical Officer (Media Operator)
8. Sh.S.Rajukumar, Technical Officer (Electronic Computer Operator)
9. Sh.S.Sunder Raj, Technical Officer (Media Operator)
10. Sh.T.Laxman, Technical Officer (Driver)

Grade T-4 (Category-II)

1. Sh.A.C.P.Rama Nageswara Rao, Sr.Technical Assistant (Lineman)
2. Sh.Pitla Srinivas, Sr.Technical Assistant (Proof Reader)
3. Sh.G.Muthyalu, Sr.Technical Assistant (Driver)
4. Sh.N.Ashok, Sr.Technical Assistant (Driver)
5. Sh.N.Prabhakar Rao, Sr.Technical Assistant (Plumber)
6. Sh.M.Srinivasa Rao, Sr.Technical Assistant (Pump Driver)
7. Sh.K.Veera Narasaiah, Sr.Technical Assistant (Carpenter-cum-Painter)

Grade T-3 (Category-II)

1. Sh.B.K.Venkatram, Technical Assistant (Pump Driver)
2. Sh.R.Siva Prasad, Technical Assistant (Driver)

SKILLED SUPPORT STAFF

1. Sh. S. Swamy, Asst .Gestt. Operator
2. Sh. Sirigiri Venkatesham, Xerox Machine Operator
3. Sh.Phool Kumar
4. Sh.P.Balraj
5. Ms.B.Santhamma
6. Ms.S.Shakuntala
7. Ms.B.Bharathamma
8. Sh.S.Jangaiah
9. Sh.Kumba Satyanarayana
10. Sh.K.Pentaiah



11. Sh.P.Yadaiah
12. Sh.G.Pentaiah
13. Sh.J.Chandraiah
14. Sh.M.Krishnaiah
15. Sh.R.Sattaiah
16. Ms.Indirapati Bharatamma
17. Sh.C.Chandramouli
18. Sh.G.Dasaratha
19. Ms.Nazia Parveen
20. Sh.L.Satyanarayana
21. Ms.N.Sukunamma
22. Ms.M.Yadamma
23. Ms.C.Narsubai
24. Ms.C.Kausalya
25. Sh.M.Ganesh Kumar
26. Sh.Chilumula Venkatesham
27. Sh.Sirigiri Narsimha
28. Sh.Kyasam Satyanarayana
29. Sh.B.Premdas
30. Mrs.A.Saroja
31. Sh.S.Nayab Rasool
32. Sh.B.Ashok
33. Sh.K.Daniel

APPOINTMENTS / NEW JOININGS

1. Shri J.N.L.Das, Sr.Admn.Officer joined on transfer from ICAR-IIHR,

- Bengaluru w.e.f. 10.04.2017 (FN)
2. Dr.Chelukumalli Srinivasa Rao, Director, ICAR-CRIDA, Hyderabad joined as Director on 28.04.2017 (AN)

PROMOTIONS

1. Shri M.Dinesh, Assistant promoted as Assistant Administrative Officer w.e.f. 01.08.2017
2. Shri M.Sridhar, UDC promoted as Assistant w.e.f. 19.12.2017
3. Smt.Savithri Murali, T-5 Technical Officer (Catering Incharge) has been promoted to the grade of T-6 Senior Technical Officer w.e.f. 01.01.2012
4. Dr.P.Vijender Reddy, T-7-8 Assistant Chief Technical Officer (Research Assistant) has been promoted to the grade of T-9 Chief Technical Officer w.e.f. 10.08.2017
5. Dr.P.D.Sreekanth, Sr.Scientist has been promoted as Principal Scientist w.e.f. 28.02.2017 under CAS
6. Dr.Alok Kumar, Principal Scientist has been promoted as Principal Scientist w.e.f. 02.09.2016 under CAS

TRANSFERS

1. Dr.Cheeli Janardhan Rao, Chief Technical Officer (Video Cameraman) – Relieved w.e.f. 10.04.2017 for joining the post of Reader at University of Hyderabad.

2. Dr.(Mrs.)J.Renuka, Assistant Director (OL) – Relieved in the AN of 15.04.2017 on her promotion as Dy.Director (OL) at ICAR-CIFT, Cochin
3. Shri Subash, S.P., Scientist – Transferred to ICAR-NIAEPR, New Delhi w.e.f. 08.05.2017 (AN)
4. Shri P.Venkatesh, UDC – Absorbed as Assistant at ICAR-ATARI, Hyderabad w.e.f. 14.12.2017

RETIREMENTS

1. Shri P.Neelakantam, Asst.Admn. Officer – superannuated on 31.07.2017 (AN)
2. Shri P.Anand Kumar, Private Secretary – Retired Voluntarily w.e.f. 05.10.2017 (FN)
3. Shri K.Radha Sujatha, Stenographer Grade-III – Retired Voluntarily w.e.f. 05.10.2017 (FN)
4. Dr.(Mrs.)R.Kalpna Sastry, Joint Director – Retired Voluntarily w.e.f. 19.11.2017 (FN)
5. Shri Prashant Gaikwad, T-4 Senior Technical Assistant (Binder) – Retired Voluntarily w.e.f. 18.12.2017 (FN)

OBITUARY

Dr.G.P.Reddy, Principal Scientist (Agricultural Economics) expired on 26.04.2017

8.2 List of the Committees (as on 31.03.2018)

Vigilance

1. Dr. S.K. Soam, Vigilance Officer (wef 11-03-2014)

Higher Education Committee

2. Dr. S.K. Soam, Joint Director (Incharge) Chairman
3. Dr. Bharat Sontakki, Head, XSM Division Member
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